# Operations & Examination Management Portal

**Project Plan and Functional Blueprint**

***Final Module-wise Functional Requirements, Fields, Permissions & Role-based Flow***

## 1. Executive Summary

The Operations & Examination Management Portal will replace fragmented Excel, email, WhatsApp and manual follow-up processes with a centralized, role-based system of record. It will provide controlled workflows for project setup, centre booking, consent, documents, audit, allocation, readiness, exam-day operations, commercial processing, invoicing, payments and closure.

> The operating principle is: CAPTURE ONCE → VALIDATE → APPROVE → EXECUTE → TRACK → ESCALATE → CLOSE → RETAIN HISTORY

The portal is intended to eliminate duplicate centre bookings, scattered centre information, document gaps, commercial ambiguity, weak accountability and limited management visibility.

## 2. Business Objectives

- Create one authoritative source for projects, centres, bookings, documents, audits and financial status.
- Prevent conflicting centre bookings through soft locks and database-level validation.
- Separate submission from verification and approval.
- Retain complete history for cancellations, replacements, revisions and changes.
- Provide ownership, due dates, reminders, escalations and audit trails.
- Give management real-time visibility from requirement receipt through payment and closure.

## 3. Success Measures

| Area | Target outcome |
|---|---|
| Booking control | No two users can successfully create conflicting bookings. |
| Data governance | Centre Master remains authoritative and maintains change history. |
| Compliance | Mandatory documents, eligibility and audit validity are enforced. |
| Commercial control | Approved cost sheets are versioned and require re-approval after change. |
| Accountability | Tasks, callbacks, incidents and tickets have owners and timestamps. |
| Management visibility | Dashboards show stage, pending work, overdue work, readiness, billing and payment. |

## 4. End-to-End Process

Purpose: provide a single, controlled flow from client requirement through project closure, with validation, approvals, ownership, escalation and retained history at every critical stage.

### a. Executive Flowmap

| Stage 1 | Stage 2 | Stage 3 | Stage 4 | Stage 5 |
|---|---|---|---|---|
| 01<br>Client Requirement | 02<br>Project Creation | 03<br>Exam Setup | 04<br>Centre Identification | 05<br>Eligibility + Blacklist Check |
| 06<br>Centre Lock | 07<br>Booking | 08<br>Centre Consent | 09<br>Document Collection | 10<br>Audit Check |
| 11<br>Audit / Technical Verification | 12<br>Final Centre Approval | 13<br>Client Allocation | 14<br>Allocation Validation | 15<br>Final Allocation Lock |
| 16<br>Commercial / Cost Sheet | 17<br>Commercial Approval | 18<br>Readiness Plan | 19<br>Calling / Follow-up | 20<br>Installation / Configuration |
| 21<br>Mock Day | 22<br>Final Readiness | 23<br>Exam Day | 24<br>Incident / Ticket Management | 25<br>Post-Exam |
| 26<br>Invoice Request | 27<br>Accounts / Payment | 28<br>Project MIS | 29<br>Project Closure | |

*Flow direction: read each row from left to right, then continue to the next row. Each stage produces the inputs required by the next stage.*

## b. Detailed Stage Map

| No. | Stage | Key activity / control | Primary owner |
|---|---|---|---|
| 01 | Client Requirement | Capture requirement, scope, client, exam and expected deliverables. | PM / Operations |
| 02 | Project Creation | Create the project and assign project manager, booking representative and priority. | PM / Operations |
| 03 | Exam Setup | Define exam, date, shift, city, candidate count and buffer requirement. | PM / Operations |
| 04 | Centre Identification | Search Centre Master or create a new centre pipeline record. | Operations |
| 05 | Eligibility + Blacklist Check | Validate centre for the specific client, exam and validity period. | Manager / Super Admin |
| 06 | Centre Lock | Apply temporary soft lock while the booking is being processed. | Booking Executive |
| 07 | Booking | Create booking after eligibility validation and conflict checks. | Booking Executive |
| 08 | Centre Consent | Centre accepts, requests discussion or rejects through secure portal/link. | Centre / Operations |
| 09 | Document Collection | Collect, validate and review mandatory and optional documents. | Centre / Operations |
| 10 | Audit Check | Check previous audit validity or initiate a new audit. | Audit Manager |
| 11 | Audit / Technical Verification | Perform checklist, evidence collection, technical scan and duplicate MAC check. | Auditor / IT |
| 12 | Final Centre Approval | Review audit, documents and eligibility; approve or return for rework. | Manager / Audit Manager |
| 13 | Client Allocation | Upload/import client allocation file and map columns. | Operations / PM |
| 14 | Allocation Validation | Validate centre, date, shift, candidate and buffer capacity. | System / Operations |
| 15 | Final Allocation Lock | Approve and lock final allocation; later changes require change request. | Manager / PM |
| 16 | Commercial / Cost Sheet | Calculate rates, charges, terms and amount due by approved allocation. | Operations / Finance |
| 17 | Commercial Approval | Approve cost sheet; revisions create new versions and require re-approval. | Manager / Finance |
| 18 | Readiness Plan | Plan installation, manpower, configuration, mock and final confirmations. | Operations / Centre |
| 19 | Calling / Follow-up | Create tasks, assign owners, record outcomes and schedule callbacks. | Operations |
| 20 | Installation / Configuration | Install and configure systems, network, devices and required materials. | Operations / IT / Centre |
| 21 | Mock Day | Conduct mock readiness activity and record exceptions. | Operations / Centre |
| 22 | Final Readiness | Confirm centre, manpower, systems, allocation and issue-free readiness. | Operations / Manager |
| 23 | Exam Day | Execute the exam with live operational monitoring. | Operations / Centre |
| 24 | Incident / Ticket Management | Raise, assign, escalate, resolve and close incidents or general tickets. | Operations / Manager |
| 25 | Post-Exam | Complete reconciliation, evidence, exceptions, materials and centre closure tasks. | Operations / Centre |
| 26 | Invoice Request | Request and collect centre invoice with supporting information. | Operations / Centre |
| 27 | Accounts / Payment | Verify invoice, process payment, record reference and outstanding balance. | Accounts |
| 28 | Project MIS | Refresh operational, compliance, commercial and financial reporting. | Management / MIS |
| 29 | Project Closure | Confirm all operational, financial and compliance actions are complete; retain history. | PM / Manager |

## c. Control Gates and Decision Points

| Gate | Decision point | Validation question | System response |
|---|---|---|---|
| G1 | Requirement validation | Is the client requirement complete and approved for project creation? | Return for clarification or create project. |
| G2 | Centre eligibility | Is the centre eligible for the specific client, exam and validity period? | Continue, reverify or seek authorized override. |
| G3 | Booking concurrency | Is there an active conflicting lock or booking for the same centre, project/exam/date/shift? | Block conflicting booking and show current processing status. |
| G4 | Centre consent | Has the authorized centre contact accepted the requirement? | Continue, follow up, discuss or cancel booking. |
| G5 | Document completeness | Are all mandatory documents submitted and internally approved? | Approve, request rework or reject; do not skip mandatory documents. |
| G6 | Audit validity | Is a previous audit valid, or has a new audit passed review? | Proceed or assign audit/re-audit. |
| G7 | Final centre approval | Have eligibility, consent, documents and audit requirements passed? | Approve centre or send to rework. |
| G8 | Allocation validation | Do date, shift, candidate count, buffer and centre capacity reconcile? | Show errors/warnings for correction and review. |
| G9 | Allocation lock | Has the authorized approver approved the final allocation? | Lock allocation; subsequent changes require controlled change request. |
| G10 | Commercial approval | Does the cost sheet match approved centre and allocation information? | Approve version or create revised version. |
| G11 | Final readiness | Are installation, configuration, manpower, mock and exceptions complete? | Mark ready or escalate blockers. |
| G12 | Financial closure | Are invoice and payment reconciled with no outstanding action? | Close financials or keep project open. |

## d. Exception and History Flows

| Exception | Controlled flow |
|---|---|
| Cancelled booking | Cancellation reason → retain original booking → link replacement centre → create new booking → preserve both records. |
| Document rework | Rejected/rework status → reason → centre resubmission → new version → review → approval. |
| Audit failure | Rework → corrective action → re-audit → review → approval or rejection. |
| Allocation change after lock | Change request → reason → impact validation → approval → new allocation version → new lock. |
| Cost change after approval | Create revised cost sheet version → commercial review → re-approval → retain prior approved version. |
| Incident escalation | Incident raised → severity → owner → first response → escalation timer → resolution → verification → closure. |
| Overdue task | Due date missed → reminder → escalation to manager → management visibility until closure. |

## e. Required System Outputs

| Output category | Expected output |
|---|---|
| Operational record | Project, centre, booking, consent, document, audit, allocation, readiness, incident, ticket and logistics records. |
| Approval record | Submitter, reviewer, decision, timestamp, rejection/rework reason and approver. |
| History record | Old value, new value, changed by, changed time and change reason. |
| Task record | Owner, due date, reminder, outcome, callback and escalation status. |
| Management view | Project stage, centre coverage, pending work, overdue work, readiness, exam-day issues, invoices, payment outstanding and closure readiness. |
| Security record | Role, scope, access decision and critical activity audit trail. |

## f. Final Status Flow

Requirement Received → Validated → Project Created → Centre Identified → Eligible → Soft Locked → Booked → Consent Received → Documents Approved → Audit Approved → Centre Approved → Allocation Validated → Allocation Locked → Cost Sheet Approved → Readiness Planned → Ready → Exam Executed → Post-Exam Complete → Invoice Verified → Payment Cleared → MIS Updated → Project Closed.

Design principle: no critical record is silently deleted or overwritten. Every significant transition, exception, approval and revision remains traceable.

### Cancellation and replacement flow

Cancelled Booking → Cancellation Reason → Original Record Retained → Replacement Centre → New Booking → Both Records Linked. Cancellation must never delete or overwrite the original booking.

# 5. Development Priorities

| Priority | Scope | Purpose |
|---|---|---|
| P1 – Critical | Project Management; Centre Master; Eligibility; Booking & Locking; Centre Consent; Document Management; Commercial / Cost Sheet; Invoice Request; Centralized Portal | Core MVP and single source of truth. |
| P2 – Important | Calling; Accountability; Audit Compliance; Incident; Ticket; Cancellation & Replacement; Mock / Readiness; Enhanced MIS | Operational control and process maturity. |
| P3 – Future | WhatsApp API; Email-to-Ticket; Telephony integration | Integration extensions that should not delay MVP. |

# 6. Solution Architecture

Management / CEO → MIS / BI → Operations, Finance and Compliance → Centre Master Database → Centre / Venue Portal. Cross-cutting capabilities: RBAC, audit trail, notifications, escalation, documents, versioning, history and security.

### Data layers

MASTER DATA → TRANSACTION DATA → WORKFLOW / APPROVAL → DOCUMENTS → AUDIT LOG → REPORTING. Avoid one giant table; maintain logically separate entities and relationships.

# 7. Functional Module Plan

## M01 – Authentication, Users & Roles

| Category | Details |
|---|---|
| **Module Name** | Authentication, Users & Roles |
| **Primary User** | Super Admin |
| **Purpose** | Manage users, roles, permissions, access scopes and user lifecycle. |
| **Process Flow** | User Creation → Role Assignment → Scope Assignment → Activation → Login → Access → Activity Logging → Deactivation / Role Change |
| **Mandatory / Key Fields** | User ID; Employee Name; Official Email; Mobile; Department; Role; Client Scope; Project Scope; Centre Scope; Status; Created By; Created Date/Time |
| **Super Admin Permissions** | Full user, role, scope and permission administration. |
| **Manager Permissions** | View team members and approve or request delegated access where permitted. |
| **Normal User Permissions** | Access only assigned modules and assigned data. |
| **Business Rules** | A user must have an assigned role and applicable scope before activation. Deactivated users cannot log in. Role or scope changes must be recorded in the audit trail. |
| **System Outputs** | User account; assigned role; access scope; activation status; activity log; deactivation or role-change history. |

**Permission Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User |
|---|---|---|---|---|---|
| Create user | Full | No | No | No | No |
| Edit user | Full | No | No | No | No |
| Activate / deactivate user | Full | No | No | No | No |
| Assign role | Full | No | No | No | No |
| Assign client scope | Full | View assigned scope | No | No | No |
| Assign project scope | Full | Assign / recommend | No | View assigned projects | Own centre scope only |
| Assign centre scope | Full | Recommend | Assigned audit centres | Assigned centres | Own centre only |
| Configure permissions | Full | No | No | No | No |
| View activity logs | Full | Project scope | Audit scope | Assigned activity | Own activity where permitted |

**Field-Level Table:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| **User ID** | System-generated | Text / ID | Unique identifier for the user. |
| **Employee Name** | Yes | Text | Full name of the employee. |
| **Official Email** | Yes | Email | Official login and communication email. |
| **Mobile** | Yes | Number | Registered contact number. |
| **Department** | Yes | Dropdown | Department to which the user belongs. |
| **Role** | Yes | Dropdown | Assigned system role. |

| | | | |
|---|---|---|---|
| **Client Scope** | Conditional | Multi-select | Clients accessible to the user. |
| **Project Scope** | Conditional | Multi-select | Projects accessible to the user. |
| **Centre Scope** | Conditional | Multi-select | Centres accessible to the user. |
| **Status** | Yes | Dropdown | Draft, Active, Suspended or Deactivated. |
| **Created By** | System-generated | User reference | User who created the account. |
| **Created Date/Time** | System-generated | Date/time | Date and time of account creation. |

# M02 – Client & Project Management

| Category | Details |
|---|---|
| **Module Name** | Client & Project Management |
| **Primary User** | Project Manager / Operations |
| **Purpose** | Serve as the central parent module for managing the complete project lifecycle from client requirement through project closure. |
| **Process Flow** | Client Requirement → Validation → Project Creation → Exam Setup → Cities → Candidate Requirement → Team Assignment → Execution |
| **Mandatory / Key Fields** | Project ID; Client; Exam Name; Exam Code; Requirement Received Date; Exam Date; Shift; City; Candidate Requirement; Buffer Requirement; Booking Representative; Project Manager; Priority; Project Stage; Project Status; SLA / Deadline |
| **Business Rules** | Every project must have a unique Project ID. The project must be validated before execution begins. A Project Manager and Booking Representative must be assigned before centre identification or booking. Project stage and status must be updated throughout the lifecycle. |
| **System Outputs** | Validated project record; exam setup; city-wise requirement; candidate and buffer requirement; assigned project team; project stage; project status; SLA tracking; management visibility. |

**Project Lifecycle Stages:**

| Stage No. | Project Stage | Main Activities | Expected Output |
|---|---|---|---|
| 1 | **Requirement** | Receive and record client requirement. | Requirement record |
| 2 | **Centre Identification** | Identify suitable centres based on city, capacity and eligibility. | Centre pipeline |
| 3 | **Booking** | Lock and book eligible centres. | Confirmed booking |
| 4 | **Documentation** | Collect and verify required centre documents. | Approved document set |
| 5 | **Audit** | Complete audit and technical verification. | Audit-approved centre |
| 6 | **Allocation** | Upload and validate client allocation. | Validated allocation |
| 7 | **Readiness** | Plan installation, manpower, configuration and confirmations. | Readiness plan |
| 8 | **Mock** | Conduct mock activity and record exceptions. | Mock status |
| 9 | **Exam** | Execute the exam and monitor operations. | Exam execution record |
| 10 | **Post Exam** | Complete reconciliation and post-exam activities. | Post-exam completion |
| 11 | **Billing** | Manage invoice request, verification and payment tracking. | Billing status |
| 12 | **Closure** | Confirm all operational, financial and compliance actions are complete. | Closed project |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| **Project ID** | System-generated | Text / ID | Unique identifier for the project. |
| **Client** | Yes | Dropdown / Lookup | Client associated with the examination project. |
| **Exam Name** | Yes | Text / Lookup | Name of the examination. |
| **Exam Code** | Yes | Text | Unique code assigned to the examination. |
| **Requirement Received Date** | Yes | Date | Date on which the client requirement was received. |
| **Exam Date** | Yes | Date | Planned date of examination. |
| **Shift** | Yes | Dropdown / Multi-select | Examination shift or shifts. |
| **City** | Yes | Dropdown / Multi-select | City or cities where the exam will be conducted. |
| **Candidate Requirement** | Yes | Number | Total number of candidates to be accommodated. |
| **Buffer Requirement** | Yes | Number / Percentage | Additional seats required as operational buffer. |
| **Booking Representative** | Yes | User Lookup | Employee responsible for centre booking activities. |
| **Project Manager** | Yes | User Lookup | Manager accountable for the project. |
| **Priority** | Yes | Dropdown | Low, Medium, High or Critical. |
| **Project Stage** | System / Controlled | Dropdown | Current stage in the project lifecycle. |
| **Project Status** | Yes | Dropdown | Draft, Active, On Hold, Completed, Cancelled or Closed. |
| **SLA / Deadline** | Conditional | Date / Date-Time | Deadline for the project or specific milestone. |

**Role and Permission Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User |
|---|---|---|---|---|---|
| Create project | Full | Create | No | Request / support | No |
| Edit project details | Full | Full assigned scope | View relevant project | Edit assigned operational data | No |
| Set exam details | Full | Create / edit | View | View | No |
| Add cities | Full | Create / edit | View | Suggest / update assigned scope | No |
| Enter candidate requirement | Full | Create / edit | View | Update assigned scope | No |
| Assign project team | Full | Assign / recommend | No | View own assignment | No |
| Update project stage | Full | Approve / update | Update audit stage | Update assigned activities | Update centre activities only |
| Update project status | Full | Approve / update | Recommend audit status | Request status change | No |
| View SLA / | Full | Full assigned | Audit deadlines | Assigned tasks | Centre-specific |

| | | | | | |
|---|---|---|---|---|---|
| deadlines | | project | | | deadlines |
| View project MIS | Full | Full project scope | Audit-related scope | Assigned project scope | Own centre scope |

**Validation and Control Rules:**

| Control Area | Requirement |
|---|---|
| **Project creation** | A project cannot be activated without a client, exam, exam date, city, candidate requirement, Project Manager and priority. |
| **Exam setup** | Exam date, shift and city must be defined before centre identification begins. |
| **Candidate capacity** | Candidate requirement plus buffer must be considered during centre selection and allocation validation. |
| **Team assignment** | A Booking Representative and Project Manager must be assigned before booking activities begin. |
| **Stage control** | Users cannot skip required stages unless an authorized role performs an approved override. |
| **Status control** | Project status changes must be recorded with user, timestamp and reason where applicable. |
| **SLA monitoring** | Overdue project milestones should generate reminders and escalations. |
| **History retention** | Previous values, changes, approvals and cancelled project records must remain available for audit and reporting. |

# M03 – Centre Master

| Category | Details |
|---|---|
| **Module Name** | Centre Master |
| **Primary User** | Operations / Admin |
| **Purpose** | Maintain the single authoritative database of all examination centres, their details, capacity, infrastructure, contacts, usage history and audit status. |
| **Process Flow** | Search Existing Centre → Eligibility Check → Use Master OR Pipeline / New Centre → Verification → Approval → Centre Master |
| **Mandatory / Key Fields** | Centre ID; Centre Name; Legal/Registered Name; Address; City; State; PIN; Capacity; System Count; Infrastructure Type; Contact Person; Mobile; Email; Centre Status; Source; Last Used Date; Last Audit Date |
| **Business Rules** | Centre Master is the authoritative source for approved centre information. A centre submission does not automatically become master data. New or updated centre information must pass verification and approval before becoming authoritative. |
| **System Outputs** | Approved centre master record; centre search result; new-centre pipeline record; verified capacity and infrastructure details; centre status; audit and usage history. |

**Centre Lifecycle:**

| Stage No. | Centre Stage | Main Activities | Expected Output |
|---|---|---|---|
| **1** | **Search Existing Centre** | Search Centre Master using centre name, city, PIN, contact or Centre ID. | Existing-centre result or no-match result |
| **2** | **Eligibility Check** | Check centre eligibility for the relevant client, exam, city, date and capacity. | Eligible, restricted or under-review status |
| **3** | **Use Master** | Select an existing approved centre from Centre Master. | Centre available for booking |
| **4** | **Pipeline / New Centre** | Create a submission for a centre not found in the master database. | New-centre pipeline record |
| **5** | **Verification** | Validate centre details, legal identity, address, capacity, systems, infrastructure and contact information. | Verified centre submission |
| **6** | **Approval** | Authorized manager reviews and approves or rejects the submission. | Approval decision |
| **7** | **Centre Master** | Convert approved submission into an authoritative master record or update the existing record. | Active Centre Master record |

**Existing Centre vs New Centre Flow:**

| Scenario | System Action | Approval Requirement | Final Result |
|---|---|---|---|
| **Existing approved centre found** | Display master record and current status. | Follow normal eligibility and booking controls. | Centre can proceed if eligible. |
| **Existing centre with outdated information** | Create update request and retain old values. | Internal verification and authorized approval. | New version of Centre Master data. |
| **Centre not found** | Create pipeline / new-centre submission. | Verification and approval required. | New Centre Master record after approval. |
| **Duplicate centre suspected** | Flag possible duplicate based on name, address, PIN, contact or other identifiers. | Admin review required. | Merge, reject or continue as separate centre. |
| **Centre rejected** | Retain submission, rejection reason and evidence. | No master creation. | Rework or resubmission |

| | | | |
|---|---|---|---|
| | | | permitted. |
| **Centre deactivated** | Preserve historical usage and booking records. | Authorized role required. | Centre unavailable for new booking. |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| **Centre ID** | System-generated | Text / ID | Unique identifier for the centre. Generated when the centre is approved or created according to the configured process. |
| **Centre Name** | Yes | Text | Common or operating name of the centre. |
| **Legal / Registered Name** | Yes | Text | Official registered name of the centre or legal entity. |
| **Address** | Yes | Long text | Complete physical address of the centre. |
| **City** | Yes | Dropdown / Lookup | City in which the centre is located. |
| **State** | Yes | Dropdown / Lookup | State in which the centre is located. |
| **PIN** | Yes | Numeric | Postal identification number for the centre address. |
| **Capacity** | Yes | Number | Maximum approved candidate capacity. |
| **System Count** | Yes | Number | Number of available examination systems or machines. |
| **Infrastructure Type** | Yes | Dropdown / Multi-select | Infrastructure classification, such as owned, leased, school, college or commercial facility. |
| **Contact Person** | Yes | Text | Authorized centre contact person. |
| **Mobile** | Yes | Mobile number | Primary contact number for centre coordination. |
| **Email** | Yes | Email | Official centre communication email address. |
| **Centre Status** | Yes | Dropdown | Pipeline, Submitted, Under Verification, Approved, Active, On Hold, Restricted, Inactive or Rejected. |
| **Source** | Yes | Dropdown / Text | Source through which the centre was identified or submitted. |
| **Last Used Date** | System / Conditional | Date | Date on which the centre was last used for an examination. |
| **Last Audit Date** | System / Conditional | Date | Date of the most recent completed audit. |

**Role and Permission Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User |
|---|---|---|---|---|---|
| Search Centre Master | Full | Full | Audit scope | Full | Own centre |
| Create new-centre submission | Full | Approve / review | Suggest based on audit | Create / suggest | Submit own details |
| Edit centre information | Full | Approve / edit | Update technical findings | Suggest / request change | Edit submitted details |
| Approve centre record | Full | Full approval | No | No | No |
| Update capacity | Full | Approve / review | Verify during audit | Suggest update | Submit revised data |
| Update infrastructure | Full | Approve / review | Verify during audit | Suggest update | Submit revised data |
| Change centre status | Full | Approve / | Recommend | Request change | No |

| | | | | | |
|---|---|---|---|---|---|
| | | request | restriction | | |
| View centre history | Full | Full | Assigned audit scope | Assigned centre scope | Own centre history |
| Deactivate centre | Full | Approve / request | Recommend | Request | No |

**Verification Checklist:**

| Verification Area | Checks |
|---|---|
| **Centre identity** | Centre name, legal name and duplicate-centre search. |
| **Address** | Full address, city, state and PIN validation. |
| **Capacity** | Candidate capacity, system count and buffer suitability. |
| **Infrastructure** | Building type, rooms, power, network, security and other required facilities. |
| **Contact details** | Authorized contact person, mobile number and email address. |
| **Documents** | Registration, authorization, ownership/lease and other required evidence. |
| **Audit status** | Last audit date, audit validity and outstanding observations. |
| **Commercial readiness** | Required rate or commercial information, where applicable. |
| **Eligibility** | Client-specific, exam-specific and date-specific eligibility. |

**Power and Backup Infrastructure:**

| Category | Field | Mandatory | Field Type | Description |
|---|---|---|---|---|
| **UPS** | UPS Available | Yes | Yes / No | Confirms whether UPS backup is available at the centre. |
| **UPS** | UPS Quantity | Conditional | Number | Number of UPS units available. Required when UPS Available is Yes. |
| **UPS** | UPS Capacity | Conditional | Text / Number | Capacity of UPS, such as kVA, kW or VA. |
| **UPS** | UPS Coverage | Conditional | Dropdown / Text | Systems or rooms supported by the UPS. |
| **UPS** | UPS Backup Duration | Conditional | Number | Expected backup duration in minutes or hours. |
| **UPS** | UPS Type | Conditional | Dropdown | Online, offline, line-interactive or other. |
| **UPS** | UPS Condition | Conditional | Dropdown | Excellent, Good, Requires Service or Not Usable. |
| **UPS** | UPS Last Service Date | Conditional | Date | Date of the latest UPS service or maintenance. |
| **UPS** | UPS Next Service Due Date | Conditional | Date | Planned or scheduled next service date. |
| **UPS** | UPS Service Report | Conditional | File Upload | Supporting maintenance or service document. |
| **UPS** | UPS Remarks | No | Long Text | Additional information or limitations. |
| **DG** | DG Available | Yes | Yes / No | Confirms whether a diesel generator is available. |
| **DG** | DG Quantity | Conditional | Number | Number of generators available. |
| **DG** | DG Capacity | Conditional | Text / Number | DG capacity in kVA or other configured unit. |

| | | | | |
|---|---|---|---|---|
| **DG** | DG Coverage | Conditional | Dropdown / Text | Systems, rooms or building areas supported by the DG. |
| **DG** | DG Last Service Date | Conditional | Date | Date of the latest DG service. |
| **DG** | DG Next Service Due Date | Conditional | Date | Date on which the next DG service is due. |
| **DG** | DG Service Report | Conditional | File Upload | Latest DG maintenance or service report. |
| **DG** | Diesel Tank Capacity | Conditional | Number | Total diesel storage capacity in litres. |
| **DG** | Current Diesel Availability | Conditional | Number | Diesel available at the time of submission or inspection, in litres. |
| **DG** | Expected Backup Duration | Conditional | Number | Estimated backup duration based on available diesel and load. |
| **DG** | DG Auto Start | Conditional | Yes / No | Indicates whether the DG starts automatically during power failure. |
| **DG** | DG Changeover Time | Conditional | Number | Approximate time required to switch to DG supply. |
| **DG** | DG Condition | Conditional | Dropdown | Excellent, Good, Requires Service or Not Usable. |
| **DG** | DG Remarks | No | Long Text | Additional operational or maintenance information. |
| **Electricity** | Electricity Available | Yes | Yes / No | Confirms availability of regular electricity supply. |
| **Electricity** | Electricity Connection Type | Yes | Dropdown | Single phase, three phase or other. |
| **Electricity** | Sanctioned Load | Yes | Number | Load sanctioned by the electricity provider, preferably in kVA or kW. |
| **Electricity** | Available Load | Yes | Number | Usable load currently available at the centre. |
| **Electricity** | Electricity Provider | No | Text | Name of the electricity distribution company or provider. |
| **Electricity** | Meter Number | No | Text | Electricity meter identification number. |
| **Electricity** | Power Supply Reliability | Yes | Dropdown | Reliable, Occasionally interrupted or Frequently interrupted. |
| **Electricity** | Recent Power Failure | Yes | Yes / No | Whether the centre has experienced a recent power failure. |
| **Electricity** | Average Outage Duration | Conditional | Number | Average duration of a power outage, in minutes or hours. |
| **Electricity** | Separate Exam Circuit | Yes | Yes / No | Confirms whether examination systems can operate on a dedicated circuit. |
| **Electricity** | Electrical Earthing Available | Yes | Yes / No | Confirms proper electrical earthing for systems and equipment. |
| **Electricity** | Electrical Inspection Report | Conditional | File Upload | Supporting electrical inspection or safety report. |
| **Electricity** | Electricity Remarks | No | Long Text | Additional details, limitations or planned improvements. |

**Centre Power Details Form**

**The centre-facing form can be organized into four sections:**

**1. UPS Details**

| Question | Response |
|---|---|
| Is UPS backup available? | Yes / No |
| Number of UPS units | Numeric input |
| Total UPS capacity | Numeric input with unit |
| Which systems or rooms are covered? | Text / selection |
| Expected backup duration | Minutes / hours |
| UPS last service date | Date |
| Upload latest UPS service report | File upload |
| Current UPS condition | Dropdown |
| Remarks | Text area |

**2. DG Details:**

| Question | Response |
|---|---|
| **Is DG backup available?** | Yes / No |
| **DG capacity** | Numeric input with unit |
| **DG last service date** | Date |
| **DG next service due date** | Date |
| **Diesel tank capacity** | Litres |
| **Current diesel available** | Litres |
| **Expected backup duration** | Minutes / hours |
| **Does DG start automatically?** | Yes / No |
| **DG changeover time** | Minutes |
| **Upload latest DG service report** | File upload |
| **Current DG condition** | Dropdown |
| **Remarks** | Text area |

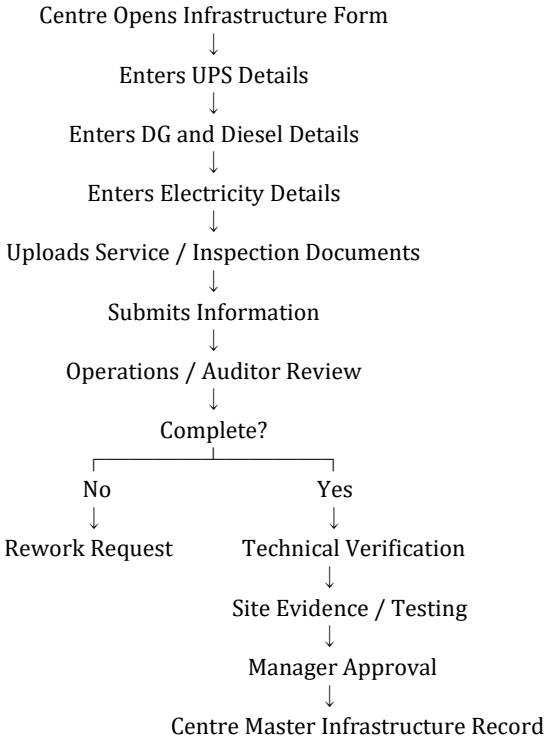
**3. Electricity Details:**

| Question | Response |
|---|---|
| **Is regular electricity available?** | Yes / No |
| **Connection type** | Single phase / Three phase |
| **Sanctioned load** | kVA / kW |
| **Available usable load** | kVA / kW |
| **Power supply reliability** | Reliable / Occasional interruptions / Frequent interruptions |
| **Average outage duration** | Minutes / hours |
| **Is there a separate exam circuit?** | Yes / No |
| **Is proper earthing available?** | Yes / No |
| **Upload electrical inspection report** | File upload |
| **Remarks** | Text area |

**4. Verification Details:**

| Field | Description |
|---|---|
| **Submitted By** | Centre user who provided the information. |
| **Submission Date/Time** | System-generated timestamp. |
| **Verified By** | Operations, Audit or Technical verifier. |
| **Verification Date/Time** | System-generated timestamp. |
| **Verification Status** | Pending, Verified, Rework or Rejected. |
| **Verification Remarks** | Comments from the verifier. |
| **Evidence Photos** | Photos of UPS, DG, diesel tank, meter and electrical panels. |
| **Service Documents** | Uploaded maintenance and inspection records. |
| **Approved By** | Authorized manager or audit approver. |
| **Approval Date/Time** | System-generated timestamp. |

**Verification and Approval Flow:**




**Important Business Rules:**

| Rule | Requirement |
|---|---|
| **UPS dependency** | If UPS Available = Yes, UPS capacity, coverage, condition and service date should be captured. |
| **DG dependency** | If DG Available = Yes, DG capacity, last service date, diesel tank capacity and current diesel availability should be mandatory. |
| **Diesel monitoring** | Diesel capacity and current diesel level should be recorded separately. Tank capacity alone does not confirm exam-day readiness. |
| **Service expiry** | The system should flag DG or UPS service dates that are overdue or approaching. |
| **Electricity validation** | Sanctioned load and usable load should be recorded separately. |
| **Technical verification** | Operations or an auditor should verify the submitted details against documents, photographs or physical inspection. |

| | |
|---|---|
| **Exam readiness** | The centre should not receive final readiness approval if required backup power details are missing or unresolved. |
| **Change history** | Any change to capacity, service date, diesel availability or power status must retain old value, new value, user, timestamp and reason. |
| **Confidential access** | Centre users may update their own information, while internal remarks and risk assessments should be restricted by role. |
| **Evidence retention** | Service reports, inspection certificates and photographs must remain linked to the centre and relevant audit. |

The Centre Master should contain not only general centre details but also verified operational infrastructure information:

**Centre Master = Centre Identity + Capacity + Systems + Infrastructure + Power Backup + Electricity + Contact Details + Eligibility + Audit History**

# M04 – Eligibility & Blacklist

| Category | Details |
|---|---|
| **Module Name** | Centre Eligibility & Blacklist |
| **Primary User** | Manager / Super Admin |
| **Purpose** | Determine whether a centre can be used for a specific client, examination, project and validity period while controlling restrictions, re-verification and authorized overrides. |
| **Process Flow** | Centre Selected → Eligibility / Blacklist Check → Eligible OR Reverification → Authorized Override → Approval |
| **Mandatory / Key Fields** | Centre ID; Client ID; Exam ID; Restriction Type; Reason Category; Confidential Reason; Effective Date; Review Date; Expiry Date; Status; Created By; Approved By; Override Reason; Override Authority |
| **Business Rule** | Eligibility is scoped by Centre + Client + Exam + Restriction + Validity. It must not be implemented as one global blacklisted = true/false field. |
| **System Outputs** | Eligibility result; restriction status; re-verification task; approved override record; approval history; centre availability for the relevant scope. |

**Eligibility Decision Flow:**

| Step | Activity | System Action | Output |
|---|---|---|---|
| 1 | Centre Selected | User selects a centre for a project or booking. | Centre and project context captured. |
| 2 | Eligibility Check | System checks centre, client, exam, date and applicable restrictions. | Eligibility result generated. |
| 3 | Eligible | No active restriction applies to the selected scope. | Centre may proceed to booking or the next workflow stage. |
| 4 | Not Eligible / Restricted | An active restriction applies. | Booking or use is blocked. |
| 5 | Reverification | Authorized user initiates review of the restriction or centre status. | Reverification task and supporting evidence request. |
| 6 | Authorized Override | Authorized authority provides a justified exception, where permitted. | Override request with reason and authority. |
| 7 | Approval | Designated manager reviews the eligibility decision or override. | Approved, rejected or sent for rework. |
| 8 | Final Result | System applies the approved decision for the defined scope and validity. | Eligible, Restricted, Rejected or Under Review. |

**Scoped Eligibility Examples:**

| Centre | Client | Exam | Restriction | Result |
|---|---|---|---|---|
| Centre A | SSC | SSC Examination | Blacklisted | Not eligible for SSC scope. |
| Centre A | NTA | NTA Examination | None | Eligible for NTA scope, subject to other checks. |
| Centre A | Other Client | Other Examination | None | Eligible if all other requirements pass. |
| Centre B | SSC | SSC Examination | Under Review | Cannot proceed until review is completed. |
| Centre C | NTA | NTA Examination | Temporary restriction expired | Eligible if the expiry has been processed. |

A restriction for one client or examination must not automatically make the centre ineligible for every other client or examination unless a separate organization-wide restriction exists.

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| **Centre ID** | Yes | Lookup / ID | Centre to which the eligibility or restriction record applies. |
| **Client ID** | Yes | Lookup / ID | Client-specific scope of the eligibility decision. |
| **Exam ID** | Yes | Lookup / ID | Examination-specific scope of the decision. |
| **Restriction Type** | Yes | Dropdown | Blacklist, Temporary Hold, Under Review, Conditional Eligibility, Compliance Restriction or Organization-wide Restriction. |
| **Reason Category** | Yes | Dropdown | Document issue, audit failure, integrity concern, capacity issue, commercial issue, operational issue or other configured category. |
| **Confidential Reason** | Conditional | Long text / Restricted | Detailed explanation visible only to authorized roles. |
| **Effective Date** | Yes | Date | Date from which the restriction or eligibility decision applies. |
| **Review Date** | Conditional | Date | Planned date for re-evaluation of the restriction. |
| **Expiry Date** | Conditional | Date | Date on which the restriction automatically expires, subject to configured rules. |
| **Status** | Yes | Dropdown | Draft, Active, Under Review, Approved, Rejected, Expired, Overridden or Closed. |
| **Created By** | System-generated | User reference | User who created the eligibility or restriction record. |
| **Approved By** | Conditional | User reference | Authorized user who approved the decision or override. |
| **Override Reason** | Conditional | Long text | Business justification for allowing the centre despite an active restriction. |
| **Override Authority** | Conditional | User / Role reference | Authority permitted to approve the override. |

**Role and Permission Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User |
|---|---|---|---|---|---|
| Search Centre Master | Full | Full | Audit scope | Full | Own centre |
| Create new-centre submission | Full | Approve / review | Suggest based on audit | Create / suggest | Submit own details |
| Edit centre information | Full | Approve / edit | Update technical findings | Suggest / request change | Edit submitted details |
| Approve centre record | Full | Full approval | No | No | No |
| Update capacity | Full | Approve / review | Verify during audit | Suggest update | Submit revised data |
| Update infrastructure | Full | Approve / review | Verify during audit | Suggest update | Submit revised data |
| Change centre status | Full | Approve / request | Recommend restriction | Request change | No |
| View centre history | Full | Full | Assigned audit scope | Assigned centre scope | Own centre history |

| | | | | | |
|---|---|---|---|---|---|
| Deactivate centre | Full | Approve / request | Recommend | Request | No |

**Restriction Status Flow:**

| Current Status | Action | Next Status |
|---|---|---|
| **Draft** | Restriction or eligibility record is prepared. | Under Review |
| **Under Review** | Manager or Super Admin evaluates evidence. | Approved, Rejected or Rework |
| **Approved** | Restriction is active for the configured scope. | Active |
| **Active** | Centre is restricted for the applicable client, exam or period. | Under Review, Overridden, Expired or Closed |
| **Under Review** | Reverification is in progress. | Eligible, Restricted or Rejected |
| **Overridden** | Authorized exception has been approved. | Eligible for defined scope |
| **Expired** | Restriction validity period has ended. | Review Required or Eligible |
| **Rejected** | Restriction or override request was not approved. | Closed or Rework |

**Visibility Rules:**

| User Type | Information Visible |
|---|---|
| **Admin** | Full eligibility records, restriction history, confidential reason, approvals and overrides. |
| **Project Manager** | Eligibility status, relevant reason category, review details and authorized confidential information. |
| **Operations Executive** | Eligible, Not Eligible, Under Review or Override Approved. Confidential reasons may be hidden. |
| **Auditor** | Eligibility status and information relevant to assigned audit work. |
| **Accounts** | No eligibility detail unless required for a financial workflow. |
| **Centre User** | Only requests, instructions or non-confidential information needed to correct or submit centre details. |

For example, an Operations user may see:

**Centre Not Eligible – SSC Examination**

An authorized Manager may additionally see:

**Confidential reason: previous integrity incident; review due on 15 October 2026.**

**Key Business Controls:**

| Control | Requirement |
|---|---|
| **Scoped decision** | Every eligibility decision must contain Centre ID, Client ID and Exam ID. |
| **Validity control** | Effective, review and expiry dates must be supported where applicable. |
| **No global blacklist flag** | Do not use a single centre-level Boolean field as the complete eligibility model. |
| **Booking block** | A centre with an active applicable restriction cannot be booked without an authorized override. |
| **Override control** | Overrides require a reason, authority, approval and timestamp. |

| | |
|---|---|
| **Confidentiality** | Detailed restriction reasons must be protected through role-based access. |
| **Automatic reminders** | Upcoming review dates and expiring restrictions should generate notifications. |
| **Expiry handling** | Expired restrictions should not silently become eligible; the system should apply the configured review rule. |
| **Audit trail** | Creation, modification, approval, override and closure must be logged. |
| **History retention** | Previous restrictions and decisions must remain available for reporting and audit. |

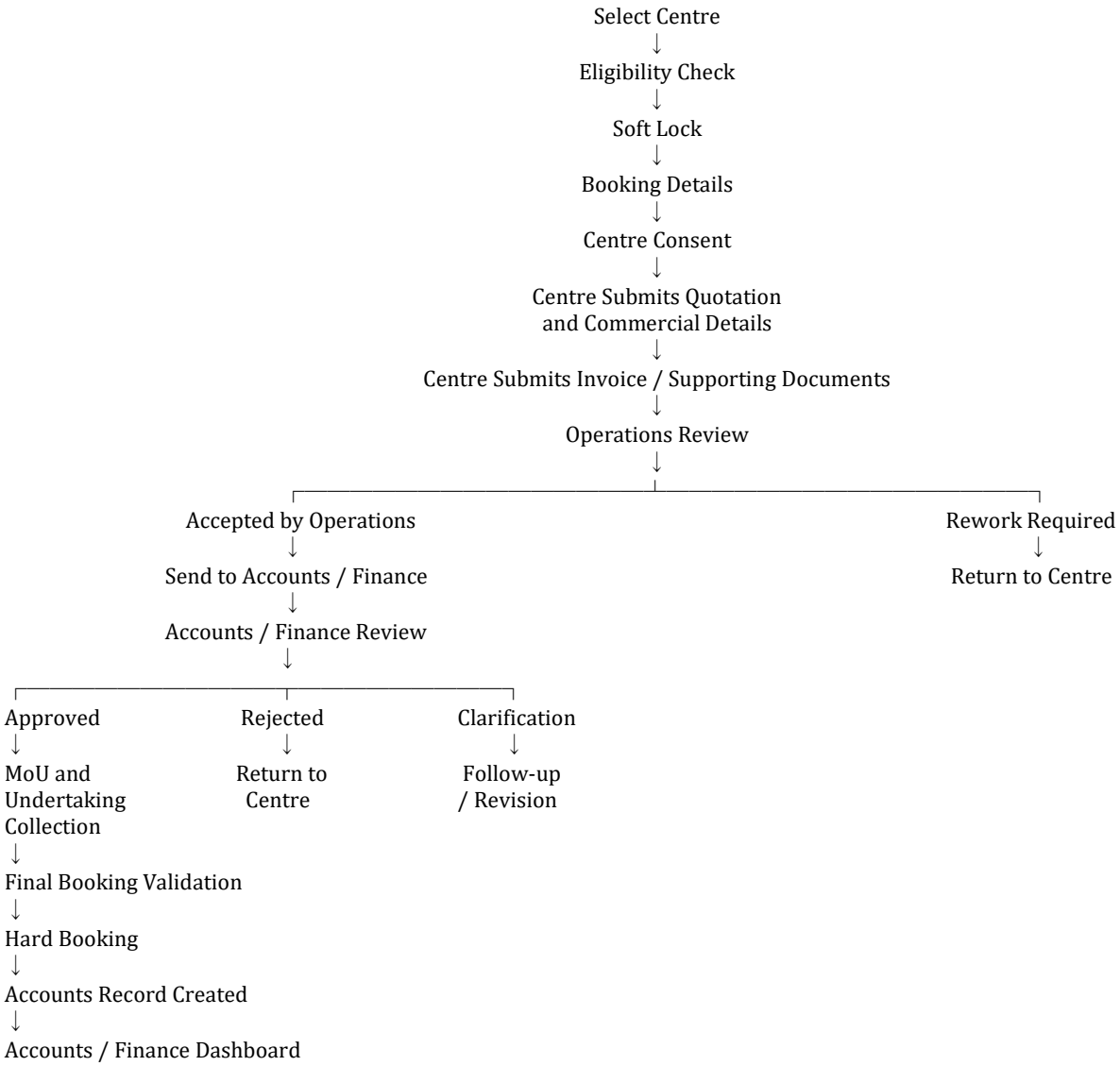
**System Outputs:**

| Output | Description |
|---|---|
| **Eligibility status** | Eligible, Not Eligible, Under Review, Conditionally Eligible or Override Approved. |
| **Restriction record** | Client- and exam-specific restriction with validity dates. |
| **Reverification task** | Task assigned to an authorized user for review. |
| **Override request** | Controlled exception request with reason and authority. |
| **Approval history** | Complete reviewer, decision and timestamp record. |
| **Booking control** | Automated permission or block before centre booking. |
| **Management report** | Centre eligibility and restriction report filtered by client, exam, city, date and status. |

# M05 – Booking & Locking

| Category | Details |
|---|---|
| Module Name | M05 – Centre Booking & Locking |
| Primary Users | Operations Executive, Project Manager, Accounts / Finance |
| Supporting User | Centre User |
| Purpose | Identify, validate, reserve and book an eligible centre while managing centre commercial terms, quotations, invoices, approvals, signed agreements and financial records. |
| Commercial Scope | Centre quotation, negotiated rate, applicable charges, invoice, payment terms, GST details, MoU, undertakings and approval status. |
| Booking Flow | Select Centre → Eligibility → Soft Lock → Booking Details → Centre Consent → Centre Quotation / Commercial Submission → Operations Review → Accounts / Finance Review → Approval / Rejection → MoU and Undertaking Collection → Final Validation → Hard Booking |
| Important Rule | A centre should not be finally booked until all required operational, commercial and approval conditions are complete. |

**End-to-End Flow:**



**Role of Accounts / Finance:**

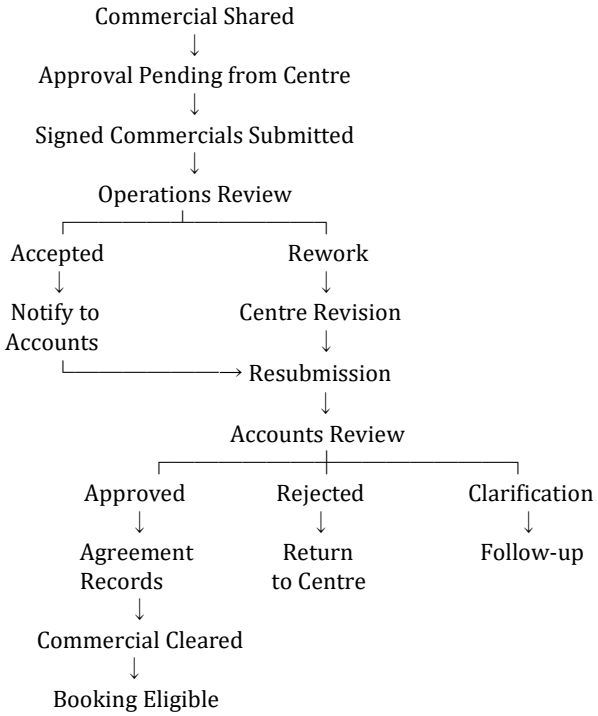
| Activity | Operations Executive | Project Manager | Accounts / Finance | Centre User |
|---|---|---|---|---|
| Select centre | Create | Review | View | No |
| Initiate soft lock | Create | Monitor | View | No |
| Capture booking details | Create / Edit | Approve / Review | View | View relevant details |
| Obtain centre consent | Coordinate | Monitor | View | Accept / reject |
| Submit quotation | Request / review | View / approve operationally | Review commercially | Submit |
| Review quotation | Initial review | Escalation / approval | Final commercial review | Respond to queries |
| Negotiate commercial terms | Coordinate | Approve exceptions | Own commercial approval | Provide revised quotation |
| Submit invoice | Request | View | Review | Submit |
| Approve quotation | No final authority | Approve operational exception | Final commercial approval | No |
| Reject quotation | Recommend / return | Review escalation | Approve rejection | Correct / resubmit |
| Collect signed MoU | Coordinate | Monitor | Verify and maintain record | Upload / sign |
| Collect undertaking | Coordinate | Monitor | Verify and maintain record | Upload / sign |
| Finalize commercial record | No | Review | Full responsibility | Provide information |
| Hard booking | Complete after approvals | Approve / authorize | Commercial clearance | No |
| Maintain approved commercial record | View | View | Full responsibility | View permitted information |

**Commercial and Finance Workflow:**

Commercial Review Flow

| Step | Stage | Responsible Role | Description |
|---|---|---|---|
| 1 | Commercial Request | Operations Executive | Request quotation and commercial information from the centre. |
| 2 | Quotation Submission | Centre User | Upload quotation and provide rates, charges, taxes and payment terms. |
| 3 | Operations Review | Operations Executive | Check that quotation matches the centre, project, exam, date, shift and expected services. |
| 4 | Operations Recommendation | Operations Executive | Mark quotation as Recommended, Rework Required or Not Recommended. |
| 5 | Commercial Review | Accounts / Finance | Review rates, taxes, payment terms, invoice details and commercial policy. |
| 6 | Clarification / Negotiation | Accounts / Finance and Operations | Request clarification or revised quotation where needed. |
| 7 | Commercial Decision | Accounts / Finance | Approve or reject the quotation. |
| 8 | Agreement Collection | Accounts / Finance | Collect and verify signed MoU and undertakings after approval. |
| 9 | Commercial Clearance | Accounts / Finance | Mark commercial requirements complete. |
| 10 | Booking Release | System / Authorized User | Allow final validation and hard booking. |

**Commercial Status Flow:**




**Commercial Fields:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Commercial Record ID | System-generated | ID | Unique commercial record linked to the booking. |
| Booking ID | Yes | Lookup | Booking associated with the commercial record. |
| Centre ID | Yes | Lookup | Centre submitting the quotation. |
| Project ID | Yes | Lookup | Associated project. |
| Client ID | Yes | Lookup | Associated client. |
| Exam ID | Yes | Lookup | Associated examination. |
| Quotation Number | Yes | Text | Centre quotation reference number. |
| Quotation Date | Yes | Date | Date of quotation. |
| Quotation File | Yes | File upload | Uploaded quotation document. |
| Rate Category | Yes | Dropdown | Day rate, shift rate, candidate rate, system rate or other configured category. |
| Centre Rate | Yes | Amount | Base rate quoted by the centre. |
| Candidate Rate | Conditional | Amount | Rate calculated per candidate, where applicable. |
| Day / Shift Rate | Conditional | Amount | Rate applicable to the examination day or shift. |
| Other Charges | Conditional | Amount | Additional charges such as manpower, power backup, logistics or consumables. |
| Tax / GST | Conditional | Amount / Percentage | Applicable taxes. |
| Total Quoted Amount | Yes | Amount | Total amount from the centre quotation. |
| Negotiated Amount | Conditional | Amount | Final negotiated amount, if different from quoted amount. |

| | | | |
|---|---|---|---|
| Payment Terms | Yes | Dropdown / Text | Payment terms agreed with the centre. |
| Commercial Validity | Yes | Date | Validity period of the quotation. |
| Operations Review Status | Yes | Dropdown | Pending, Recommended, Rework Required or Not Recommended. |
| Accounts Review Status | Yes | Dropdown | Pending, Under Review, Approved, Rejected or Clarification Required. |
| Rejection / Rework Reason | Conditional | Long text | Reason for rejection or requested correction. |
| Approved Amount | Conditional | Amount | Final approved amount. |
| Approved By | Conditional | User reference | Accounts / Finance approver. |
| Approval Date/Time | Conditional | Date/time | Date and time of commercial approval. |

**Invoice Fields:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Invoice ID | System-generated | ID | Unique invoice record. |
| Invoice Number | Conditional | Text | Invoice number submitted by the centre. |
| Invoice Date | Conditional | Date | Date on the invoice. |
| Invoice Amount | Conditional | Amount | Amount shown on the invoice. |
| GST Number | Conditional | Text | Centre GST identification number. |
| Invoice File | Conditional | File upload | Uploaded invoice copy. |
| Invoice Matching Status | Conditional | Dropdown | Matched, Mismatch, Pending Review or Rejected. |
| Invoice Review Remarks | Conditional | Long text | Review comments. |
| Invoice Approval Status | Conditional | Dropdown | Pending, Approved, Rejected or Rework Required. |

**MoU and Undertaking Fields:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| MoU Required | Yes | Yes / No | Indicates whether an MoU is required for the centre. |
| MoU Document | Conditional | File upload | Signed MoU uploaded by the centre. |
| MoU Version | Conditional | Number / Text | Version of the agreement. |
| MoU Signed Date | Conditional | Date | Date on which the MoU was signed. |
| MoU Effective Date | Conditional | Date | Date from which the agreement becomes effective. |
| MoU Expiry Date | Conditional | Date | Agreement expiry date, if applicable. |
| MoU Verification Status | Conditional | Dropdown | Pending, Under Review, Approved, Rework or Rejected. |
| Undertaking Required | Yes | Yes / No | Indicates whether an undertaking is required. |
| Undertaking Document | Conditional | File upload | Signed undertaking uploaded by the centre. |

| | | | |
|---|---|---|---|
| Undertaking Signed Date | Conditional | Date | Date on which the undertaking was signed. |
| Undertaking Verification Status | Conditional | Dropdown | Pending, Under Review, Approved, Rework or Rejected. |
| Agreement Verified By | Conditional | User reference | Accounts / Finance reviewer. |
| Agreement Verification Date | Conditional | Date/time | Date and time of agreement verification. |
| Agreement Remarks | Conditional | Long text | Verification comments or exceptions. |

**Booking Approval Gates:**

| Gate | Requirement | Responsible Role |
|---|---|---|
| Centre eligibility | Centre is eligible for the selected client and exam. | Admin / Project Manager |
| Soft-lock validity | Centre remains reserved for the current booking process. | System |
| Centre consent | Centre accepts the requirement. | Centre User / Operations |
| Quotation submission | Required commercial quotation is received. | Centre User |
| Operations review | Quotation matches the project and operational requirement. | Operations Executive |
| Accounts review | Rates, taxes, payment terms and quotation are commercially acceptable. | Accounts / Finance |
| Commercial approval | Final amount and terms are approved. | Accounts / Finance |
| MoU | Required signed MoU is uploaded and verified. | Accounts / Finance |
| Undertaking | Required undertaking is uploaded and verified. | Accounts / Finance |
| Capacity and allocation | Candidate count, buffer and centre capacity are validated. | Operations / Project Manager |
| Final validation | No booking conflict or unresolved critical issue exists. | System / Project Manager |
| Hard booking | Booking is formally confirmed. | Operations / Authorized approver |

**Booking Status Flow:**

| Status | Meaning | Next Status |
|---|---|---|
| Centre Selected | Centre has been selected for consideration. | Eligibility Checked |
| Soft Locked | Centre temporarily reserved for the booking user. | Consent Pending or Released |
| Consent Pending | Centre has not yet accepted the booking requirement. | Consent Accepted, Discussion or Rejected |
| Consent Accepted | Centre has accepted the operational requirement. | Quotation Pending |
| Quotation Pending | Commercial quotation is awaited from the centre. | Quotation Submitted |
| Quotation Submitted | Centre has provided quotation and commercial details. | Operations Review |
| Operations Review | Operations is validating the quotation. | Accounts Review, Rework or Rejected |

| | | |
|---|---|---|
| Accounts Review | Accounts / Finance is reviewing commercials. | Approved, Rejected or Clarification |
| Commercial Approved | Commercial terms and amount are approved. | Agreement Pending |
| Agreement Pending | Signed MoU or undertaking is awaited. | Agreement Verified or Rework |
| Agreement Verified | Required agreements are verified. | Final Validation |
| Final Validation | System checks all booking conditions. | Confirmed or Rework |
| Confirmed | Hard booking is complete. | Active Booking |
| Rejected | Commercial or operational request is rejected. | Closed or Replacement |
| Cancelled | Booking is cancelled with a recorded reason. | Replacement Linked or Closed |

**Accounts / Finance Dashboard:**

A separate dashboard should provide Accounts and Finance with visibility across all centre-related commercial and financial activities.

**Dashboard Purpose**

| Objective | Dashboard Capability |
|---|---|
| Commercial monitoring | Track quotations, negotiations, approvals and rejections. |
| Agreement monitoring | Track MoUs and undertakings pending, verified, expired or incomplete. |
| Invoice monitoring | Track invoice requests, uploaded invoices, mismatches and approvals. |
| Payment monitoring | Track amounts due, paid, outstanding and overdue. |
| Centre-wise visibility | View all commercial and financial records by centre. |
| Project-wise visibility | Monitor commercial status by project and exam. |
| Approval monitoring | Identify records awaiting Accounts / Finance action. |
| SLA control | Highlight overdue quotations, invoices, agreements and payments. |
| Audit readiness | Provide complete document and approval history. |

**Key Dashboard KPIs:**

| KPI | Description |
|---|---|
| Total active projects | Projects with ongoing commercial or financial activity. |
| Centres requiring quotation | Centres where commercial submission is pending. |
| Quotations received | Total quotations submitted by centres. |
| Quotations pending Operations review | Quotations awaiting operational validation. |
| Quotations pending Finance approval | Quotations awaiting commercial decision. |
| Quotations approved | Approved quotation count and total value. |
| Quotations rejected | Rejected quotations with reasons. |
| Clarifications pending | Records awaiting additional information. |
| MoUs pending | Approved commercials with MoU not yet verified. |
| Undertakings pending | Approved commercials with undertaking not yet verified. |
| Agreements expiring soon | MoUs or undertakings approaching expiry. |
| Invoices pending | Invoice requests not yet submitted or reviewed. |
| Invoice mismatches | Invoice amount or details do not match the approved commercial |

| | |
|---|---|
| | record. |
| Total approved value | Value of approved centre commercials. |
| Amount due | Total amount payable to centres. |
| Amount paid | Total payment made. |
| Outstanding amount | Unpaid amount remaining. |
| Overdue payments | Payments past the due date. |
| Booking blocked for commercial reasons | Bookings that cannot proceed due to pending commercial approval or agreements. |

**Dashboard Filters:**

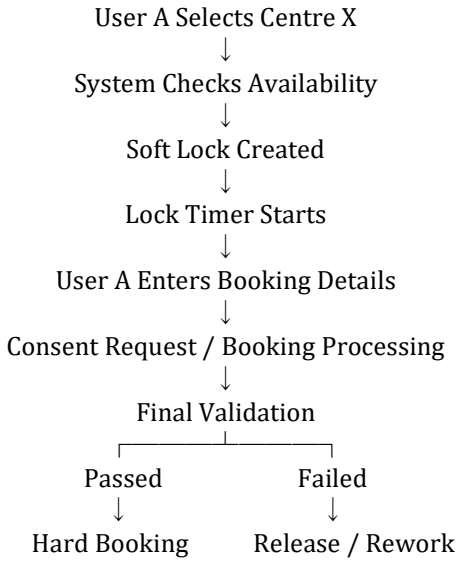
| Filter | Examples |
|---|---|
| Client | Client-wise commercial status. |
| Project | Project-specific view. |
| Exam | Examination-wise rates and invoices. |
| Centre | Centre-specific history. |
| City / State | Geographic analysis. |
| Project Manager | Ownership and escalation. |
| Operations Executive | Submission and follow-up responsibility. |
| Accounts / Finance owner | Approval workload. |
| Quotation status | Pending, Submitted, Rework, Approved, Rejected. |
| Agreement status | Pending, Verified, Expiring, Expired. |
| Invoice status | Pending, Submitted, Matched, Mismatch, Approved, Rejected. |
| Payment status | Due, Partially Paid, Paid, Outstanding, Overdue. |
| Date range | Requirement, quotation, approval, invoice or payment date. |

**Accounts / Finance Permission Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Accounts / Finance | Centre User |
|---|---|---|---|---|---|---|
| View quotation | Full | Project scope | No | Assigned project | Full | Own centre |
| Upload quotation | Full | No | No | Request / support | Upload if received through internal channel | Own centre |
| Review quotation operationally | Full | Approve / review | No | Initial review | View Operations recommendation | Respond to queries |
| Approve commercial terms | Full | Approve exception / recommend | No | No | Full approval | No |
| Reject quotation | Full | Recommend / approve | No | Recommend / return | Full authority | Correct / resubmit |
| Edit approved amount | Full through versioning | Approve revised version | No | Request revision | Create / approve revision | Submit revised quotation |
| View commercial history | Full | Project scope | No | Assigned scope | Full | Own centre history |
| Upload invoice | Full | No | No | Upload / | Upload / review | Own invoice |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | support | | |
| Approve invoice | Full | Review / approve where configured | No | Recommend | Full approval | No |
| Verify MoU | Full | View / monitor | No | Coordinate | Full verification | Upload signed document |
| Verify undertaking | Full | View / monitor | No | Coordinate | Full verification | Upload signed document |
| Maintain agreement records | Full | View | No | View status | Full responsibility | View own documents |
| View payment status | Full | Project scope | No | Assigned project status | Full | Own centre status |
| Create payment record | Full | Approve / authorize | No | Request / support | Full | No |
| Edit payment record | Full | Authorized approval | No | No direct edit | Full | No |
| View Finance dashboard | Full | Summary / project scope | No | Limited status view | Full | No |
| Export finance reports | Full | Assigned project scope | No | Assigned scope | Full | No |

**Soft Lock Flow:**




**If User B selects the same centre during the active lock:**

Centre X Currently Being Processed by Another User.

User B must not be allowed to proceed with the same conflicting booking until the lock expires, is released, or the first process is completed.

**Booking Validation Rules:**

| Validation Area | Requirement |
|---|---|
| **Centre status** | Centre must be Active and approved in Centre Master. |
| **Eligibility** | Centre must be eligible for the selected client, examination and validity period. |
| **Capacity** | Candidate count, buffer requirement and centre capacity must be reconciled. |
| **Date conflict** | System must check for an existing booking for the same centre and examination date. |
| **Shift conflict** | System must check whether the selected shift conflicts with another booking. |
| **Project conflict** | System must prevent duplicate booking for the same project, centre, exam, date and shift. |
| **Consent** | Required centre consent must be received before hard booking, unless an authorized exception exists. |
| **Commercial reference** | Commercial reference should be linked where required by the configured workflow. |
| **Lock validity** | Expired soft locks must not be converted into hard bookings without a new availability check. |
| **Final save** | Database transaction and unique constraints must provide final protection against concurrency. |

**Conflict Prevention Logic:**

The system should validate the following combination before creating a hard booking:

**Centre ID + Exam ID + Exam Date + Shift + Active Booking Status**

Depending on business rules, the validation may also include:

**Project ID + Client ID + Centre ID + Exam ID + Exam Date + Shift**

If an active conflicting record exists, the system must reject the save and display a clear message such as:

**This centre is already booked or being processed for the selected examination date and shift.**

# M06 – Digital Centre Consent

| Category | Details |
|---|---|
| **Module Name** | Digital Centre Consent |
| **Primary User** | Centre / Operations |
| **Purpose** | Capture formal centre acceptance, discussion requests or rejection of a booking requirement through a secure and traceable digital workflow. |
| **Process Flow** | Booking → Secure Link / Login → Review Requirement → Accept / Need Discussion / Reject → Timestamp → Internal Review |
| **Mandatory / Key Fields** | Consent ID; Booking ID; Centre ID; Authorized Contact; Consent Decision; Comment; Timestamp; Consent Version; Reviewer; Review Timestamp |
| **Business Rule** | Consent history and timestamps must be retained. Earlier consent decisions must not be overwritten when a new version or response is submitted. |
| **System Outputs** | Secure consent request; centre response; consent version; internal review record; timestamped consent history; booking status update; follow-up task where required. |

**Consent Lifecycle:**

| Step | Activity | System Action | Expected Output |
|---|---|---|---|
| **1** | **Booking Created** | Booking Executive completes the initial booking details. | Booking reference generated. |
| **2** | **Secure Invitation** | System sends a secure link or login invitation to the authorized centre contact. | Consent request issued. |
| **3** | **Centre Authentication** | Centre user opens the link or logs into the Centre Portal. | Authorized access established. |
| **4** | **Review Requirement** | Centre reviews exam date, shift, candidate count, facilities, commercials and other booking conditions. | Requirement review completed. |
| **5** | **Consent Decision** | Centre selects Accept, Need Discussion or Reject. | Consent response recorded. |
| **6** | **Comments / Evidence** | Centre adds remarks or supporting documents where applicable. | Decision context captured. |
| **7** | **Internal Review** | Operations reviews the centre's response and determines the next action. | Approved, follow-up, rework or rejected workflow. |
| **8** | **Booking Update** | System updates the booking based on the reviewed consent decision. | Booking proceeds, remains pending or is closed. |

**Consent Decision Matrix:**

| Decision | Meaning | System Action | Next Status |
|---|---|---|---|
| Accept | Centre agrees to the booking requirement. | Record acceptance and notify Operations. | Consent Accepted |
| Need Discussion | Centre requires clarification, negotiation or additional information. | Create follow-up task and preserve comments. | Discussion Required |
| Reject | Centre declines the booking. | Require rejection comment and notify Operations. | Consent Rejected |
| Pending | Centre has not yet responded. | Send reminders according to configured schedule. | Consent Pending |
| Revised Accept | Centre accepts a revised requirement after discussion. | Create a new consent version and retain previous response. | Consent Accepted |
| Withdrawn | Centre withdraws an earlier response before final booking. | Record reason and route for internal review. | Under Review |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| **Consent ID** | System-generated | Text / ID | Unique identifier for the consent record. |
| **Booking ID** | Yes | Lookup / ID | Booking associated with the consent request. |
| **Centre ID** | Yes | Lookup / ID | Centre providing the consent. |
| **Authorized Contact** | Yes | User / Contact Lookup | Authorized person responsible for responding on behalf of the centre. |
| **Consent Decision** | Yes | Dropdown | Pending, Accept, Need Discussion, Reject or other configured status. |
| **Comment** | Conditional | Long text | Centre remarks, questions, conditions or rejection reason. Required for Need Discussion and Reject. |
| **Timestamp** | System-generated | Date/time | Date and time when the centre submitted the decision. |
| **Consent Version** | System-generated | Number / Text | Version number for the consent request or response. |
| **Reviewer** | Conditional | User reference | Internal Operations or Manager reviewing the consent. |
| **Review Timestamp** | Conditional | Date/time | Date and time of internal review. |

**Consent Request Details:**

The centre should be able to review the following booking information before responding:

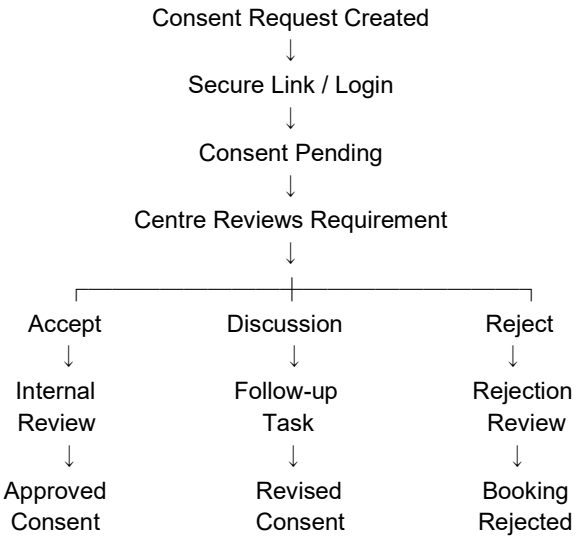
| Information | Source |
|---|---|
| Client | Project / Client Master |
| Examination name and code | Project / Exam setup |
| Examination date | Project setup |
| Shift | Project / Booking |
| City and centre address | Centre Master |
| Candidate count | Booking |
| Buffer requirement | Project / Allocation |
| Expected systems and infrastructure | Centre Master |
| UPS, DG and electricity details | Verified centre infrastructure |
| Commercial reference or rate information | Cost Sheet, where permitted |
| Required documents | Document checklist |
| Important instructions | Communication template / project requirement |
| Response deadline | Booking SLA / consent configuration |

**Role and Permission Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User |
|---|---|---|---|---|---|
| Create consent request | Full | Approve / monitor | No | Create / send | No |
| View consent status | Full | Full project scope | View relevant scope | Assigned booking scope | Own centre |
| Send reminder | Full | Monitor / trigger | No | Trigger | No |
| Submit consent | No | No | No | No | Own centre |

| | | | | | |
|---|---|---|---|---|---|
| decision | | | | | |
| Add internal comments | Full | Full | Audit comments only | Operational comments | Centre comments |
| Request discussion | Full | Approve / monitor | No | Handle follow-up | Submit request |
| Accept / reject on behalf of centre | Authorized exception | Authorized exception | No | No | Own centre decision |
| Review consent | Full | Full | No | Prepare / view | No |
| View consent history | Full | Full project | Relevant scope | Assigned scope | Own centre versions |

**Consent Status Flow:**



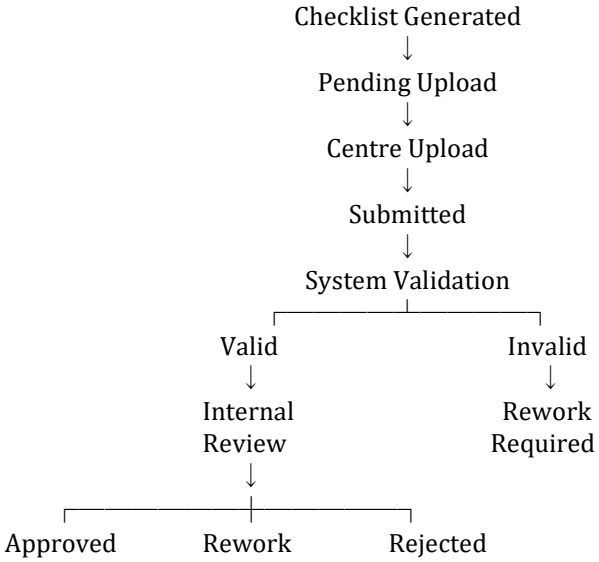
# M07 – Document Management

| Category | Details |
|---|---|
| **Module Name** | Document Management |
| **Primary User** | Centre / Operations |
| **Purpose** | Manage document checklists, centre uploads, validation, internal review, approvals, rework and document history for every project and booking. |
| **Process Flow** | Checklist → Centre Upload → Validation → Internal Review → Approved / Rework / Rejected |
| **Mandatory / Key Fields** | Document ID; Document Type; Centre; Project; Booking; Required/Optional; File; Document Number; Issue Date; Expiry Date; Uploaded By; Upload Time; Version; Verification Status; Reviewer; Review Time; Rejection Reason |
| **Business Rules** | Mandatory documents cannot be skipped. Rejected documents and previous versions must remain available. A document becomes approved only after internal verification and review. |
| **System Outputs** | Document checklist; upload request; validation result; approved document; rework request; rejection record; expiry alert; version history; document completeness status. |

**Document Lifecycle:**

| Step | Activity | System Action | Expected Output |
|---|---|---|---|
| **1** | **Checklist** | System generates required and optional documents based on client, exam, centre and project rules. | Booking-specific document checklist |
| **2** | **Centre Upload** | Centre uploads the required files and enters document details. | Submitted document record |
| **3** | **Validation** | System checks file type, size, mandatory fields, duplicate file and expiry information. | Valid or invalid submission |
| **4** | **Internal Review** | Operations or authorized reviewer checks authenticity, relevance and completeness. | Review decision |
| **5** | **Approved** | Document passes verification. | Approved document version |
| **6** | **Rework** | Document needs correction or additional information. | Rework request with reason |
| **7** | **Rejected** | Document is unacceptable or invalid. | Rejected record retained with reason |

**Document Status Flow:**



**Document Status Definitions:**

| Status | Meaning | Next Action |
|---|---|---|
| **Required** | Document is mandatory for the project, centre or booking. | Centre must upload it. |
| **Optional** | Document may be uploaded but is not mandatory. | Upload if applicable. |
| **Pending Upload** | Document is expected but has not been submitted. | Centre upload required. |
| **Submitted** | Centre has uploaded the document. | System validation and internal review. |
| **Validation Failed** | File or metadata does not meet system rules. | Centre must correct and resubmit. |
| **Under Review** | Internal reviewer is assessing the document. | Approve, request rework or reject. |
| **Approved** | Document is accepted for the applicable scope. | Remains available for use. |
| **Rework Required** | Document needs correction or additional evidence. | Centre uploads a new version. |
| **Rejected** | Document is not accepted. | Reason retained; replacement version may be submitted. |
| **Expired** | Document expiry date has passed. | New valid version required. |
| **Superseded** | A newer version has replaced the document for current use. | Old version retained for history. |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| **Document ID** | System-generated | Text / ID | Unique identifier for the document record. |
| **Document Type** | Yes | Dropdown | Type of document, such as registration certificate, authorization, GST certificate, lease document or audit evidence. |
| **Centre** | Yes | Lookup / ID | Centre to which the document belongs. |
| **Project** | Conditional | Lookup / ID | Project for which the document is being submitted. |
| **Booking** | Conditional | Lookup / ID | Specific booking associated with the document. |
| **Required / Optional** | Yes | Dropdown | Indicates whether the document is mandatory or optional. |
| **File** | Yes for submission | File upload | Uploaded document file. |
| **Document Number** | Conditional | Text | Official document number, where applicable. |
| **Issue Date** | Conditional | Date | Date on which the document was issued. |
| **Expiry Date** | Conditional | Date | Date on which the document expires. |
| **Uploaded By** | System-generated | User reference | Centre or internal user who uploaded the document. |
| **Upload Time** | System-generated | Date/time | Date and time of upload. |
| **Version** | System- | Number / | Version number of the document. |

| | | | |
|---|---|---|---|
| | generated | Text | |
| **Verification Status** | Yes | Dropdown | Pending, Submitted, Under Review, Approved, Rework Required, Rejected, Expired or Superseded. |
| **Reviewer** | Conditional | User reference | Internal user who reviewed the document. |
| **Review Time** | Conditional | Date/time | Date and time of review. |
| **Rejection Reason** | Conditional | Long text | Reason for rejection or rework request. Required when applicable. |

**Document Checklist Example:**

| Document Type | Required? | Submitted? | Verification Status | Expiry Tracking | Action |
|---|---|---|---|---|---|
| **Centre registration certificate** | Yes | Yes | Approved | Yes | No action |
| **GST certificate** | Yes | Yes | Under Review | No / configured | Internal review |
| **Centre authorization letter** | Yes | No | Pending Upload | No | Centre upload required |
| **Electricity inspection report** | Yes | Yes | Rework Required | Yes | Upload corrected version |
| **UPS service report** | Conditional | Yes | Approved | Yes | No action |
| **DG service report** | Conditional | No | Pending Upload | Yes | Centre upload required |
| **Additional infrastructure photos** | Optional | Yes | Approved | No | No action |

**Role and Permission Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User |
|---|---|---|---|---|---|
| Configure document types | Full | Review / request | Technical document input | No | No |
| Create checklist | Full | Approve / configure | Audit checklist | Create project checklist | View assigned checklist |
| Upload document | Full | Full | Audit evidence | Upload / support | Own centre documents |
| Edit document metadata | Full | Approve / edit | Audit-related data | Assigned records | Submitted details before review |
| Validate document | Full | Monitor / approve | Audit evidence | Validate | No |
| Approve document | Full | Full approval | Audit documents | Verify / recommend | No |
| Request rework | Full | Full | Audit rework | Request | Respond / resubmit |
| Reject document | Full | Approve rejection | Audit evidence | Recommend | No |
| View document history | Full | Project scope | Audit scope | Assigned scope | Own centre |
| Delete document | No silent deletion | No silent deletion | No silent deletion | No silent deletion | No silent deletion |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Validate file** | Full access | Approve / Monitor | Validate | Validate audit evidence | Validate commercial documents | No access |
| **Approve document** | Full access | Full approval | Verify / Recommend | Approve audit documents where authorized | Approve finance documents | No access |
| **Request rework** | Full access | Full access | Request / Recommend | Request audit rework | Request invoice rework | Respond and resubmit |
| **Reject document** | Full access | Approve rejection | Recommend rejection | Reject audit evidence where authorized | Reject finance document | No access |
| **View document history** | Full access | Full project scope | Assigned scope | Assigned audit scope | Finance scope | Own centre documents |
| **Delete document** | No silent deletion | No silent deletion | No silent deletion | No silent deletion | No silent deletion | No silent deletion |

**Upload and Validation Rules:**

| Validation Area | Requirement |
|---|---|
| **File type** | Allow only configured file formats, such as PDF, JPG, PNG or XLSX where applicable. |
| **File size** | Enforce configurable maximum file size. |
| **Mandatory metadata** | Document number, issue date or expiry date must be captured when applicable. |
| **Expiry date** | The system should flag expired and soon-to-expire documents. |
| **Duplicate check** | Detect duplicate uploads or identical document numbers where possible. |
| **Readability** | Reviewer must be able to mark a file as unreadable and request rework. |
| **Document scope** | Document must be linked to the correct centre, project and booking. |
| **Mandatory completion** | Project or booking cannot proceed to the configured next stage if mandatory documents are not approved. |
| **Security** | Only permitted roles can view, download or replace documents. |
| **Versioning** | Each replacement upload creates a new version; the previous version remains retained. |

**Rework and Rejection Flow:**

**Document Under Review**
↓
**Issue Identified**
↓
**Reviewer Selects Rework or Reject**
↓
**Reason Becomes Mandatory**
↓
**Centre / Operations Notified**
↓
**Corrected Document Uploaded**
↓
**New Version Created**
↓
**Validation**
↓
**Internal Review**
↓
**Approved / Rework / Rejected**

**Document Version History:**

| Version | Uploaded By | Upload Date/Time | Status | Reviewer | Review Date/Time | Reason |
|---|---|---|---|---|---|---|
| V1 | Centre User | System-generated | Rejected | Operations User | System-generated | Document was expired. |
| V2 | Centre User | System-generated | Rework Required | Operations User | System-generated | Stamp or signature unclear. |
| V3 | Centre User | System-generated | Approved | Operations Manager | System-generated | Valid document accepted. |

*The system should identify V3 as the current approved version while retaining V1 and V2 for audit and historical reference.*

**Automated Notifications:**

| Trigger | Recipient | Notification |
|---|---|---|
| Checklist generated | Centre User & Operations Team | Document checklist is available. |
| Document pending | Centre User | Required document upload reminder. |
| Upload completed | Operations Team | Document is ready for validation. |
| Validation failure | Centre User | File or metadata correction required. |
| Rework requested | Centre User & Operations Team | Document rework reason and due date. |
| Document rejected | Centre User & Project Manager | Rejection reason and next action. |
| Document approved | Centre User & Operations Team | Document accepted. |
| Expiry approaching | Operations Team | Renewal or updated document required. |
| Mandatory document overdue | Centre User & Project Manager | Escalation for incomplete documentation. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| Mandatory-document gate | Required documents must be approved before the booking or project can move to the configured next stage. |
| Approval separation | Centre upload does not equal document approval. |
| Rejection history | Rejected documents and rejection reasons remain available. |
| Version control | Re-uploading creates a new version rather than overwriting the old file. |
| Expiry monitoring | Expiring and expired documents generate alerts and may block readiness or audit completion. |
| Reviewer accountability | Every review decision records reviewer and review time. |
| Reason enforcement | Rework and rejection decisions require a clear reason. |
| Scope control | Documents remain linked to the correct centre, project and booking. |
| Access control | Confidential or financial documents are restricted by role. |
| Audit trail | Upload, view, download, review, approval, rejection and replacement activities are logged. |

**System Outputs:**

| Output | Description |
|---|---|
| **Document checklist** | Required and optional documents for the selected centre, project and booking. |
| **Completeness indicator** | Shows submitted, pending, approved, rework and rejected documents. |
| **Validation result** | Indicates whether the file and metadata passed system checks. |
| **Review decision** | Approved, Rework Required or Rejected. |
| **Rework task** | Assigned action with reason, owner and due date. |
| **Expiry alert** | Notification for upcoming or completed document expiry. |
| **Current approved version** | Latest approved document used for operational decisions. |
| **Historical versions** | Previous documents, decisions and rejection reasons retained. |
| **Compliance report** | Centre- and project-wise documentation status for management and audit. |

# M08 – Audit & Compliance

| Category | Details |
|---|---|
| Module Name | Audit & Compliance |
| Primary User | Auditor / Audit Manager |
| Purpose | Verify and approve the operational, infrastructure, technical and compliance readiness of a centre before it is released for booking or further execution. |
| Process Flow | Booking → Previous Audit Check → Assign Auditor → Visit if Needed → Checklist → Evidence / Photos / Signature → IP Scan → Submit → Review → Approved / Rework / Re-Audit → Booking |
| Mandatory / Key Fields | Audit ID; Centre; Project; Audit Type; Audit Reason; Previous Audit Date; Validity; Auditor; Visit Date; Checklist; Evidence; Photos; Signature; Observations; Scan Reference; Status; Reviewer; Approval |
| Business Rules | Audit validity is configurable by audit type, client, exam, project or risk level. The auditor who conducts and submits an audit cannot approve the same audit. |
| System Outputs | Previous audit assessment; auditor assignment; visit requirement decision; completed checklist; evidence package; IP scan reference; audit review decision; approved audit; rework task; linked re-audit; booking release. |

**Detailed Process Table:**

| Step | Stage | Process Description | System Output |
|---|---|---|---|
| **1** | **Booking** | A centre booking or booking request initiates the audit requirement. | Audit requirement linked to booking. |
| **2** | **Previous Audit Check** | System checks whether a valid and applicable audit already exists. | Previous audit result and validity status. |
| **3** | **Assign Auditor** | Audit Manager assigns an eligible auditor and due date. | Auditor assignment. |
| **4** | **Visit if Needed** | Physical visit is conducted only when required by validity, risk, policy or open observations. | Visit record, date and evidence. |
| **5** | **Checklist** | Auditor completes the configured operational, infrastructure, electrical, UPS, DG, security and compliance checklist. | Completed checklist. |
| **6** | **Evidence / Photos / Signature** | Auditor uploads documents, photographs, observations and required signatures. | Evidence package. |
| **7** | **IP Scan** | Technical scan is performed and linked to the audit. | Scan reference, system details and exceptions. |
| **8** | **Submit** | Auditor submits the completed audit for independent review. | Status changes to Submitted. |
| **9** | **Review** | Audit Manager or authorized reviewer evaluates the checklist, evidence and observations. | Approval decision. |
| **10A** | **Approved** | Audit meets all applicable requirements. | Audit approved and centre released for booking. |
| **10B** | **Rework** | Correctable gaps or missing evidence are found. | Rework task with reason and due date. |
| **10C** | **Re-Audit** | Material or unresolved findings require a new | New audit linked to original |

| | | | |
|---|---|---|---|
| | | audit cycle. | audit. |
| **11** | **Booking** | Approved audit result permits booking to proceed, subject to eligibility and concurrency checks. | Booking confirmed or released for final booking validation. |

**Previous Audit Check:**

| Previous Audit Condition | Visit Requirement | System Action |
|---|---|---|
| Valid and applicable audit | Not normally required | Reuse audit subject to configured policy. |
| Audit expired | Required | Assign auditor and create new audit cycle. |
| No previous audit | Required | Assign auditor and schedule visit. |
| Audit belongs to another client or exam | Conditional | Apply configured scope-reuse rules. |
| Open critical observation | Required | Require corrective action and re-audit. |
| Centre infrastructure changed | Conditional | Require targeted verification or full visit. |
| Client policy requires physical audit | Required | Schedule visit regardless of previous audit. |
| Audit validity unclear | Required | Route to Audit Manager for decision. |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| **Audit ID** | System-generated | Text / ID | Unique identifier for the audit record. |
| **Centre** | Yes | Lookup / ID | Centre being audited. |
| **Project** | Yes | Lookup / ID | Project associated with the audit. |
| **Audit Type** | Yes | Dropdown | Initial, Renewal, Re-Audit, Technical, Compliance, Surprise or client-specific audit. |
| **Audit Reason** | Yes | Dropdown / Long text | Reason why the audit is required. |
| **Previous Audit Date** | Conditional | Date | Date of the last relevant audit. |
| **Validity** | Yes | Date range / Configuration | Effective period of the approved audit. |
| **Auditor** | Yes | User Lookup | Auditor assigned to perform and submit the audit. |
| **Visit Date** | Conditional | Date / Date-time | Date and time of the physical audit visit, if required. |
| **Checklist** | Yes | Structured form | Configured checklist and response values. |
| **Evidence** | Conditional | File / Text / Link | Supporting documents, readings and other evidence. |
| **Photos** | Conditional | Image upload | Photos supporting centre and infrastructure verification. |
| **Signature** | Conditional | Digital signature / File | Auditor and centre acknowledgment or verification signature. |
| **Observations** | Yes | Long text | Findings, risks, deficiencies and recommendations. |
| **Scan Reference** | Conditional | Lookup / ID | Link to the related IP or technical scan. |
| **Status** | Yes | Dropdown | Pending, Assigned, Visit Required, In Progress, Submitted, Under Review, Approved, Rework, Re-Audit, Rejected or Closed. |
| **Reviewer** | Conditional | User reference | Audit Manager or authorized independent reviewer. |
| **Approval** | Conditional | Decision field | Approved, Rework, Re-Audit or Rejected, with date |

| | | | |
|---|---|---|---|
| | | | and comments. |

**Audit Checklist Areas:**

| Checklist Area | Example Verification Items |
|---|---|
| Centre identity | Legal name, address, registration and centre authorization. |
| Capacity | Seating capacity, system count, rooms and buffer availability. |
| Electricity | Supply reliability, sanctioned load, usable load, earthing and dedicated circuit. |
| UPS | Availability, capacity, coverage, condition and last service date. |
| DG | Availability, capacity, last service date, diesel tank capacity and current diesel level. |
| Infrastructure | Rooms, furniture, lighting, ventilation, accessibility and signage. |
| Network | Internet connectivity, bandwidth, redundancy. Switches, Racks with safety and network layout. |
| Systems | CPU, CMOS Battery, RAM, storage, OS, camera, keyboard, mouse and related devices. |
| Security | CCTV, access control, fire safety and restricted-area controls. |
| Documents | Registration, authorization, service and inspection documents. |
| Manpower | Technical, security, invigilator and centre support availability. |
| Technical compliance | IP scan, device classification and duplicate MAC detection. |
| Risk and exceptions | Open issues, severity, corrective action, owner and due date. |

**Audit Status Flow:**

| Status | Meaning | Next Status |
|---|---|---|
| Audit Pending | Audit requirement has been generated from the booking. | Previous Audit Checked |
| Previous Audit Checked | Existing audit has been evaluated for validity and scope. | Audit Reusable, Auditor Assigned or Visit Required |
| Auditor Assigned | Auditor and due date have been recorded. | Visit Required or In Progress |
| Visit Required | Physical centre visit is required. | Visit Completed |
| Visit Completed | Auditor has completed the visit. | Checklist In Progress |
| In Progress | Checklist, evidence or technical scan is being completed. | Submitted, Rework or Exception |
| Submitted | Auditor has submitted the audit. | Under Review |
| Under Review | Independent reviewer is assessing the audit. | Approved, Rework or Re-Audit |
| Approved | Audit is accepted for the configured validity period. | Booking |
| Rework | Correctable issue requires action. | Resubmitted or Re-Audit |
| Re-Audit | New audit cycle is required. | Auditor Assigned |
| Rejected | Centre or audit fails required controls. | Closed or Reverification |

**Role and Permission Matrix:**

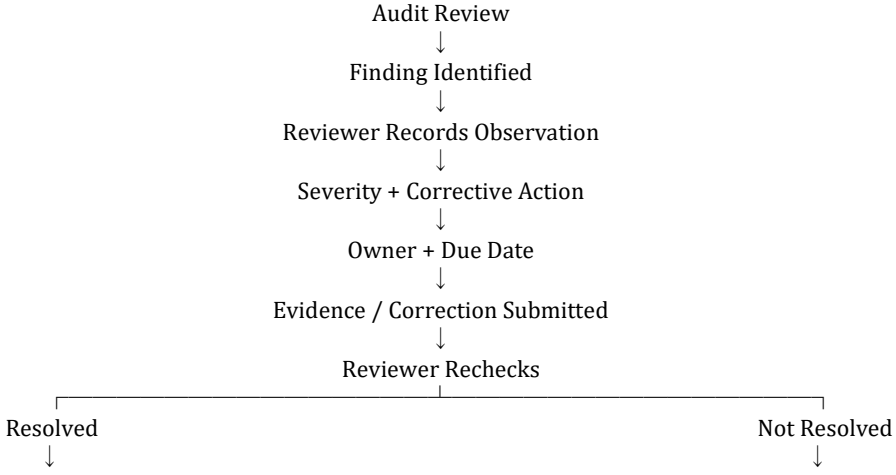
| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User |
|---|---|---|---|---|---|
| Configure audit types | Full | Review / request | Technical input | No | No |
| Configure audit | Full | Approve / | Recommend | No | No |

| | | | | | |
|---|---|---|---|---|---|
| validity | | request | | | |
| Create audit requirement | Full | Create / request | Request / support | View | No |
| Check previous audit | Full | Full project scope | Full audit scope | View status | No |
| Assign auditor | Full | Request / monitor | No | No | No |
| Decide visit requirement | Full | Approve / monitor | Recommend | No | Support |
| Conduct audit | Authorized | No | Create / submit | No | Support |
| Complete checklist | Full | View / review | Create / submit | View | Provide information |
| Upload evidence | Full | Review | Upload | Support | Upload support |
| Run / link IP scan | Full | Monitor | Create / submit | View | Support |
| Submit audit | Authorized exception | No | Submit own audit | No | No |
| Review audit | Full | Review | No | View | No |
| Approve audit | Full | Approve | Prohibited for own audit | No | No |
| Request rework | Full | Full | No | View / coordinate | Respond |
| Initiate re-audit | Full | Approve / request | Recommend | Coordinate | Support |
| View audit history | Full | Full project | Assigned audit scope | Assigned project scope | Own centre, permitted details |

**Auditor Independence Rule:**

| Rule | System Requirement |
|---|---|
| Auditor submission | Auditor may complete and submit the audit. |
| Independent review | Audit must be assigned to an Audit Manager or another authorized reviewer. |
| Self-approval prevention | The auditor who submitted the audit cannot approve it. |
| Override handling | Any exceptional approval must record authority, reason and timestamp. |
| Audit logging | Assignment, visit, submission, review, rework, re-audit and approval must be logged. |

**Rework and Re-Audit Flow:**



Approve Audit

Rework / Re-Audit
↓
New Linked Audit Record

*The re-audit must have its own Audit ID while retaining a link to the original audit. The original audit must remain unchanged for historical reference.*

**Booking Release Conditions:**

The centre may proceed to booking only when:

| Booking Gate | Required Result |
|---|---|
| Previous audit check | Completed. |
| Audit applicability | Valid audit reused or new audit completed. |
| Visit requirement | Completed when required. |
| Checklist | Complete with no unresolved critical items. |
| Evidence | Required documents, photos and signatures available. |
| IP scan | Complete or authorized exception recorded. |
| Audit review | Approved by an independent reviewer. |
| Audit validity | Current and applicable to the client, exam and project. |
| Eligibility | Centre eligible for the selected client and examination. |
| Capacity | Candidate count and buffer fit within approved centre capacity. |
| Booking conflict | No active conflicting lock or booking exists. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| **Previous audit first** | The system must check for a valid applicable audit before automatically creating a new visit. |
| **Conditional visit** | Physical visit is required only when validity, risk, policy or findings require it. |
| **Configurable validity** | Audit validity must be configurable rather than hard-coded. |
| **Mandatory checklist** | Required checklist questions cannot be skipped. |
| **Evidence control** | Critical findings must include supporting evidence or an approved exception. |
| **Technical linkage** | The IP scan must be linked to the relevant audit. |
| **Independent approval** | Auditor cannot approve own audit. |
| **Rework tracking** | Every rework item requires a reason, owner and due date. |
| **Re-audit linkage** | Re-audits must link to the original audit without overwriting it. |
| **Booking dependency** | Booking must be blocked until the audit is approved, unless an authorized exception is recorded. |
| **Expiry monitoring** | Upcoming and expired audits should generate alerts and escalation. |
| **History retention** | Original audits, evidence, decisions and corrections must remain available. |

**System Outputs:**

| Output | Description |
|---|---|
| Previous audit assessment | Validity, scope, expiry and open-observation result. |
| Auditor assignment | Assigned auditor, due date and visit requirement. |
| Visit record | Visit date, attendance, location and evidence. |

| | |
|---|---|
| Audit checklist | Structured operational and compliance assessment. |
| Evidence package | Documents, photos, signatures and observations. |
| Technical scan result | Device and network details, including duplicate MAC exceptions. |
| Review decision | Approved, Rework, Re-Audit or Rejected. |
| Corrective-action task | Issue, severity, owner, due date and resolution evidence. |
| Approved audit result | Validity period and booking release status. |
| Booking release | Centre made available for the next booking validation step. |
| Audit history | Complete history of original audit, revisions, rework and re-audits. |

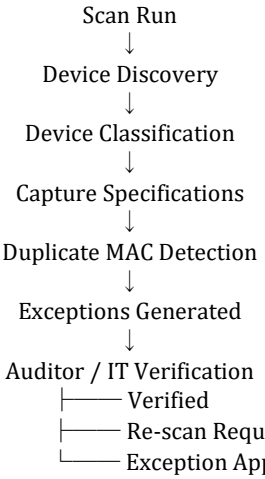
# M09 – Technical / IP Scan

| Category | Details |
|---|---|
| **Module Name** | Technical / IP Scan |
| **Primary User** | Auditor / IT |
| **Purpose** | Discover, classify and verify examination devices and network information at a centre, while automatically identifying duplicate MAC addresses and other technical exceptions. |
| **Process Flow** | Scan Run → Device Discovery → Classification → Specification → Duplicate MAC Detection → Exception → Verification |
| **Mandatory / Key Fields** | Scan ID; Centre; Audit; IP Address; MAC Address; Hostname; Device Type; OS; CPU; RAM; Storage; Graphics; Camera; Keyboard; Mouse; Scan Date/Time; Duplicate MAC Flag |
| **Business Rule** | Duplicate MAC addresses must be automatically flagged for review. A scan with unresolved critical exceptions should not be treated as technically verified without authorized approval. |
| **System Outputs** | Scan run record; discovered device list; hardware and software specifications; duplicate MAC report; exception list; verification result; audit-linked technical report. |

**Technical Scan Lifecycle:**

| Step | Activity | System Action | Expected Output |
|---|---|---|---|
| 1 | Scan Run | Auditor or IT user initiates a scan for a centre and audit. | Scan ID and scan session created. |
| 2 | Device Discovery | System identifies devices connected to the centre network. | Device inventory generated. |
| 3 | Classification | System classifies each discovered device by type and role. | Device type assigned. |
| 4 | Specification | System captures IP, MAC, hostname, OS and hardware details. | Device specification record. |
| 5 | Duplicate MAC Detection | System compares MAC addresses within the scan and against configured history. | Duplicate MAC flags generated. |
| 6 | Exception | System records duplicate MACs, missing specifications, unsupported devices or other anomalies. | Exception list and severity. |
| 7 | Verification | Auditor or IT reviewer examines results and records a verification decision. | Verified scan, re-scan request or approved exception. |

**Scan Flow:**



**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Scan ID | System-generated | Text / ID | Unique identifier for the scan run. |
| Centre | Yes | Lookup / ID | Centre where the scan is performed. |
| Audit | Yes | Lookup / ID | Audit record to which the scan is linked. |
| IP Address | Yes | Text / Network value | IP address detected for the device. |
| MAC Address | Yes | Text | Hardware network identifier of the device. |
| Hostname | Conditional | Text | Device hostname detected during discovery. |
| Device Type | Yes | Dropdown | Desktop, laptop, server, network device, printer, router or other configured type. |
| OS | Conditional | Text / Dropdown | Operating system and version. |
| CPU | Conditional | Text | Processor details. |
| RAM | Conditional | Number / Text | Installed memory. |
| Storage | Conditional | Number / Text | Storage capacity and type. |
| Graphics | Conditional | Text | Graphics adapter or display capability. |
| Camera | Conditional | Yes / No / Details | Camera availability and detected details. |
| Keyboard | Conditional | Yes / No / Details | Keyboard availability and device details. |
| Mouse | Conditional | Yes / No / Details | Mouse availability and device details. |
| Scan Date/Time | System-generated | Date/time | Date and time of scan execution. |
| Duplicate MAC Flag | System-generated | Yes / No | Indicates whether the MAC address is duplicated. |

**Scan Run Fields:**

In addition to device-level fields, the system should maintain scan-session information:

| Field | Mandatory | Description |
|---|---|---|
| Scan ID | System-generated | Unique scan-session identifier. |
| Centre | Yes | Centre associated with the scan. |
| Audit ID | Yes | Audit to which the scan belongs. |
| Scan Type | Yes | Initial, Re-scan, Pre-exam, Mock, Post-change or Verification. |
| Scan Initiated By | System-generated | User or system process that initiated the scan. |
| Scan Start Time | System-generated | Start timestamp. |
| Scan End Time | System-generated | Completion timestamp. |
| Devices Discovered | System- | Total number of discovered devices. |

| | | |
|---|---|---|
| | generated | |
| Devices Classified | System-generated | Number of devices successfully classified. |
| Duplicate MAC Count | System-generated | Number of duplicate MAC exceptions. |
| Missing Data Count | System-generated | Devices with incomplete specifications. |
| Scan Status | Yes | Initiated, Running, Completed, Failed, Partial or Cancelled. |
| Verification Status | Yes | Pending, Under Review, Verified, Re-scan Required or Exception Approved. |
| Reviewer | Conditional | Auditor or IT reviewer. |
| Review Date/Time | Conditional | Date and time of technical review. |
| Review Remarks | Conditional | Verification notes and exception comments. |

**Device Classification:**

| Device Type | Example Details to Capture |
|---|---|
| Examination workstation | IP, MAC, hostname, OS, CPU, RAM, storage, graphics, camera, keyboard and mouse. |
| Server | IP, MAC, hostname, OS, CPU, RAM, storage, role and uptime where available. |
| Network device | IP, MAC, hostname, device type, manufacturer and network role. |
| Router / switch | Management IP, MAC, model, ports and connectivity status. |
| Printer or peripheral | IP, MAC, hostname, type and operational status. |
| Unknown device | Detected identifiers, available specifications and exception reason. |

**Duplicate MAC Detection:**

The system should automatically compare:

**MAC Address within the current scan**
**+**
**MAC Address from previous scans for the same centre**
**+**
**MAC Address associated with another centre, where permitted**

| Condition | System Result |
|---|---|
| MAC address appears once in current scan | No duplicate flag. |
| MAC address appears more than once in current scan | Duplicate MAC Flag = Yes; critical exception created. |
| MAC address was previously assigned to another device at the same centre | Historical duplicate or reassignment warning. |
| MAC address appears at another centre | Cross-centre duplicate exception, subject to policy. |
| MAC address is missing or malformed | Technical data exception. |
| Device changes after approved scan | Change detected; re-scan may be required. |

**Exception Management:**

| Exception Type | Example | Suggested Severity | Action |
|---|---|---|---|
| Duplicate MAC | Same MAC assigned to multiple | Critical | Investigate, correct and re- |

| | | | |
|---|---|---|---|
| | devices. | | scan. |
| Missing MAC | Device has no readable MAC address. | High | Manual verification or technical correction. |
| Missing hardware data | CPU, RAM or storage not detected. | Medium | Manual entry or re-scan. |
| Unsupported OS | Device operating system does not meet requirement. | High | Remediation or replacement. |
| Insufficient RAM | Device does not meet configured minimum. | High | Upgrade or replace device. |
| Missing camera | Required camera is unavailable. | High | Install or replace device. |
| Unrecognized device | Device classification is not possible. | Medium | IT review required. |
| Device count mismatch | Discovered devices differ from declared count. | High | Reconcile inventory and re-scan. |
| Network connectivity failure | Device cannot communicate with scan service. | High | Network correction or manual verification. |
| Stale scan | Scan is outside configured validity period. | Medium | Run a new scan. |

**Exception Status Flow:**

| Status | Meaning | Next Status |
|---|---|---|
| Detected | System has identified an exception. | Assigned |
| Assigned | Exception has an owner. | In Progress |
| In Progress | Investigation or corrective action is underway. | Resolved, Re-scan Required or Escalated |
| Re-scan Required | A new technical scan is needed. | Scanned / Under Review |
| Resolved | Corrective action is complete. | Verified |
| Exception Approved | Authorized reviewer has accepted the exception. | Verified with Exception |
| Escalated | Issue requires manager or client attention. | Resolved or Rejected |
| Closed | Exception is resolved and verified. | Historical record retained |

**Role and Permission Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User |
|---|---|---|---|---|---|
| Configure scan rules | Full | Review / request | Technical recommendation | No | No |
| Initiate scan | Full | Request / monitor | Create | Support / view | Support |
| View scan results | Full | Summary / project scope | Full audit scope | Project summary | Own centre, permitted view |
| Classify device | Full | No | Create / edit | No | No |
| Review duplicate MAC | Full | Escalate / monitor | Investigate / resolve | View exception status | Support |
| Edit technical details | Full | No | Update technical data | No | Submit correction |
| Create exception | Full | Request / escalate | Create | Request / escalate | Report issue |
| Approve exception | Full | Authorized approval | Recommend | No | No |
| Approve scan | Full | Review / approve | Verify / submit | No | No |

| | | | | | |
|---|---|---|---|---|---|
| Request re-scan | Full | Request | Recommend / initiate | Coordinate | Support |
| Delete scan | No silent deletion | No silent deletion | No silent deletion | No silent deletion | No |

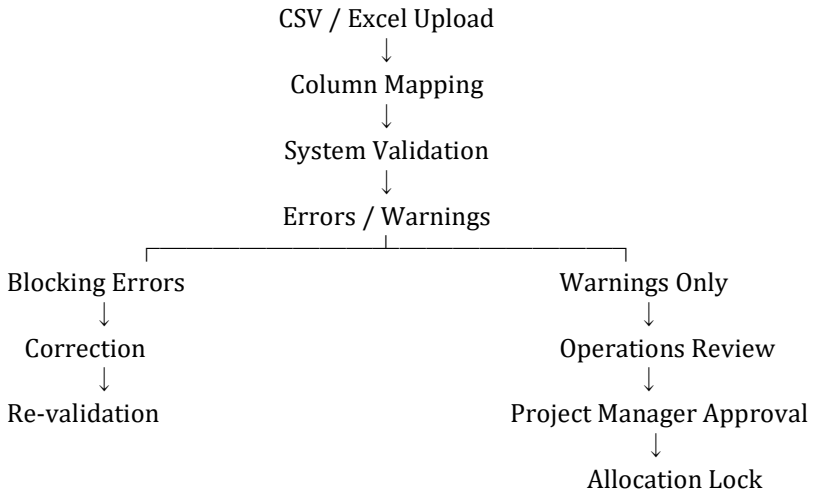
# M10 – Allocation Management

| Category | Details |
|---|---|
| Module Name | Allocation Management |
| Primary User | Operations / Project Manager |
| Purpose | Import, map, validate, review, approve and lock client allocation data while ensuring seat capacity, buffer requirements and allocation changes are controlled. |
| Process Flow | CSV/Excel Upload → Column Mapping → Validation → Errors/Warnings → Review → Approval → Lock |
| Mandatory / Key Fields | Allocation ID; Version; Project; Centre; Date; Shift; Candidate Count; Buffer Seats; File; Uploaded By; Validation Result; Approval Status; Lock Timestamp |
| Business Rules | Allocation versions are retained. A locked allocation cannot be directly edited. Any change after locking requires a documented change request, reason, validation and approval. |
| System Outputs | Uploaded allocation file; mapped columns; validation report; errors and warnings; reviewed allocation; approved version; locked allocation; change request and version history. |

**Allocation Lifecycle:**

| Step | Activity | System Action | Expected Output |
|---|---|---|---|
| 1 | CSV / Excel Upload | Operations or PM uploads the client allocation file. | Allocation record and file version created. |
| 2 | Column Mapping | User maps source-file columns to portal fields. | Mapping configuration saved. |
| 3 | Validation | System validates project, centre, date, shift, candidate count and buffer seats. | Validation results generated. |
| 4 | Errors / Warnings | System separates blocking errors from non-blocking warnings. | Error and warning report. |
| 5 | Review | Operations reviews the results and corrects or confirms exceptions. | Reviewed allocation version. |
| 6 | Approval | Project Manager checks and approves the allocation. | Approved allocation version. |
| 7 | Lock | Approved allocation is locked for operational execution. | Locked allocation with timestamp. |

**Allocation Flow:**



**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Allocation ID | System-generated | Text / ID | Unique identifier for the allocation record. |
| Version | System-generated | Number / Text | Version number of the allocation file or approved allocation. |
| Project | Yes | Lookup / ID | Project associated with the allocation. |
| Centre | Yes | Lookup / ID | Centre assigned in the allocation. |
| Date | Yes | Date | Examination date for the allocation. |
| Shift | Yes | Dropdown | Shift assigned to the candidates. |
| Candidate Count | Yes | Number | Number of candidates allocated to the centre and shift. |
| Buffer Seats | Yes | Number | Additional seats reserved as operational buffer. |
| File | Yes | File upload | Original CSV or Excel file uploaded by the user. |
| Uploaded By | System-generated | User reference | User who uploaded the allocation file. |
| Validation Result | System-generated | Status / Summary | Passed, Passed with Warnings, Failed or Pending Review. |
| Approval Status | Yes | Dropdown | Draft, Under Review, Approved, Rejected, Change Requested or Locked. |
| Lock Timestamp | Conditional | Date/time | Date and time when the allocation was locked. |

**Recommended Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Source File Name | System-generated | Original uploaded filename. |
| Upload Date/Time | System-generated | Date and time of file upload. |
| Column Mapping Version | System-generated | Mapping template used for the upload. |
| Total Rows | System-generated | Number of records in the uploaded file. |
| Valid Rows | System-generated | Number of records that passed validation. |
| Error Rows | System-generated | Number of records with blocking errors. |
| Warning Rows | System-generated | Number of records with non-blocking warnings. |
| Total Candidates | System-generated | Total candidates across the allocation. |
| Total Buffer Seats | System-generated | Total buffer seats across the allocation. |
| Review Remarks | Conditional | Comments from the Operations reviewer. |
| Approval Remarks | Conditional | Comments from the Project Manager. |
| Rejection Reason | Conditional | Required when allocation is rejected. |
| Change Request ID | Conditional | Link to an approved change after locking. |
| Previous Version | Conditional | Link to the previous allocation version. |
| Locked By | Conditional | User who locked the allocation. |
| Unlock Reason | Conditional | Required if exceptional unlocking is permitted. |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts / Finance |
|---|---|---|---|---|---|---|
| Upload allocation file | Full | Full | No | Create / upload | View relevant details | View where commercially relevant |
| Configure import templates | Full | Request / approve | No | Suggest mapping | No | No |

| | | | | | | |
|---|---|---|---|---|---|---|
| Map columns | Full | Review | No | Create / edit | No | No |
| Run validation | Full | Review | No | Execute | No | No |
| Correct allocation errors | Full | Approve changes | No | Edit before approval | Provide centre information | No |
| Review warnings | Full | Full review | No | Initial review | Provide clarification | View if cost impact exists |
| Approve allocation | Full | Full approval | No | Submit for approval | No | No |
| Lock allocation | Full | Full authority | No | No, unless delegated | View locked allocation | View if cost impact exists |
| Request allocation change | Full | Approve / review | No | Create request | Request correction for own centre | Request where commercial impact exists |
| Approve post-lock change | Full | Full approval | No | Recommend | No | Approve commercial impact if applicable |
| View allocation | Full | Full project scope | Audit scope if needed | Assigned project scope | Own centre allocation | Financially relevant view |
| Export allocation | Full | Assigned project scope | Permitted audit scope | Assigned scope | Own centre where permitted | Financially relevant scope |
| View version history | Full | Full | Relevant scope | Assigned scope | Own centre | Relevant financial scope |

**Role Responsibilities:**

**Admin**

| Responsibility | Description |
|---|---|
| Configuration | Maintain allocation templates, field rules and validation parameters. |
| Access control | Manage user access, project scope and approval permissions. |
| Exception control | Handle emergency overrides or exceptional unlock requests. |
| Audit trail | Monitor allocation changes and administrative actions. |
| Reporting | Access all allocation records and historical versions. |

**Project Manager**

| Responsibility | Description |
|---|---|
| Project validation | Confirm that the uploaded allocation belongs to the correct project. |
| Business review | Review candidate counts, centre assignments, shifts and buffer requirements. |
| Approval | Approve or reject the allocation before locking. |
| Exception approval | Approve authorized warnings or exceptions. |
| Change control | Approve change requests after allocation lock. |
| Operational accountability | Ensure the approved allocation is used for readiness and exam execution. |

**Auditor / IT Team Member**

| Responsibility | Description |
|---|---|
| Technical support | Verify technical implications of allocation changes where required. |
| Centre-system review | Confirm that allocated candidate volume is supported by available systems. |

| | |
|---|---|
| Exception input | Provide technical comments for capacity, system or network exceptions. |
| Access limitation | Cannot approve or lock the allocation as the primary allocation owner. |

**Operations Executive**

| **Responsibility** | Accounts / Finance |
|---|---|
| File management | Accounts / Finance |
| Column mapping | Accounts / Finance |
| Data correction | Accounts / Finance |
| Review preparation | Accounts / Finance |
| Centre coordination | Accounts / Finance |
| Change request | Accounts / Finance |
| Execution | Accounts / Finance |

**Centre User**

| **Responsibility** | **Description** |
|---|---|
| Allocation visibility | View the allocation relevant to the centre. |
| Capacity confirmation | Confirm that candidate count, shift and buffer can be supported. |
| Clarification | Report discrepancies or operational concerns. |
| Change request support | Provide information where a centre-specific change is required. |
| Restricted access | Cannot edit, approve or lock the allocation. |

**Accounts / Finance**

| **Responsibility** | **Description** |
|---|---|
| Commercial impact review | Review allocation changes that affect centre rates, candidate charges or invoice amounts. |
| Financial visibility | View approved allocation details needed for cost-sheet and invoice reconciliation. |
| Approval input | Provide commercial confirmation where a change increases financial liability. |
| Access limitation | Cannot modify operational allocation fields or lock the allocation. |

**Validation Rules:**

| **Validation Area** | **Requirement** | **Error / Warning** |
|---|---|---|
| File format | File must be CSV or supported Excel format. | Error |
| Mandatory columns | Required columns must be mapped. | Error |
| Project match | Project must exist and be active. | Error |
| Centre match | Centre must exist in Centre Master or approved booking records. | Error |
| Date match | Allocation date must match the project or booking date. | Error |
| Shift match | Shift must be configured for the project and centre booking. | Error |
| Candidate count | Candidate count must be numeric and greater than zero. | Error |
| Buffer seats | Buffer must be numeric and within configured limits. | Error / Warning |
| Capacity | Candidate count plus buffer must not exceed approved centre capacity. | Error |
| System capacity | Candidate count must not exceed verified system count where applicable. | Error / Warning |
| Duplicate candidate | Candidate identifier should not appear more than once | Error |

| | | |
|---|---|---|
| | unless permitted. | |
| Duplicate centre assignment | Conflicting centre/date/shift assignment must be blocked. | Error |
| Missing candidate data | Required candidate details must be present. | Error |
| Extra columns | Unmapped columns should be reported. | Warning |
| Existing allocation | A new upload must create a new version rather than overwrite the current version. | Warning |
| Commercial impact | Change in candidate count or centre may affect commercials. | Warning / Review Required |

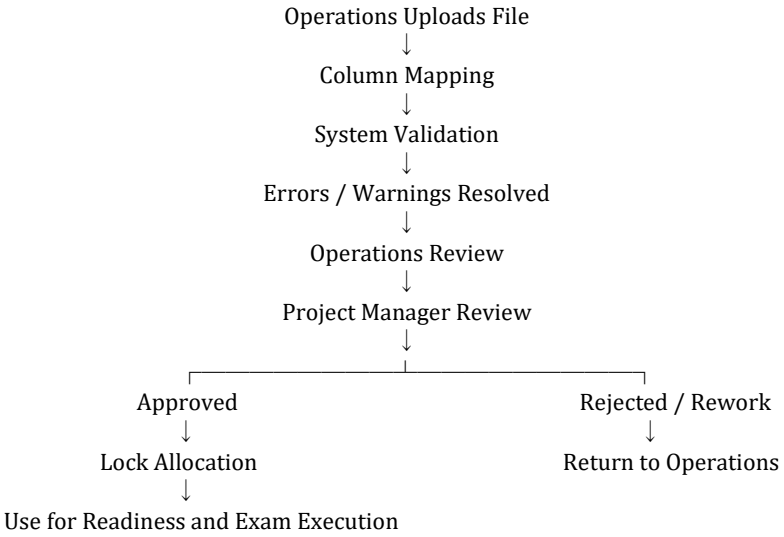
**Validation Result Categories:**

| Result | Meaning | User Action |
|---|---|---|
| Passed | All records and rules passed. | Submit for review. |
| Passed with Warnings | No blocking errors, but review items exist. | Review and provide confirmation. |
| Failed | One or more blocking errors exist. | Correct and re-upload or edit. |
| Pending Review | Validation is complete and awaits Operations or PM review. | Review required. |
| Approved | Project Manager has accepted the allocation. | Proceed to lock. |
| Locked | Allocation is final for operational execution. | Changes require change request. |

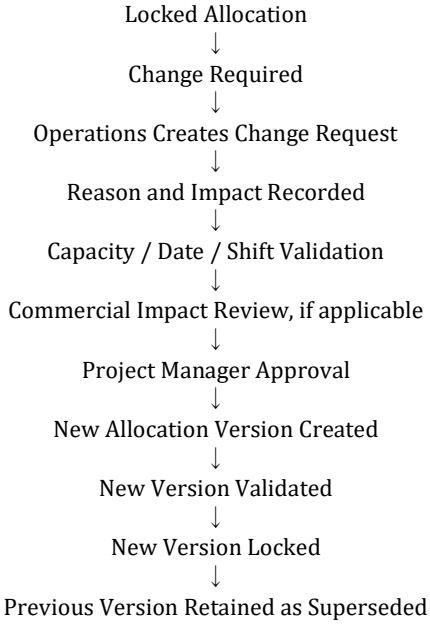
**Allocation Status Flow:**

| Status | Meaning | Next Status |
|---|---|---|
| Draft | Allocation record has been created. | File Uploaded or Cancelled |
| File Uploaded | Source file has been attached. | Mapping Required |
| Mapping Required | Source columns require mapping. | Validation Pending |
| Validation Pending | System validation is running or queued. | Passed, Warning or Failed |
| Failed | Blocking errors were found. | Correction / Re-upload |
| Passed with Warnings | Validation passed with review items. | Under Review |
| Under Review | Operations or PM is reviewing the result. | Approved, Rejected or Rework |
| Approved | PM has approved the allocation. | Locked |
| Locked | Final allocation is frozen. | Change Request or Completed |
| Change Requested | Post-lock change has been initiated. | Approved, Rejected or New Version |
| Superseded | A newer version has replaced the allocation. | Historical record retained |

**Approval Flow:**




**Locked Allocation Change Flow:**




**Change Request Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Change Request ID | System-generated | Unique identifier for the change request. |
| Allocation ID | Yes | Allocation being changed. |
| Current Version | Yes | Currently locked allocation version. |
| Proposed Version | System-generated | New version created after approval. |
| Change Reason | Yes | Business reason for the requested change. |
| Requested By | System-generated | User who initiated the request. |
| Request Date/Time | System-generated | Date and time of request. |
| Changed Centres | Conditional | Centres affected by the change. |
| Changed Dates / Shifts | Conditional | Date or shift changes. |
| Candidate Count Impact | Conditional | Increase or decrease in candidates. |

| | | |
|---|---|---|
| Buffer Impact | Conditional | Change in buffer requirement. |
| Commercial Impact | Conditional | Impact on centre rates or total cost. |
| Operations Recommendation | Yes | Operations review and recommendation. |
| Finance Review | Conditional | Required where the change affects commercials. |
| PM Decision | Yes | Approved, Rejected or Rework. |
| Approval Date/Time | Conditional | Date and time of approval. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| Version retention | Every uploaded and approved allocation version must be retained. |
| No overwrite | A new allocation upload must not overwrite the previous version. |
| Validation gate | Allocation cannot be approved if blocking validation errors remain. |
| Warning review | Warnings require review and recorded action before approval. |
| Approval separation | Operations prepares and reviews; Project Manager approves. |
| Final lock | Approved allocation must be locked before readiness and exam execution. |
| Post-lock control | Locked allocation cannot be edited directly. |
| Change request | Every post-lock change requires reason, impact review and approval. |
| Commercial impact | Accounts / Finance must review changes affecting rates, invoices or centre payments. |
| Centre confirmation | Centre User can view and confirm relevant allocation but cannot edit or approve it. |
| Audit trail | Upload, mapping, validation, correction, approval, lock and change actions must be logged. |
| History retention | Previous versions must remain available for reporting, reconciliation and audit. |

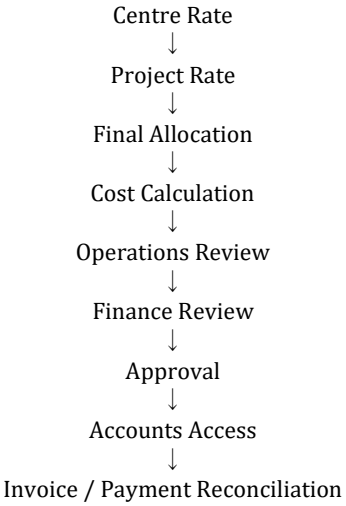
**System Outputs:**

| Output | Description |
|---|---|
| Allocation record | Project- and centre-linked allocation record. |
| Mapping template | Saved column mapping for future uploads. |
| Validation report | Errors, warnings, valid rows and rejected rows. |
| Capacity report | Candidate count, buffer and centre/system capacity comparison. |
| Review record | Operations comments, corrections and recommendations. |
| Approval record | Project Manager decision, timestamp and comments. |
| Locked allocation | Final version used for readiness and exam execution. |
| Change request | Controlled post-lock modification request. |
| Version history | Original, revised, approved, locked and superseded versions. |
| Commercial impact notification | Alert to Accounts / Finance when allocation affects cost or invoice values. |
| MIS update | Allocation status reflected in project and management dashboards. |

## M11 – Commercial / Cost Sheet

| Category | Details |
|---|---|
| Module Name | Commercial / Cost Sheet |
| Primary Users | Operations / Finance |
| Purpose | Prepare, calculate, review, approve and maintain centre-wise commercial costs based on project requirements, final allocation and agreed rates. |
| Process Flow | Centre Rate → Project Rate → Final Allocation → Calculation → Review → Approval → Accounts Access |
| Mandatory / Key Fields | Cost Sheet ID; Version; Project; Centre; Client; Exam; Rate Category; Day Rate; Shift Rate; Candidate Rate; Other Charges; Payment Terms; Validity; Amount Due; Approval Proof; Approved By; Approved Date/Time |
| Business Rules | Approved cost sheets cannot be silently edited. Any change must create a new version, retain the previous approved version and go through review and re-approval. |
| System Outputs | Draft cost sheet; allocation-linked calculation; commercial review record; approved cost sheet; version history; amount due; Accounts access; invoice and payment reference. |

**Commercial Cost Flow:**




**Cost Sheet Lifecycle:**

| Step | Activity | System Action | Expected Output |
|---|---|---|---|
| 1 | Centre Rate | Capture the rate quoted or agreed with the centre. | Centre commercial rate |
| 2 | Project Rate | Apply project, client and examination-specific rate rules. | Project-specific rate |
| 3 | Final Allocation | Link approved candidate count, centre, date and shift. | Allocation-linked cost basis |
| 4 | Calculation | Calculate day, shift, candidate and additional charges. | Total amount due |
| 5 | Review | Operations checks commercial data against booking and allocation. | Operations recommendation |
| 6 | Approval | Finance and authorized approver review and approve or reject. | Approved or rejected cost sheet |
| 7 | Accounts Access | Accounts uses the approved cost sheet for invoice and payment processing. | Financial record and payable amount |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Cost Sheet ID | System-generated | Text / ID | Unique identifier for the cost sheet. |
| Version | System-generated | Number / Text | Version number of the cost sheet. |
| Project | Yes | Lookup / ID | Project associated with the commercial record. |
| Centre | Yes | Lookup / ID | Centre for which the cost is being prepared. |
| Client | Yes | Lookup / ID | Client associated with the examination. |
| Exam | Yes | Lookup / ID | Examination associated with the cost sheet. |
| Rate Category | Yes | Dropdown | Day Rate, Shift Rate, Candidate Rate, System Rate, Manpower Rate, Logistics or Other. |
| Day Rate | Conditional | Amount | Amount applicable per examination day. |
| Shift Rate | Conditional | Amount | Amount applicable per shift. |
| Candidate Rate | Conditional | Amount | Amount calculated per candidate. |
| Other Charges | Conditional | Amount / Breakdown | Additional approved charges. |
| Payment Terms | Yes | Dropdown / Text | Agreed payment terms and conditions. |
| Validity | Yes | Date range | Period during which the commercial terms are valid. |
| Amount Due | System-calculated | Amount | Final amount payable based on approved rates and allocation. |
| Approval Proof | Conditional | File / Reference | Supporting approval email, quotation, signed approval or other evidence. |
| Approved By | Conditional | User reference | Authorized user who approved the cost sheet. |
| Approved Date/Time | Conditional | Date/time | Date and time of approval. |

**Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Booking ID | Yes | Link to the approved centre booking. |
| Allocation ID | Yes | Link to the final allocation used for calculation. |
| Quotation Number | Conditional | Centre quotation reference. |
| Quotation Date | Conditional | Date of centre quotation. |
| Currency | Yes | Currency used for the commercial calculation. |
| Candidate Count | Yes | Approved candidate count from final allocation. |
| Buffer Seats | Yes | Buffer seats considered in the commercial calculation. |
| Total Seats | System-generated | Candidate count plus buffer seats, where chargeable. |
| Tax / GST | Conditional | Applicable tax amount or percentage. |
| Discount | Conditional | Approved discount or adjustment. |
| Total Quoted Amount | Conditional | Amount originally quoted by the centre. |
| Negotiated Amount | Conditional | Amount after negotiation. |
| Approved Amount | Yes | Final amount approved for the project and centre. |
| Commercial Status | Yes | Draft, Submitted, Under Review, Approved, Rejected, Rework, Expired or Superseded. |
| Prepared By | System- | User who prepared the cost sheet. |

| | | |
|---|---|---|
| | generated | |
| Prepared Date/Time | System-generated | Creation timestamp. |
| Reviewed By | Conditional | Operations or Finance reviewer. |
| Review Date/Time | Conditional | Review timestamp. |
| Rejection / Rework Reason | Conditional | Reason for rejection or requested correction. |
| Previous Version | Conditional | Link to previous cost sheet version. |
| Change Request ID | Conditional | Link to approved change request after approval. |

**Rate Components:**

| Component | Calculation Basis | Example |
|---|---|---|
| Day Rate | Fixed amount per examination day | One amount for the centre per day. |
| Shift Rate | Fixed amount per shift | Amount multiplied by number of shifts. |
| Candidate Rate | Rate multiplied by approved candidates | Candidate Count × Candidate Rate. |
| System Rate | Rate based on systems used | Verified systems × approved rate. |
| Manpower Rate | Rate based on staff requirement | Approved manpower count × rate. |
| Logistics Charge | Fixed or variable charge | Courier, transport or material movement. |
| Power Backup Charge | Approved additional charge | UPS, DG or diesel-related charge where applicable. |
| Other Charges | Approved additional items | Consumables, testing or special requirements. |
| Tax / GST | Percentage or fixed amount | Applied according to configured tax rules. |
| Discount | Approved reduction | Deducted only with documented approval. |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Operations Executive | Finance | Auditor / IT | Centre User |
|---|---|---|---|---|---|---|
| Configure rate categories | Full | Review / request | Suggest | Configure / approve | No | No |
| Create cost sheet | Full | Review | Review | Prepare / review | No | Submit quotation |
| Enter centre rate | Full | Review | Capture quotation | Verify | No | Provide quotation |
| Link final allocation | Full | Approve | Link / validate | View | No | View relevant details |
| Calculate amount | Full | Review | Prepare | Verify / calculate | No | No |
| Review commercial details | Full | Review | Initial review | Final commercial review | No | Clarify information |
| Approve cost sheet | Full | Approve where configured | Submit only | Final approval | No | No |
| Reject / request rework | Full | Approve / recommend | Recommend / return | Approve / return | No | Correct / resubmit |
| Edit approved cost sheet | New version only | Approve revised version | Request revision | Create / approve revision | No | Submit revised quotation |
| Maintain | Full | View / | View assigned | Full | No | View |

| | | | | | | |
|---|---|---|---|---|---|---|
| approved records | | monitor | scope | responsibility | | permitted information |
| Access payment-linked amount | Full | Project scope | Status only | Full | No | Own centre view |
| View version history | Full | Full project scope | Assigned scope | Full | No | Own centre history |
| Export commercial reports | Full | Assigned project scope | Assigned scope | Full | No | No |

**Role Responsibilities**

**Operations Executive**

- Collect centre quotation and commercial details.
- Link the quotation to the correct booking, project, exam and allocation.
- Confirm that rates match operational requirements.
- Check candidate count, date, shift, buffer and applicable charges.
- Prepare the cost sheet for Finance review.
- Request corrections from the centre where details are incomplete.
- Initiate a change request when an approved commercial record requires modification.

**Project Manager**

- Review commercial alignment with the project.
- Approve operational exceptions or deviations.
- Confirm that the commercial record supports the approved allocation.
- Approve the cost sheet where the workflow requires PM approval.
- Approve changes after the cost sheet has been approved.
- Monitor commercial status and pending actions.

**Finance**

- Verify centre quotation, rates, taxes, payment terms and calculations.
- Confirm the approved amount and financial liability.
- Approve or reject the cost sheet.
- Maintain the approved commercial record.
- Verify invoice values against the approved cost sheet.
- Track payment terms, amounts due, payments and outstanding balances.
- Maintain signed MoUs and undertakings where applicable.
- Review commercial impact of allocation or booking changes.

**Centre User**

- Submit quotation and rate details.
- Upload quotation, invoice and supporting documents.
- Provide GST and payment information.
- Respond to clarification or rework requests.
- Submit revised quotation as a new version where required.
- View permitted approved commercial details.

**Auditor / IT Team Member**

- No routine cost-sheet preparation or approval responsibility.
- May provide technical input where system count, infrastructure, equipment or technical services affect the cost.
- May view relevant technical-commercial information where authorized.

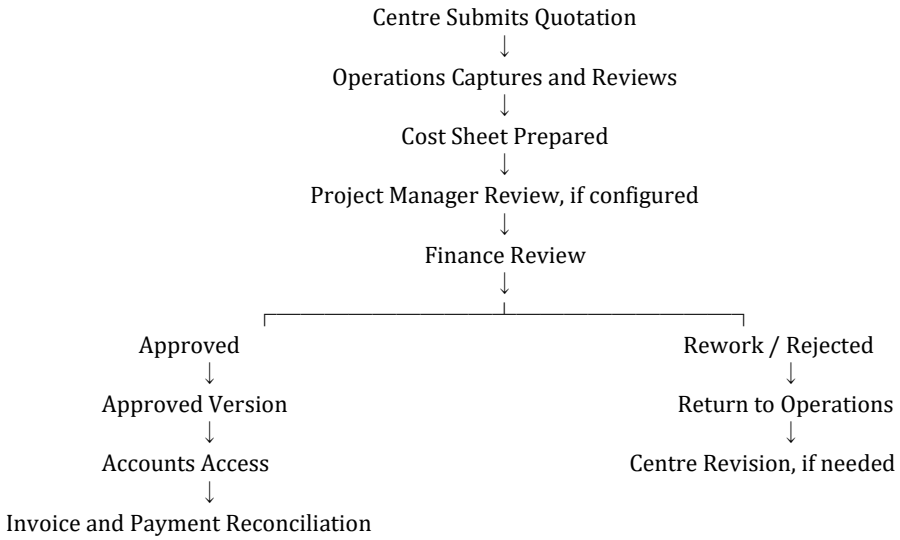
**Admin**

- Configure commercial rules, access rights and rate categories.
- View and manage all records.
- Handle exceptional corrections or administrative overrides.
- Ensure all changes and approvals are audited.

**Commercial Status Flow:**

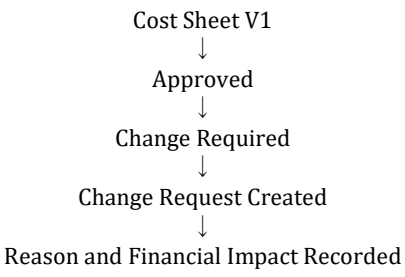
| Status | Meaning | Next Status |
|---|---|---|
| Draft | Cost sheet is being prepared. | Submitted or Cancelled |
| Submitted | Cost sheet has been submitted for review. | Operations Review |
| Operations Review | Operations is checking booking and allocation alignment. | Finance Review, Rework or Rejected |
| Finance Review | Finance is checking rates, taxes and terms. | Approved, Rework, Clarification or Rejected |
| Clarification Required | Additional information is needed. | Finance Review or Rework |
| Rework | Corrections are required before approval. | Resubmitted |
| Approved | Commercial terms and amount are approved. | Accounts Access / Active |
| Rejected | Commercial terms are not approved. | Rework, Replacement or Closed |
| Expired | Commercial validity period has ended. | Revised Version or Renewal |
| Superseded | A newer approved version replaces the current version. | Historical record retained |

**Approval Flow:**




**Version Control:**

Approved Cost Sheet Change Flow



↓
Operations Review
↓
Finance Review
↓
Cost Sheet V2 Created
↓
Recalculation
↓
Re-Approval
↓
V2 Approved
↓
V1 Retained as Historical Version

**Version History Table:**

| Version | Status | Amount | Change Reason | Prepared / Changed By | Approved By |
|---|---|---|---|---|---|
| V1 | Approved | Original approved amount | Initial quotation and allocation | System user | Finance |
| V2 | Superseded | Revised amount | Candidate count or rate changed | System user | Finance |
| V3 | Current Approved | Final amount | Approved commercial correction | System user | Finance / PM |

*The system must always identify the current approved version while preserving all earlier versions.*

**Commercial Validation Rules:**

| Validation Area | Requirement |
|---|---|
| Booking linkage | Cost sheet must be linked to an approved or commercially valid booking. |
| Allocation linkage | Candidate count, date, shift and centre must match the final allocation. |
| Rate validity | Centre rate must fall within its quotation or agreement validity period. |
| Calculation | Amount due must be system-calculated from approved rate components. |
| Quotation match | Quoted amount and calculated amount must be reconciled. |
| Tax validation | GST or applicable tax must be separately captured and validated. |
| Payment terms | Payment terms must be recorded before approval. |
| Agreement requirement | MoU or undertaking must be linked where required by policy. |
| Duplicate prevention | Duplicate commercial records for the same booking and version must be prevented. |
| Approval proof | Approval evidence must be attached or referenced. |
| Revision control | Approved records cannot be directly edited. |
| Financial impact | Any allocation or booking change affecting cost must trigger Finance review. |

**Accounts Access:**

| Information | Purpose |
|---|---|
| Approved cost sheet | Establish payable amount. |
| Current approved version | Ensure latest terms are used. |
| Centre quotation | Validate invoice against quotation. |
| Final allocation | Reconcile candidate count, centre and shift. |
| Payment terms | Determine due date and payment conditions. |

| | |
|---|---|
| MoU and undertaking | Confirm contractual documentation. |
| Invoice request | Match billed amount with approved amount. |
| Payment status | Track due, paid, partially paid and outstanding amounts. |
| Version history | Investigate changes and commercial disputes. |
| Approval proof | Support audit and financial controls. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| No silent editing | Approved cost sheets cannot be changed directly. |
| New version required | Any approved commercial change must create a new version. |
| Re-approval | Revised versions must pass the same configured approval process. |
| Previous version retention | Earlier versions remain available for audit and reconciliation. |
| Allocation dependency | Cost calculation must use approved final allocation data. |
| Finance authority | Finance approves rates, taxes, payment terms and final payable amount. |
| Operations review | Operations confirms that commercial terms match execution requirements. |
| Change impact review | Allocation or booking changes with financial impact must be routed to Finance. |
| Agreement control | Signed MoU and undertakings must be maintained after commercial approval. |
| Invoice matching | Centre invoices must be checked against the current approved cost sheet. |
| Confidential financial access | Commercial and payment data must be restricted by role and scope. |
| Audit trail | Creation, calculation, review, approval, rejection, revision and access actions must be logged. |

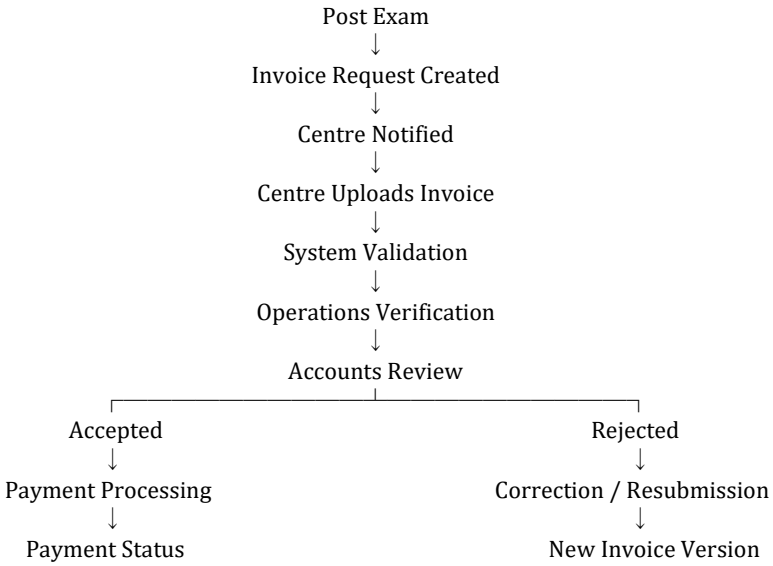
**System Outputs:**

| Output | Description |
|---|---|
| Draft cost sheet | Prepared commercial record before approval. |
| Calculation summary | Breakdown of rates, quantities, taxes, discounts and total amount. |
| Review record | Operations and Finance observations and recommendation. |
| Approved cost sheet | Current approved commercial version. |
| Rework request | Correction request with reason, owner and due date. |
| Version history | Complete history of approved and superseded versions. |
| Commercial clearance | Confirmation that the booking is financially cleared. |
| Accounts record | Finance-accessible record for invoice and payment processing. |
| Invoice matching result | Comparison of invoice amount with approved cost sheet. |
| Commercial dashboard data | Status, approval workload, amounts due and outstanding information. |

# M12 – Invoice Request

| Category | Details |
|---|---|
| Module Name | Invoice Request Management |
| Primary Users | Operations / Centre |
| Supporting User | Accounts / Finance |
| Purpose | Digitize post-exam invoice collection, verification and Accounts review while replacing manual invoice follow-up through Excel, email and WhatsApp. |
| Process Flow | Post Exam → Invoice Request → Centre Upload → Verification → Accounts Review → Accepted / Rejected |
| Mandatory / Key Fields | Invoice Request ID; Project; Centre; Booking/Allocation; Invoice Number; Invoice Date; Invoice Amount; GST Details; Invoice File; Upload Date; Verification Status; Rejection Reason; Accounts Review |
| Business Rule | The invoice workflow should replace manual invoice chasing. Every request, reminder, upload, verification, rejection and acceptance must be recorded in the portal. |
| System Outputs | Invoice request; centre upload; verification result; correction or rejection request; Accounts decision; invoice status; payment-ready record; pending-invoice dashboard. |

**Invoice Flow:**




**Detailed Process Table:**

| Step | Stage | Responsible Role | Description | System Output |
|---|---|---|---|---|
| 1 | Post Exam | Operations | Confirm that the examination and required post-exam activities are complete. | Invoice eligibility status |
| 2 | Invoice Request | Operations | Create an invoice request linked to the project, centre, booking and allocation. | Invoice Request ID |
| 3 | Centre Notification | System / Operations | Notify the centre of the invoice submission requirement and due date. | Request notification |
| 4 | Centre Upload | Centre User | Upload invoice, enter invoice details and provide GST information. | Submitted invoice record |

| | | | | |
|---|---|---|---|---|
| 5 | System Validation | System | Check mandatory fields, file format, invoice number and basic matching details. | Validation result |
| 6 | Operations Verification | Operations | Compare invoice with approved cost sheet, booking, allocation and exam completion data. | Verified, mismatch or rework status |
| 7 | Accounts Review | Accounts / Finance | Review invoice, taxes, amount, supporting documents and financial records. | Accepted, rejected or clarification required |
| 8 | Payment Processing | Accounts / Finance | Use accepted invoice for payment processing and financial tracking. | Payment-ready record |
| 9 | Closure | Operations / Finance | Link payment status and close the invoice request after reconciliation. | Closed invoice record |

**Invoice Request Lifecycle:**

| Status | Meaning | Next Status |
|---|---|---|
| Eligible for Invoice | Post-exam conditions are complete and invoice may be requested. | Request Created |
| Request Created | Invoice request has been created. | Centre Notified |
| Centre Notified | Centre has received invoice instructions. | Pending Upload |
| Pending Upload | Centre has not yet submitted the invoice. | Submitted, Reminder or Escalated |
| Submitted | Centre has uploaded invoice and entered details. | Under Verification |
| Validation Failed | System validation found missing or invalid information. | Rework Required |
| Under Verification | Operations is matching invoice against operational and commercial records. | Accounts Review, Rework or Rejected |
| Accounts Review | Accounts / Finance is reviewing the invoice. | Accepted, Clarification or Rejected |
| Clarification Required | Additional information is needed. | Resubmitted or Under Review |
| Accepted | Invoice is financially accepted. | Payment Processing |
| Rejected | Invoice is not accepted. | Correction / Resubmission or Closed |
| Payment Processing | Invoice is accepted and payment is being processed. | Paid, Partially Paid or Outstanding |
| Closed | Invoice and payment records are reconciled. | Historical record retained |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Invoice Request ID | System-generated | Text / ID | Unique identifier for the invoice request. |
| Project | Yes | Lookup / ID | Project linked to the invoice request. |
| Centre | Yes | Lookup / ID | Centre submitting the invoice. |
| Booking / Allocation | Yes | Lookup / ID | Booking or final allocation against which the invoice is raised. |
| Invoice Number | Yes after upload | Text | Invoice number issued by the centre. |
| Invoice Date | Yes after | Date | Date shown on the invoice. |

| | | | |
|---|---|---|---|
| | upload | | |
| Invoice Amount | Yes after upload | Amount | Total invoice amount. |
| GST Details | Conditional | Structured fields | GST number, taxable amount, tax rate, CGST, SGST, IGST and total tax where applicable. |
| Invoice File | Yes after upload | File upload | Scanned or digitally generated invoice file. |
| Upload Date | System-generated | Date/time | Date and time when the invoice was uploaded. |
| Verification Status | Yes | Dropdown | Pending, Submitted, Under Verification, Verified, Rework, Rejected or Accepted. |
| Rejection Reason | Conditional | Long text | Required when invoice is rejected or returned for correction. |
| Accounts Review | Conditional | Decision / Long text | Accounts decision, remarks, reviewer and review date. |

**GST Fields:**

| Field | Mandatory | Description |
|---|---|---|
| GST Registered | Yes | Indicates whether the centre is GST registered. |
| GSTIN | Conditional | GST identification number. |
| Legal Name on Invoice | Yes | Name appearing on the invoice. |
| Invoice Address | Yes | Billing address appearing on the invoice. |
| Taxable Amount | Conditional | Amount before GST. |
| CGST Amount | Conditional | Central GST amount. |
| SGST Amount | Conditional | State GST amount. |
| IGST Amount | Conditional | Integrated GST amount. |
| Total Tax Amount | Conditional | Total applicable tax. |
| Total Invoice Amount | Yes | Tax-inclusive invoice amount. |
| Place of Supply | Conditional | Applicable supply location. |
| Reverse Charge | Conditional | Yes / No, where applicable. |

**Invoice Verification Checklist:**

| Verification Area | Check | Result |
|---|---|---|
| Centre identity | Invoice is issued by the approved centre. | Pass / Fail |
| Project linkage | Invoice references the correct project or service period. | Pass / Fail |
| Booking linkage | Invoice relates to the correct booking and centre. | Pass / Fail |
| Allocation linkage | Candidate count, date and shift match the approved allocation. | Pass / Fail |
| Cost-sheet match | Invoice amount matches the latest approved cost sheet. | Pass / Mismatch |
| Rate validation | Day, shift, candidate and other rates match approved terms. | Pass / Mismatch |
| GST validation | GST details are complete and mathematically correct. | Pass / Fail |
| Invoice number | Invoice number is present and not duplicated. | Pass / Fail |
| Invoice date | Invoice date is valid and within the required period. | Pass / Fail |
| Supporting documents | MoU, undertaking or other records are available where required. | Pass / Fail |
| File quality | Invoice is readable and complete. | Pass / Fail |
| Approval authority | Invoice is reviewed by authorized Operations and Accounts users. | Pass / Fail |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts / Finance |
|---|---|---|---|---|---|---|
| Configure invoice rules | Full | Review / request | No | Suggest | No | Configure / approve |
| Create invoice request | Full | Approve / monitor | No | Create | No | Create if authorized |
| Send invoice request | Full | Monitor | No | Send / follow up | Receive | Monitor |
| Upload invoice | Full | No | No | Upload / support | Own centre invoice | Upload if received internally |
| Edit invoice details | Full | Review | No | Edit before review | Edit own submission before review | Edit finance fields |
| Validate invoice | Full | Review | No | Initial verification | No | Final verification |
| Match with cost sheet | Full | Review | No | Check operational match | Provide clarification | Full financial match |
| Review GST details | Full | View | No | Preliminary check | Provide details | Full review |
| Accept invoice | Full | Approve where configured | No | Recommend | No | Final acceptance |
| Reject invoice | Full | Approve / escalate | No | Recommend / return | Correct and resubmit | Final rejection |
| Request correction | Full | Full | No | Request | Respond / resubmit | Request |
| View invoice status | Full | Project scope | No | Assigned project | Own invoice | Full |
| View payment status | Full | Project scope | No | Assigned project status | Own centre status | Full |
| Close invoice request | Full | Approve / monitor | No | Request closure | No | Final financial closure |
| Export invoice report | Full | Assigned project scope | No | Assigned scope | No | Full finance scope |

**Role Responsibilities**

**Operations Executive**

- Confirm post-exam completion before creating the invoice request.
- Create and link the request to the correct project, centre, booking and allocation.
- Notify and follow up with the centre.
- Verify invoice details against the approved cost sheet and allocation.
- Identify operational mismatches.
- Return incomplete or incorrect invoices for correction.
- Notify Accounts / Finance when the invoice is ready for review.
- Monitor pending and overdue invoices.

**Centre User**

- Receive the invoice request.
- Upload the invoice and enter invoice details.
- Provide correct GST and bank or payment information where required.
- Respond to clarification or rework requests.
- Resubmit a corrected invoice as a new version.
- View the status of the centre's own invoice request.

**Accounts / Finance**

- Review the invoice against the approved cost sheet.
- Validate amount, tax, GST and payment terms.
- Check invoice duplication and financial completeness.
- Accept, reject or request clarification.
- Maintain accepted invoice and payment records.
- Track amount due, amount paid and outstanding balance.
- Close the invoice after payment reconciliation.

**Project Manager**

- Monitor invoice completion as part of project closure.
- Review escalated disputes, mismatches or overdue invoices.
- Approve exceptions where configured.
- Ensure project closure is not completed while critical invoice or payment actions remain unresolved.

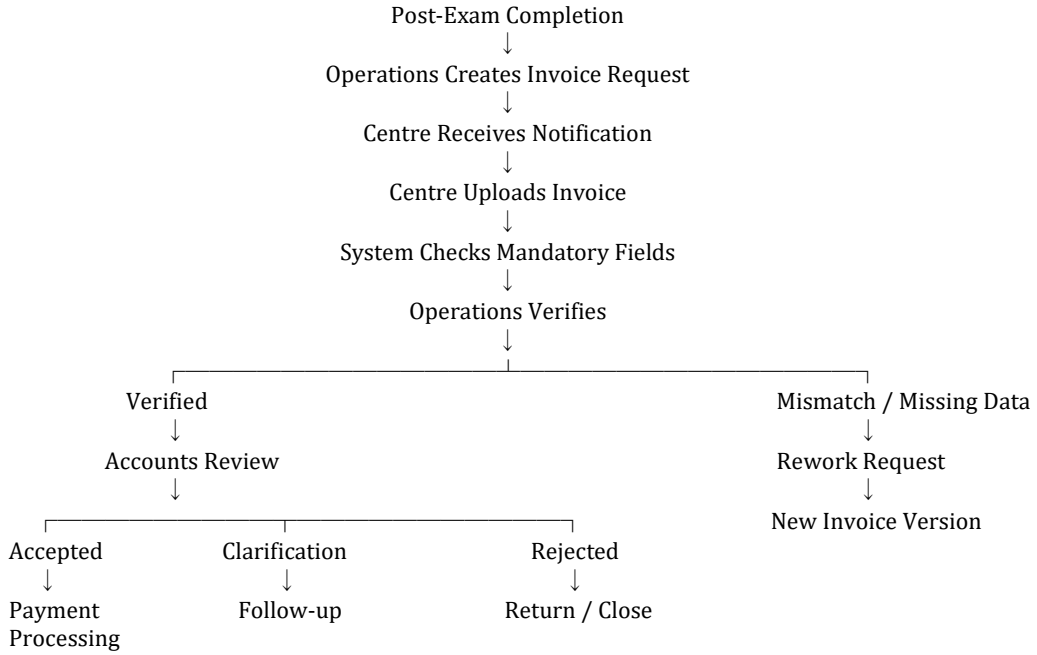
**Auditor / IT Team Member**

- No routine invoice approval responsibility.
- May provide technical confirmation if an invoice includes approved technical services, equipment or infrastructure-related charges.
- May view relevant supporting records where authorized.

**Admin**

- Configure invoice workflows, statuses, reminders and permissions.
- Maintain access control and audit logs.
- Handle administrative corrections through controlled versioning.
- View all invoice and payment records.

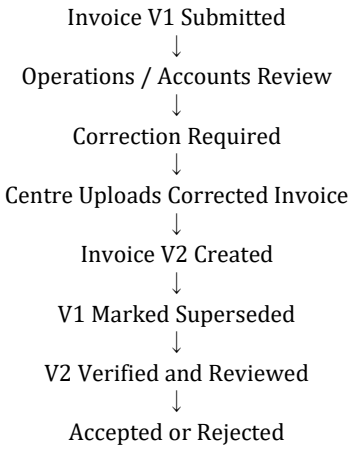
**<u>Invoice Request and Accounts Review Flow:</u>**



**Rejection and Rework Rules:**

| Rejection / Rework Reason | Required Action |
|---|---|
| Invoice amount does not match cost sheet | Return for clarification or approved commercial correction. |
| Incorrect candidate count | Reconcile with final allocation before resubmission. |
| Incorrect centre or project | Centre submits corrected invoice with correct reference. |
| Missing GST details | Centre updates tax information and uploads corrected invoice. |
| Invalid GSTIN | Verify centre master and request correction. |
| Duplicate invoice number | Block acceptance and investigate duplicate submission. |
| Unreadable invoice | Request a clearer file. |
| Missing signature or authorization | Request corrected invoice. |
| Invoice outside payment terms | Route to Finance / Project Manager for decision. |
| Missing supporting agreement | Request MoU, undertaking or related document where required. |
| Invoice submitted after deadline | Record overdue reason and escalate where configured. |

**Invoice Version Control:**




| Version Rule | Requirement |
|---|---|
| New upload | Every corrected upload creates a new invoice version. |
| Previous version | Previous invoice remains retained and marked Superseded or Rejected. |
| Current version | Only the accepted version may be used for payment processing. |
| Approval history | Each version retains reviewer, decision, reason and timestamps. |
| Amount changes | Any amount change must be reconciled against the approved cost sheet. |
| Duplicate prevention | The same invoice number should not be accepted twice for the same centre and project. |

**Automated Notifications:**

| Trigger | Recipient | Notification |
|---|---|---|
| Invoice request created | Centre User | Invoice submission request and due date. |
| Invoice not uploaded | Centre User / Operations | Reminder before due date. |
| Invoice overdue | Operations Executive / Project Manager | Escalation for pending submission. |
| Invoice uploaded | Operations Executive | Invoice ready for verification. |
| Verification failed | Centre User | Correction or rework reason. |
| Sent to Accounts | Accounts / Finance | Invoice ready for financial review. |
| Clarification required | Centre User / Operations | Additional information required. |

| | | |
|---|---|---|
| Invoice accepted | Operations / Centre | Invoice accepted and moved to payment processing. |
| Invoice rejected | Centre / Operations / PM | Rejection reason and next action. |
| Payment overdue | Accounts / PM | Outstanding payment escalation. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| Post-exam trigger | Invoice request should be initiated only after configured post-exam conditions are complete. |
| Centre submission | Centre User submits invoice and supporting information through the portal. |
| Mandatory fields | Invoice number, date, amount, GST information and invoice file must be completed before submission. |
| Operations verification | Operations must verify the invoice against booking, allocation and approved commercial records. |
| Accounts approval | Accounts / Finance performs final financial review and acceptance. |
| Invoice matching | Invoice amount must be compared with the current approved cost sheet. |
| Rework reason | Any correction or rejection must include a reason. |
| Versioning | Corrected invoices create new versions; old versions remain available. |
| Duplicate prevention | Duplicate invoice numbers and duplicate payment claims must be flagged. |
| Payment dependency | Only accepted invoices should proceed to payment processing. |
| Project closure dependency | Project closure should be blocked or flagged when critical invoice or payment actions remain pending. |
| Audit trail | Request, notification, upload, verification, review, acceptance, rejection and payment actions must be logged. |
| Role-based access | Centres see only their own invoices; financial data is restricted to authorized users. |

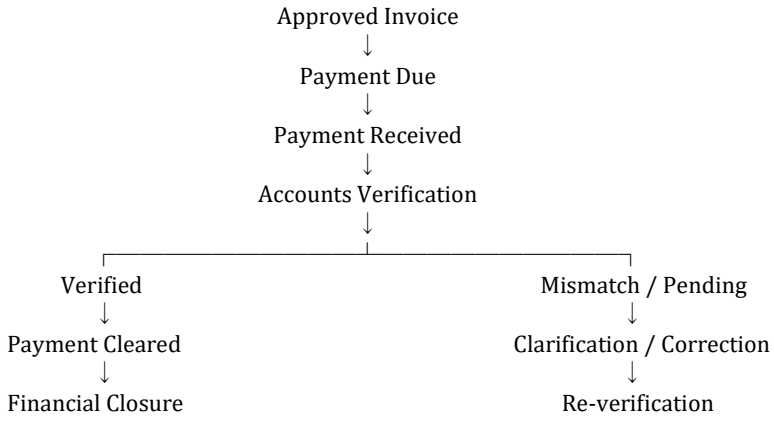
**System Outputs:**

| Output | Description |
|---|---|
| Invoice request | Project- and centre-linked request generated after post-exam completion. |
| Centre upload record | Invoice file, number, date, amount and GST details. |
| Validation result | Completeness, format, duplicate and matching checks. |
| Operations verification | Operational match with booking, allocation and cost sheet. |
| Accounts review record | Acceptance, rejection, clarification, reviewer and timestamp. |
| Rework task | Correction request with reason, owner and due date. |
| Invoice version history | Original and corrected invoice versions retained. |
| Payment-ready record | Accepted invoice available for payment processing. |
| Pending invoice report | Centre- and project-wise pending, overdue and rejected invoices. |
| Finance dashboard update | Invoice value, accepted amount, outstanding amount and payment status. |

# M13 – Accounts & Payment

| Category | Details |
|---|---|
| Module Name | Accounts & Payment Management |
| Primary User | Accounts |
| Supporting Users | Project Manager, Operations Executive and Centre User |
| Purpose | Manage approved centre invoices, payment due dates, payment receipts, verification, outstanding amounts and financial closure. |
| Process Flow | Approved Invoice → Due → Payment Received → Verification → Cleared → Closure |
| Mandatory / Key Fields | Transaction ID; Project; Centre; Invoice; Amount Due; Amount Paid; Payment Date; Payment Reference; Cleared Date/Time; Payment Status; Outstanding Amount; Remarks |
| Business Rules | Operations may view payment status but cannot create or edit financial information. Accounts creates and edits financial records. Payment clearance must be based on verified payment details. |
| System Outputs | Payment transaction; due amount; payment entry; verification result; cleared status; outstanding balance; payment history; financial closure status; Accounts dashboard update. |

**Payment Flow:**




**Detailed Process Table:**

| Step | Stage | Responsible Role | Description | System Output |
|---|---|---|---|---|
| 1 | Approved Invoice | Accounts | Receive an invoice accepted through the invoice workflow. | Payment liability created. |
| 2 | Due | Accounts | Record payment terms, due date and amount payable. | Payment due record. |
| 3 | Payment Received | Accounts | Enter amount paid, payment date and payment reference. | Payment transaction. |
| 4 | Verification | Accounts | Verify payment with bank record, transaction proof or approved payment evidence. | Verified or exception status. |
| 5 | Cleared | Accounts | Confirm that the payment is fully reconciled with the invoice. | Cleared payment. |
| 6 | Closure | Accounts / Project Manager | Close the financial record when no amount remains outstanding. | Financial closure status. |

**Payment Status Lifecycle:**

| Status | Meaning | Next Status |
|---|---|---|
| Invoice Approved | Invoice has passed Accounts review. | Payment Due |
| Payment Due | Amount is payable according to agreed terms. | Payment Received, Overdue or On Hold |
| Partially Paid | Some amount has been paid but balance remains. | Payment Received or Outstanding |
| Payment Received | Payment transaction has been entered. | Under Verification |
| Under Verification | Accounts is validating payment details. | Cleared, Exception or Rework |
| Exception | Payment reference, amount or invoice matching requires clarification. | Under Verification or On Hold |
| Cleared | Payment is fully verified and no amount remains payable. | Financial Closure |
| Outstanding | Amount remains unpaid after partial payment or due date. | Payment Received or Escalated |
| Overdue | Due date has passed and payment remains unpaid. | Escalated, Paid or On Hold |
| On Hold | Payment is temporarily blocked due to a dispute, missing document or approval. | Released, Paid or Cancelled |
| Closed | Financial record is complete and reconciled. | Historical record retained |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Transaction ID | System-generated | Text / ID | Unique identifier for the payment transaction. |
| Project | Yes | Lookup / ID | Project associated with the payment. |
| Centre | Yes | Lookup / ID | Centre receiving the payment. |
| Invoice | Yes | Lookup / ID | Approved invoice against which payment is processed. |
| Amount Due | Yes | Amount | Total amount payable under the approved invoice. |
| Amount Paid | Yes after payment | Amount | Amount paid in the transaction. |
| Payment Date | Conditional | Date | Date on which payment was made. |
| Payment Reference | Conditional | Text | Bank reference, transaction ID, UTR or other payment reference. |
| Cleared Date/Time | Conditional | Date/time | Date and time when Accounts verified and cleared the payment. |
| Payment Status | Yes | Dropdown | Due, Partially Paid, Received, Under Verification, Cleared, Outstanding, Overdue, On Hold or Closed. |
| Outstanding Amount | System-calculated | Amount | Amount Due minus verified Amount Paid. |
| Remarks | Conditional | Long text | Payment notes, exceptions, disputes or reconciliation comments. |

**Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Invoice Request ID | Yes | Link to the invoice request workflow. |

| | | |
|---|---|---|
| Cost Sheet ID | Yes | Link to the approved commercial cost sheet. |
| Approved Cost Sheet Version | Yes | Version used for invoice and payment reconciliation. |
| Payment Terms | Yes | Agreed payment terms. |
| Due Date | Yes | Date by which payment is expected. |
| Payment Mode | Conditional | Bank Transfer, NEFT, RTGS, IMPS, Cheque or other configured mode. |
| Bank Account Name | Conditional | Recipient account name, subject to security policy. |
| Bank Name | Conditional | Receiving or paying bank. |
| UTR / Transaction Number | Conditional | Bank transaction reference. |
| Payment Proof | Conditional | Bank advice, transfer confirmation or payment document. |
| Currency | Yes | Currency of the transaction. |
| Tax Deducted | Conditional | TDS or other deduction. |
| Net Paid Amount | Conditional | Amount paid after deductions. |
| Verification Result | Conditional | Verified, Mismatch, Pending or Rejected. |
| Verified By | Conditional | Accounts user who verified the payment. |
| Verification Date/Time | Conditional | Date and time of verification. |
| Closure Status | Yes | Open, Ready for Closure or Closed. |
| Closure Approved By | Conditional | User approving financial closure. |
| Closure Date/Time | Conditional | Date and time of financial closure. |

**Payment Calculation:**

Outstanding Amount = Amount Due − Verified Amount Paid − Approved Adjustments

**For partial payments:**
Outstanding Amount > 0
↓
Payment Status = Partially Paid / Outstanding

**For complete payments:**
Outstanding Amount = 0
↓
Payment Status = Cleared
↓
Financial Closure
*Any adjustment, deduction, withholding or write-off must be recorded separately with approval and reason.*

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts |
|---|---|---|---|---|---|---|
| View payment record | Full | Assigned project | No | Assigned project status | Own centre status | Full |
| Create transaction | Full | Request / authorize | No | Request / support | No | Full |
| Edit amount paid | Full with audit | Approve / authorize | No | No | No | Full |
| Edit payment date | Full with audit | Approve / authorize | No | No | No | Full |
| Enter payment reference | Full | No | No | No | No | Full |

| | | | | | | |
|---|---|---|---|---|---|---|
| Upload payment proof | Full | Upload / review | No | Support upload | No | Full |
| Verify payment | Full | Review / approve where configured | No | No | No | Full |
| Mark payment cleared | Full | Approve / authorize | No | No | No | Full |
| Place payment on hold | Full | Full approval | No | Request | No | Full |
| Release payment hold | Full | Approve / authorize | No | No | No | Full |
| View outstanding amount | Full | Full assigned project | No | Assigned project | Own centre | Full |
| Close financial record | Full | Approve / monitor | No | Request closure | No | Full |
| Export payment report | Full | Assigned project scope | No | Assigned scope | No | Full |
| View payment history | Full | Project scope | No | Assigned scope | Own centre | Full |

**Role Responsibilities**

**Accounts**

- Receive approved invoice records.
- Create payment due records.
- Maintain due dates and payment terms.
- Enter payments received.
- Record payment reference and proof.
- Verify payments against bank or transaction records.
- Calculate outstanding amounts.
- Manage partial payments, deductions and exceptions.
- Mark payments as cleared.
- Maintain payment and financial history.
- Complete financial closure.
- Monitor overdue payments through the Accounts dashboard.

**Operations Executive**

- View payment status for assigned projects and centres.
- Follow up with Accounts or the Project Manager on pending payment issues.
- Support clarification between the centre and Accounts.
- Cannot create or edit financial amounts, payment references or clearance records.

**Project Manager**

- Monitor project-level payment status.
- Review escalated disputes or overdue payments.
- Approve exceptions where configured.
- Confirm financial completion before project closure.

**Centre User**

- View invoice and payment status for the centre's own records.
- Provide clarification or supporting information when requested.
- Cannot create, edit or verify payment records.

**Auditor / IT Team Member**

- No routine payment access.
- May view payment status only if required for a project or audit context.
- Cannot create, edit or approve financial records.

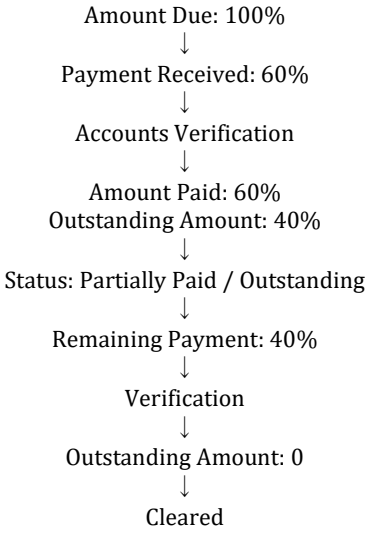
**Admin**

- Configure payment statuses, workflows, access and dashboard rules.
- View all financial records.
- Perform controlled administrative corrections.
- Ensure changes are logged and authorized.

**Payment Verification Checklist:**

| Verification Area | Check | Result |
|---|---|---|
| Invoice match | Payment is linked to the correct approved invoice. | Pass / Fail |
| Cost-sheet match | Invoice and payment amount match approved commercial terms. | Pass / Mismatch |
| Centre match | Payment beneficiary matches the approved centre. | Pass / Fail |
| Project match | Payment is linked to the correct project. | Pass / Fail |
| Amount validation | Amount paid is correctly recorded. | Pass / Fail |
| Payment reference | UTR, transaction reference or equivalent is available. | Pass / Pending |
| Payment date | Payment date is valid and recorded. | Pass / Fail |
| Payment proof | Supporting bank or payment evidence is available. | Pass / Pending |
| Tax / deduction | TDS or other deductions are correctly recorded. | Pass / Review |
| Outstanding balance | System-calculated balance is accurate. | Pass / Fail |
| Duplicate payment | Transaction has not already been recorded. | Pass / Fail |
| Approval | Payment clearance is approved by authorized Accounts user. | Pass / Fail |

**Partial Payment Flow:**




| Payment Scenario | Amount Due | Amount Paid | Outstanding | Status |
|---|---|---|---|---|
| No payment | 100,000 | 0 | 100,000 | Due / Outstanding |
| Partial payment | 100,000 | 60,000 | 40,000 | Partially Paid |
| Full payment pending verification | 100,000 | 100,000 | 0 | Under Verification |
| Full payment verified | 100,000 | 100,000 | 0 | Cleared |
| Payment exceeds due amount | 100,000 | 110,000 | Exception | Exception / Review |

**Exception and Escalation Rules:**

| Exception | System Action |
|---|---|
| Payment reference missing | Keep payment Under Verification and request details. |
| Amount paid differs from expected amount | Create payment exception and require Accounts review. |
| Duplicate reference | Block clearance and flag possible duplicate payment. |
| Payment proof unavailable | Keep status Pending Verification. |
| Partial payment | Calculate outstanding amount and retain balance. |
| Payment overdue | Generate reminder and escalate to Accounts and Project Manager. |
| Invoice mismatch | Place payment On Hold until corrected or approved. |
| Bank details mismatch | Block clearance and route for verification. |
| Payment reversal | Create reversal or adjustment record; do not delete the original transaction. |
| Tax deduction | Record deduction separately and recalculate outstanding amount according to policy. |

## Accounts Dashboard

### Dashboard Purpose

The Accounts dashboard should provide complete visibility of invoice liabilities, payments, outstanding balances and financial closure across all projects and centres.

### Key Dashboard KPIs

| KPI | Description |
|---|---|
| Approved invoices | Number and value of invoices approved for payment. |
| Total amount due | Total payable value of approved invoices. |
| Payment due this period | Amount reaching due date within the selected period. |
| Payments received | Total amount recorded as received. |
| Payments under verification | Payments awaiting confirmation. |
| Payments cleared | Fully verified payments. |
| Outstanding amount | Total unpaid balance. |
| Partially paid invoices | Invoices with remaining balances. |
| Overdue payments | Payments past their due date. |
| Payments on hold | Payments blocked due to disputes or missing information. |
| Payment exceptions | Amount mismatches, duplicate references or missing proof. |
| Financial closure pending | Projects or invoices awaiting final closure. |
| Average payment turnaround | Time from invoice acceptance to payment clearance. |

### Dashboard Filters

| Filter | Examples |
|---|---|
| Client | Client-wise payment monitoring. |
| Project | Project-level payment status. |
| Centre | Centre-wise outstanding balance. |
| Exam | Examination-wise payment analysis. |
| Project Manager | Ownership and escalation. |
| Operations Executive | Operational follow-up responsibility. |
| Invoice date | Invoice-period analysis. |
| Due date | Upcoming and overdue payments. |
| Payment date | Payment-period analysis. |

| | |
|---|---|
| Payment status | Due, Partially Paid, Received, Cleared, Outstanding or Overdue. |
| Amount range | High-value or low-value transactions. |
| Payment mode | Bank transfer, cheque or other configured mode. |

**Notifications and Escalations**

| Trigger | Recipient | Notification |
|---|---|---|
| Invoice approved | Accounts | Payment record should be created. |
| Payment due approaching | Accounts | Upcoming due-date reminder. |
| Payment due date reached | Accounts / Project Manager | Payment due notification. |
| Payment overdue | Accounts / Project Manager | Escalation notification. |
| Payment entered | Accounts | Verification task created. |
| Payment mismatch | Accounts / Project Manager | Exception alert. |
| Payment cleared | Operations / Project Manager / Centre | Payment clearance notification. |
| Outstanding balance remains | Accounts / Project Manager | Outstanding payment reminder. |
| Financial closure ready | Project Manager | Project can be reviewed for financial closure. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| Accounts ownership | Accounts creates and edits financial information. |
| Operations visibility | Operations can view payment status but cannot edit financial records. |
| Payment verification | Payment must be verified before being marked Cleared. |
| Invoice linkage | Every transaction must link to an approved invoice and project. |
| Reference requirement | Payment reference is required for verification. |
| Outstanding calculation | Outstanding amount is system-calculated, not manually typed. |
| Partial payment support | The system must support multiple payments against one invoice. |
| Duplicate prevention | Duplicate payment references and duplicate transactions must be flagged. |
| No deletion | Payment records, reversals and corrections must remain in history. |
| Adjustment control | Any adjustment or write-off requires reason and authorization. |
| Overdue escalation | Overdue payments must generate reminders and escalations. |
| Financial closure | Closure requires no unresolved payment, invoice or reconciliation issue. |
| Audit trail | Creation, editing, verification, clearance and closure actions must be logged. |
| Confidential access | Payment amounts and banking details must be restricted to authorized roles. |

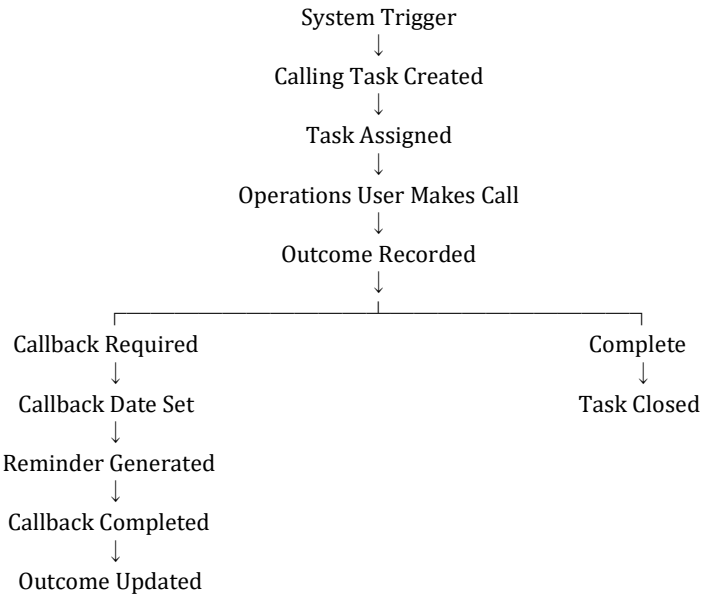
**System Outputs:**

| Output | Description |
|---|---|
| Payment due record | Approved invoice converted into a payment liability. |
| Payment transaction | Amount paid, date, reference and proof. |
| Verification result | Verified, Pending, Mismatch, Exception or Rejected. |
| Outstanding balance | System-calculated amount remaining. |
| Payment status | Due, Partially Paid, Cleared, Outstanding, Overdue or On Hold. |
| Payment clearance | Confirmation that the payment is fully verified. |
| Financial closure record | Final closure of invoice and payment obligations. |
| Accounts dashboard | Real-time monitoring of financial operations. |
| Project MIS update | Payment status reflected in project reporting. |
| Audit history | Complete record of financial actions and changes. |

# M14 – Calling & Accountability

| Category | Details |
|---|---|
| Module Name | Calling & Accountability |
| Primary User | Operations |
| Supporting Users | Project Manager, Centre User and Admin |
| Purpose | Manage centre communication and follow-up activities through assigned calling tasks, recorded outcomes, callbacks, reminders and escalations. |
| Process Flow | System Trigger → Calling Task → Assignment → Call → Outcome → Callback / Complete → Reminder |
| Mandatory / Key Fields | Task ID; Project; Centre; Task Type; Assigned User; Due Date; Call Time; Outcome; Remarks; Callback Date; Next Action; Status |
| Business Rules | Callback tasks and overdue work must be tracked automatically. Every task must have an owner, due date, outcome and next action where applicable. |
| System Outputs | Calling task; assignment record; call log; outcome; callback task; reminder; escalation; completion status; accountability dashboard. |

**Calling Flow:**




**Calling Task Lifecycle:**

| Step | Stage | Responsible Role | Description | System Output |
|---|---|---|---|---|
| 1 | System Trigger | System | Creates a task based on a workflow event, due date or pending action. | Calling task |
| 2 | Task Assignment | Project Manager / Operations | Assigns task to an Operations user. | Owner and due date |
| 3 | Call | Operations | Contacts centre or relevant stakeholder. | Call timestamp |
| 4 | Outcome | Operations | Records response, status and remarks. | Call outcome |
| 5 | Callback / Complete | Operations | Schedules callback or marks task complete. | Callback task or closure |
| 6 | Reminder | System | Sends reminders for upcoming callbacks and overdue work. | Reminder and escalation |
| 7 | Monitoring | Project Manager | Reviews task progress, pending calls and overdue activities. | Accountability dashboard |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Task ID | System-generated | Text / ID | Unique identifier for the calling task. |
| Project | Yes | Lookup / ID | Project associated with the task. |
| Centre | Yes | Lookup / ID | Centre to be contacted. |
| Task Type | Yes | Dropdown | Booking follow-up, consent, document, audit, allocation, readiness, invoice or payment follow-up. |
| Assigned User | Yes | User Lookup | Operations user responsible for completing the task. |
| Due Date | Yes | Date / Date-time | Deadline for completing the call or follow-up. |
| Call Time | Conditional | Date/time | Date and time the call was made. |
| Outcome | Conditional | Dropdown | Connected, No Answer, Busy, Callback Required, Confirmed, Rejected, Discussion Required or Invalid Contact. |
| Remarks | Conditional | Long text | Call notes, centre response and relevant details. |
| Callback Date | Conditional | Date / Date-time | Date and time for the next call. Required when callback is selected. |
| Next Action | Conditional | Long text / Dropdown | Follow-up action required after the call. |
| Status | Yes | Dropdown | New, Assigned, In Progress, Callback Scheduled, Completed, Overdue, Escalated or Cancelled. |

**Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Trigger Event | Yes | Workflow event that created the task. |
| Contact Person | Yes | Person to be contacted. |
| Contact Number | Yes | Number used for the call. |
| Alternate Contact | No | Backup contact information. |
| Priority | Yes | Low, Medium, High or Critical. |
| Attempt Number | System-generated | Number of contact attempts. |
| Last Attempt Date/Time | System-generated | Time of the most recent call attempt. |
| Call Channel | Yes | Phone, online meeting, portal message or other configured channel. |
| Response Received From | Conditional | Person who provided the response. |
| Escalation Level | System-generated | Current escalation stage. |
| Completion Date/Time | Conditional | Time at which the task was completed. |
| Created By | System-generated | User or system event that created the task. |
| Created Date/Time | System-generated | Task creation timestamp. |
| Attachments | Conditional | Supporting file, confirmation or call evidence. |
| Linked Record | Yes | Booking, consent, document, audit, allocation, readiness, invoice or incident record. |

**Task Type Matrix:**

| Task Type | Typical Trigger | Expected Outcome |
|---|---|---|
| Booking follow-up | Booking created or consent pending. | Centre confirms availability or requests discussion. |
| Consent follow-up | No response to consent request. | Accept, discuss or reject. |
| Document follow-up | Mandatory document pending or rework required. | Document uploaded or revised due date. |
| Audit follow-up | Audit visit, evidence or corrective action pending. | Visit completed or evidence submitted. |
| Allocation follow-up | Centre has not confirmed allocation. | Capacity and shift confirmed. |
| Readiness follow-up | Installation, manpower or mock confirmation pending. | Readiness status updated. |
| Exam-day follow-up | Operational issue or confirmation required. | Issue handled or escalated. |
| Invoice follow-up | Invoice not submitted or correction pending. | Invoice submitted or reworked. |
| Payment follow-up | Payment clarification or outstanding status. | Accounts or centre response recorded. |

**Task Status Flow:**

| Status | Meaning | Next Status |
|---|---|---|
| New | Task has been generated but not assigned. | Assigned or Cancelled |
| Assigned | Task owner and due date have been set. | In Progress |
| In Progress | User is attempting contact or working on follow-up. | Completed, Callback Scheduled or Escalated |
| Callback Scheduled | A future callback has been created. | In Progress, Completed or Overdue |
| Completed | Required outcome or action has been achieved. | Closed |
| Overdue | Due date has passed without completion. | In Progress or Escalated |
| Escalated | Task requires Project Manager attention. | Resolved or Closed |
| Cancelled | Task is no longer required, with reason recorded. | Historical record retained |
| Closed | Task and all required actions are complete. | Historical record retained |

**Outcome Matrix:**

| Outcome | Required Fields | System Action |
|---|---|---|
| Connected – Confirmed | Remarks, next action if any | Complete task or create next task. |
| Connected – Callback Required | Remarks, callback date, next action | Create callback task automatically. |
| No Answer | Attempt number, remarks, next action | Schedule retry according to rule. |
| Busy | Attempt number, next action | Schedule retry. |
| Need Discussion | Discussion details, owner, due date | Create follow-up or escalate. |
| Document Promised | Expected document date, remarks | Create document follow-up task. |
| Rejected | Rejection reason, remarks | Notify Project Manager and update linked workflow. |
| Invalid Contact | Contact correction details | Create contact-update task. |
| Escalation Required | Escalation reason, priority | Escalate to Project Manager. |

**Automatic Task Triggers:**

| Trigger Event | Calling Task |
|---|---|
| Centre consent request not answered | Consent follow-up task. |
| Soft lock nearing expiry | Booking confirmation task. |
| Mandatory document overdue | Document follow-up task. |
| Document marked Rework Required | Centre correction task. |
| Audit visit not confirmed | Audit coordination task. |
| Audit corrective action overdue | Audit follow-up task. |
| Allocation uploaded but centre confirmation pending | Allocation confirmation task. |
| Installation date approaching | Installation confirmation task. |
| Mock readiness incomplete | Mock follow-up task. |
| Final readiness not confirmed | Readiness escalation task. |
| Invoice request not submitted | Invoice follow-up task. |
| Payment overdue or clarification pending | Payment coordination task. |
| Incident requires external response | Incident follow-up task. |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts / Finance |
|---|---|---|---|---|---|---|
| Configure task types | Full | Review / request | Technical input | Suggest | No | Finance task input |
| Create task manually | Full | Full | Create audit tasks | Create operational tasks | Request / respond | Create finance tasks |
| Assign task | Full | Full team scope | Assign technical tasks if authorized | No, unless delegated | No | Assign finance tasks |
| View task | Full | Full assigned projects | Audit and technical scope | Assigned tasks | Own centre tasks | Finance-related tasks |
| Execute call | Full | Monitor / execute if required | Technical calls | Full responsibility | Respond | Finance calls |
| Record outcome | Full | Review | Technical outcomes | Create / edit own tasks | Respond / confirm | Finance outcomes |
| Schedule callback | Full | Monitor | Technical callback | Create / manage | Confirm availability | Finance callback |
| Close task | Full | Approve / monitor | Submit technical closure | Close assigned tasks | No | Close finance tasks |
| View overdue tasks | Full | Full team scope | Audit scope | Own tasks | Own pending responses | Finance scope |
| Escalate task | Full | Full authority | Technical escalation | Request / initiate | Report urgent issue | Finance escalation |
| Edit task history | Controlled, with audit | No overwrite | No overwrite | No overwrite | No overwrite | No overwrite |

**Role Responsibilities:**

**Operations**

- Monitor system-generated calling tasks.
- Accept assigned tasks and complete calls within the due date.
- Record call time, outcome, remarks and next action.
- Schedule callbacks where required.
- Follow up with centres for consent, documents, allocation, readiness, invoices and other activities.
- Escalate unresolved or overdue tasks.
- Ensure call notes are factual and linked to the correct project and centre.

**Project Manager**

- Assign or redistribute tasks.
- Monitor team workload and overdue calls.
- Review escalated tasks and unresolved centre issues.
- Approve changes to deadlines where required.
- Use calling performance data for project accountability.

**Centre User**

- Receive calls, reminders and follow-up requests.
- Provide responses, confirmations and documents.
- Confirm callback availability.
- View only tasks related to the centre's own records.

**Accounts / Finance**

- Manage invoice and payment-related calling tasks.
- Follow up on invoice corrections, payment references and outstanding balances.
- Record financial outcomes.
- Escalate payment issues to the Project Manager where required.

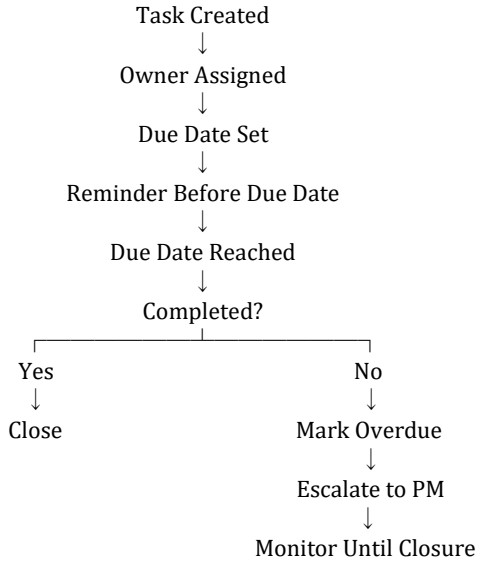
**Auditor / IT Team Member**

- Handle technical, audit, IP scan and infrastructure-related follow-up.
- Record technical outcomes and corrective-action commitments.
- Schedule re-scan or technical callback tasks.

**Admin**

- Configure task categories, reminder schedules, escalation rules and role access.
- View all calling records.
- Maintain system audit logs and administrative controls.

**Reminder and Escalation Flow:**



**Reminder Rules:**

| Condition | System Action |
|---|---|
| Task assigned | Notify assigned Operations user. |
| Due date approaching | Send reminder according to configured schedule. |
| Callback date approaching | Send callback reminder. |
| Due date missed | Mark task Overdue. |
| Task overdue beyond threshold | Escalate to Project Manager. |
| Repeated failed contact | Increase attempt count and escalate according to priority. |
| Critical task overdue | Immediate escalation to Project Manager. |
| Task completed | Notify relevant workflow owner and update linked record. |

**Calling Accountability Dashboard:**

**Key Metrics**

| KPI | Description |
|---|---|
| Total open tasks | All incomplete calling tasks. |
| Tasks due today | Tasks requiring action today. |
| Overdue tasks | Tasks past their due date. |
| Callbacks due today | Scheduled callbacks requiring action. |
| Completed calls | Calls completed in the selected period. |
| Contact success rate | Connected calls compared with total attempts. |
| Centre response rate | Centres that responded to follow-ups. |
| Average completion time | Time from task assignment to completion. |
| Tasks by employee | Workload and completion by Operations user. |
| Tasks by project | Pending follow-up by project. |
| Tasks by centre | Centre-wise communication status. |
| Escalated tasks | Tasks requiring manager attention. |
| Repeated no-answer cases | Centres with repeated unsuccessful attempts. |
| Pending outcome updates | Calls made without recorded outcomes. |

**Dashboard Filters:**

| Filter | Examples |
|---|---|
| Project | Project-specific calling activity. |
| Centre | Centre-wise communication history. |
| Task type | Consent, document, audit, readiness, invoice or payment. |
| Assigned user | Operations employee workload. |
| Priority | Low, Medium, High or Critical. |
| Status | New, Assigned, In Progress, Callback, Completed or Overdue. |
| Due date | Today, this week, overdue or custom range. |
| Outcome | Connected, no answer, callback or rejected. |
| Escalation level | Normal, manager escalation or critical escalation. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| Task ownership | Every task must have an assigned user. |
| Due date | Every task must have a due date or configured SLA. |
| Call outcome | A completed call must have an outcome and remarks. |
| Callback control | Selecting Callback Required must create a callback date and next action. |
| Overdue tracking | Missed due dates automatically mark tasks overdue. |

| | |
|---|---|
| Escalation | Overdue and critical tasks escalate to the Project Manager. |
| Linked records | Tasks must link to the correct project, centre and workflow record. |
| No silent deletion | Completed and cancelled task history must be retained. |
| Attempt tracking | Multiple call attempts must be recorded rather than overwritten. |
| Reminder automation | Reminders must be system-generated based on due and callback dates. |
| Access control | Users view and update only tasks within their assigned scope. |
| Audit trail | Assignment, reassignment, call outcome, callback, escalation and closure must be logged. |

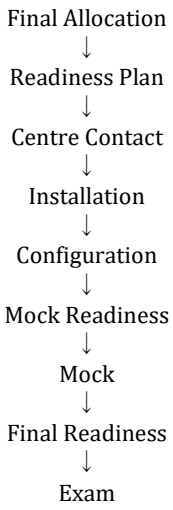
**System Outputs:**

| **Output** | **Description** |
|---|---|
| Calling task | System- or user-created follow-up task linked to a workflow event. |
| Assignment record | Owner, assignment time and due date. |
| Call log | Call time, contact method, attempt number and person contacted. |
| Outcome record | Response, remarks and next action. |
| Callback task | Automatically created follow-up task with callback date. |
| Reminder | Upcoming-task or callback notification. |
| Escalation | Manager notification for overdue or critical activity. |
| Accountability report | Employee, project, centre and task-type performance. |
| Communication history | Complete follow-up history linked to each centre and project. |
| MIS update | Pending and overdue work reflected in management dashboards. |

# M15 – Mock & Readiness

| Category | Details |
|---|---|
| Module Name | Mock & Readiness Management |
| Primary Users | Operations / Centre |
| Supporting Users | Project Manager, Auditor / IT, Admin |
| Purpose | Plan, monitor and confirm centre readiness from final allocation through installation, configuration, mock execution and final exam readiness. |
| Process Flow | Final Allocation → Readiness Plan → Centre Contact → Installation → Configuration → Mock Readiness → Mock → Final Readiness → Exam |
| Mandatory / Key Fields | Readiness ID; Project; Centre; Exam Date; Installation Date; Mock Date/Time; Centre Availability; Manpower Confirmation; Installation Status; Mock Status; Final Readiness Status; Exception; Remarks; Confirmation Time |
| Business Rules | Readiness percentage and exceptions must be visible to management. Final readiness cannot be approved while critical installation, configuration, manpower, mock or technical exceptions remain unresolved. |
| System Outputs | Readiness plan; centre confirmations; installation status; configuration status; mock result; exception tracker; readiness percentage; final readiness approval; exam-release status. |

**Readiness Flow:**




**Detailed Process Table:**

| Step | Stage | Responsible Role | Description | System Output |
|---|---|---|---|---|
| 1 | Final Allocation | Operations / PM | Use the approved and locked allocation as the basis for readiness planning. | Readiness requirement |
| 2 | Readiness Plan | Operations | Create dates, owners, tasks, dependencies and centre milestones. | Readiness plan |
| 3 | Centre Contact | Operations / Centre | Confirm centre availability, contacts, manpower and operational requirements. | Centre confirmation |
| 4 | Installation | Operations / IT / Centre | Install systems, devices, power backup, network components and materials. | Installation status |
| 5 | Configuration | Auditor / IT / Operations | Configure systems, software, network and examination settings. | Configuration status |
| 6 | Mock Readiness | Operations / IT / Centre | Confirm that the centre is prepared for mock activity. | Mock readiness decision |
| 7 | Mock | Operations / | Conduct mock activity and record findings. | Mock result and |

| | | | | |
|---|---|---|---|---|
| | | Centre | | exceptions |
| 8 | Final Readiness | PM / Operations | Review all readiness controls and approve or escalate open issues. | Final readiness status |
| 9 | Exam | Operations / Centre | Release the centre for exam execution. | Exam-ready centre |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Readiness ID | System-generated | Text / ID | Unique identifier for the readiness record. |
| Project | Yes | Lookup / ID | Project linked to readiness activities. |
| Centre | Yes | Lookup / ID | Centre being prepared for the examination. |
| Exam Date | Yes | Date | Date of the examination. |
| Installation Date | Conditional | Date / Date-time | Planned or completed installation date. |
| Mock Date/Time | Conditional | Date / Date-time | Scheduled mock date and time. |
| Centre Availability | Yes | Dropdown / Confirmation | Available, Not Available, Conditional or Awaiting Confirmation. |
| Manpower Confirmation | Yes | Dropdown / Confirmation | Confirmed, Pending, Partially Confirmed or Not Available. |
| Installation Status | Yes | Dropdown | Not Started, Planned, In Progress, Completed, Failed or Not Required. |
| Mock Status | Yes | Dropdown | Not Planned, Planned, Ready, Completed, Failed, Rework or Not Required. |
| Final Readiness Status | Yes | Dropdown | Not Ready, In Progress, Ready with Exception, Ready, Not Ready – Escalated or Approved. |
| Exception | Conditional | Structured record | Readiness issue, risk, dependency or deviation. |
| Remarks | Conditional | Long text | Additional status notes and operational comments. |
| Confirmation Time | Conditional | Date/time | Time at which centre or internal user confirmed the status. |

**Recommended Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Allocation ID | Yes | Link to the locked final allocation. |
| Allocation Version | Yes | Version used for readiness planning. |
| Readiness Owner | Yes | Operations user responsible for readiness. |
| Project Manager | Yes | PM accountable for readiness approval. |
| Centre Contact | Yes | Authorized centre contact. |
| System Count Required | Yes | Systems required according to allocation. |
| System Count Ready | Conditional | Systems installed and ready. |
| UPS Status | Yes | Verified UPS readiness status. |
| DG Status | Yes | Verified DG and diesel readiness status. |
| Electricity Status | Yes | Power supply and electrical readiness. |
| Network Status | Yes | Internet and network readiness. |
| Technical Scan Status | Conditional | Latest IP scan and technical verification status. |
| Manpower Required | Yes | Required manpower count and categories. |
| Manpower Confirmed | Conditional | Confirmed manpower count. |
| Installation Owner | Conditional | Person responsible for installation. |
| Configuration Owner | Conditional | Person responsible for configuration. |
| Mock Owner | Conditional | Person responsible for mock activity. |
| Overall Readiness Percentage | System-calculated | Percentage of completed readiness checkpoints. |

| | | |
|---|---|---|
| Exception Severity | Conditional | Low, Medium, High or Critical. |
| Exception Owner | Conditional | Person responsible for resolution. |
| Exception Due Date | Conditional | Corrective action deadline. |
| Final Readiness Approved By | Conditional | Authorized approver. |
| Final Readiness Approval Time | Conditional | Approval timestamp. |

**Readiness Status Flow:**

| Status | Meaning | Next Status |
|---|---|---|
| Not Started | Readiness record has not been initiated. | Planned |
| Planned | Readiness activities and dates have been defined. | Centre Confirmation |
| Centre Confirmation | Centre availability and manpower are being confirmed. | Installation |
| Installation | Installation is pending or in progress. | Configuration |
| Configuration | Systems and technical settings are being configured. | Mock Readiness |
| Mock Readiness | Centre is being checked for mock activity. | Mock Planned, Rework or Escalated |
| Mock Planned | Mock date and requirements are confirmed. | Mock Completed, Failed or Rescheduled |
| Mock Completed | Mock has been conducted. | Final Readiness Review |
| Rework | Issues require correction before continuing. | Installation, Configuration, Mock or Review |
| Final Readiness Review | PM / Operations is reviewing all readiness conditions. | Ready, Ready with Exception or Not Ready |
| Ready | Centre has passed readiness checks. | Exam |
| Ready with Exception | Non-critical exceptions are accepted with approval. | Exam or Rework |
| Not Ready – Escalated | Critical blocker remains unresolved. | Rework, Ready or Cancelled |
| Exam | Centre has been released for exam execution. | Exam Completed |

**Readiness Checklist:**

| Readiness Area | Check | Status |
|---|---|---|
| Final allocation | Correct locked allocation is linked. | Pending / Complete |
| Centre availability | Centre confirms availability on exam date and shift. | Pending / Complete |
| Candidate capacity | Candidate count and buffer fit approved capacity. | Pending / Complete |
| Manpower | Required technical, security and operational staff confirmed. | Pending / Complete |
| Installation | Required equipment and systems installed. | Pending / Complete |
| Configuration | Software, network and exam settings configured. | Pending / Complete |
| Electricity | Main supply, dedicated circuit and earthing verified. | Pending / Complete |
| UPS | UPS availability, capacity and service status confirmed. | Pending / Complete |
| DG | DG availability, service status and diesel capacity confirmed. | Pending / Complete |
| Network | Connectivity, bandwidth and redundancy verified. | Pending / Complete |
| Technical scan | Latest IP scan completed and exceptions resolved. | Pending / Complete |
| Documents | Required documents remain valid. | Pending / Complete |
| Mock readiness | Centre is prepared for mock execution. | Pending / Complete |
| Mock result | Mock completed with no critical unresolved issue. | Pending / Complete |
| Final confirmation | PM / authorized user approves readiness. | Pending / Complete |

**Readiness Percentage:**

Readiness should be calculated from configured checkpoints rather than manually entered.

**Readiness Percentage = (Completed Readiness Checkpoints ÷ Total Applicable Checkpoints) × 100**

**Example:**

| Checkpoint Summary | Value |
|---|---|
| Applicable checkpoints | 20 |
| Completed checkpoints | 17 |
| Readiness percentage | 85% |
| Critical exceptions | 1 |
| Final status | Not Ready – Escalated |

A centre should not automatically be marked Ready only because the percentage is high. Any unresolved critical exception should block or escalate final readiness.

**Suggested Readiness Indicators:**

| Percentage / Condition | Display |
|---|---|
| 0–49% | Red – Not Ready |
| 50–79% | Amber – In Progress |
| 80–99% | Yellow / Amber – Near Ready |
| 100% with no critical exceptions | Green – Ready |
| Any critical open exception | Red – Blocked or Escalated |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts / Finance |
|---|---|---|---|---|---|---|
| Configure readiness checklist | Full | Review / request | Technical input | Suggest | No | Commercial input where relevant |
| Create readiness plan | Full | Approve / monitor | Technical input | Create / manage | Provide confirmation | No |
| Link final allocation | Full | Approve | View technical impact | Link / validate | View own centre | View if cost impact exists |
| Confirm centre availability | Full | Monitor | No | Coordinate | Confirm own centre | No |
| Confirm manpower | Full | Review | Technical manpower input | Coordinate | Submit confirmation | No |
| Plan installation | Full | Monitor | Technical responsibility | Coordinate | Support | No |
| Update installation status | Full | Review | Update technical status | Update operational status | Confirm centre status | No |
| Update configuration status | Full | Review | Full technical update | View / coordinate | Support | No |
| Schedule mock | Full | Approve / monitor | Technical participation | Coordinate | Confirm availability | No |
| Record mock result | Full | Review / approve | Technical findings | Create / coordinate | Provide response | No |
| Create | Full | Full | Technical | Operational | Report issue | Commercial |

| | | | | | | |
|---|---|---|---|---|---|---|
| exception | | | exception | exception | | exception |
| Assign exception owner | Full | Full | Technical owner | No | No | Finance owner |
| Resolve exception | Full | Approve | Technical resolution | Operational resolution | Provide correction | Financial resolution |
| Approve final readiness | Full | Full approval | Recommend only | Submit status | Confirm own centre | No |
| Release centre for exam | Full | Full approval | Technical clearance | Execute after approval | No | No |
| View readiness dashboard | Full | Full project | Audit / technical scope | Assigned scope | Own centre | Commercial status only |

**Role Responsibilities**

**Operations**

- Create readiness records from the locked final allocation.
- Plan centre contact, installation, configuration and mock activities.
- Coordinate with centre users and the Auditor / IT team.
- Update task and milestone status.
- Record readiness exceptions and assign follow-up actions.
- Monitor readiness percentage for assigned projects and centres.
- Escalate critical or overdue blockers to the Project Manager.
- Submit the centre for final readiness approval.

**Project Manager**

- Review and approve the readiness plan.
- Monitor project-wide readiness percentage and exceptions.
- Resolve resource, date and priority conflicts.
- Approve final readiness or authorized non-critical exceptions.
- Prevent exam release when critical blockers remain open.
- Monitor readiness across all centres in the project.

**Auditor / IT Team Member**

- Validate technical installation and configuration.
- Review UPS, DG, electricity, systems, network and IP scan status.
- Participate in mock readiness and technical mock activity.
- Create and resolve technical exceptions.
- Recommend technical readiness.
- Cannot approve a non-independent audit or override unrelated operational controls.

**Centre User**

- Confirm centre availability and manpower.
- Provide access for installation, configuration and mock activity.
- Confirm UPS, DG, diesel, electricity, network and facility readiness.
- Report operational issues.
- Support corrective actions and provide evidence.
- Confirm final centre readiness information.
- Cannot approve internal readiness on behalf of the organization.

**Accounts / Finance**

- View readiness status where commercial or payment activity depends on readiness.
- Review commercial impact of changes to installation, manpower, logistics or special requirements.
- No routine authority to update operational readiness status.

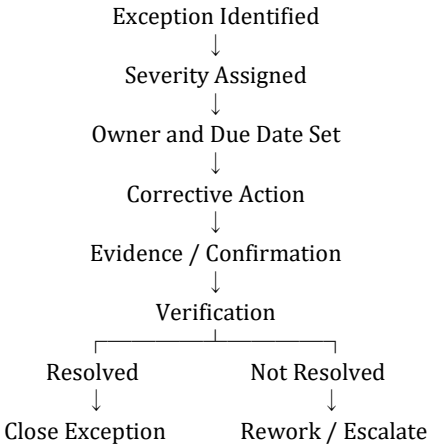
**Admin**

- Configure readiness checkpoints, statuses, escalation rules and permissions.
- View all readiness records.
- Maintain system-level controls and audit logs.

**Exception Management:**

| Exception Type | Example | Severity | Action |
|---|---|---|---|
| Centre unavailable | Centre cannot support exam date or shift. | Critical | Escalate and identify replacement. |
| Manpower pending | Required staff not confirmed. | High | Follow up and set due date. |
| Installation delayed | Systems or equipment not installed on schedule. | High | Assign owner and escalate. |
| Configuration failure | Software or settings not ready. | High | IT corrective action and re-test. |
| UPS issue | UPS unavailable or service overdue. | High | Correct or approve contingency. |
| DG issue | DG failure, low diesel or service overdue. | High | Correct and verify backup. |
| Network issue | Bandwidth or connectivity inadequate. | Critical | Technical remediation and re-test. |
| Mock failure | Mock activity failed or critical issue identified. | Critical | Rework and repeat mock. |
| Document expiry | Required document expires before exam. | Medium / High | Upload valid replacement. |
| Allocation mismatch | Centre cannot support assigned count. | Critical | Validate change and escalate. |
| Commercial dependency | Approved rate or special charge pending. | Medium | Refer to Finance. |

**Exception Flow:**




**Mock Management:**

| Mock Stage | Required Activity | Output |
|---|---|---|
| Mock planning | Confirm date, time, centre, manpower and technical participants. | Mock plan |
| Mock readiness | Confirm systems, network, power, allocation and staff are ready. | Mock readiness status |
| Mock execution | Run simulated examination activity. | Mock execution record |
| Mock | Record performance, gaps and incidents. | Observation list |

| | | |
|---|---|---|
| observation | | |
| Corrective action | Assign owners and due dates for issues. | Corrective-action tasks |
| Mock closure | Verify that critical findings are resolved. | Mock completion status |
| Final readiness | PM reviews mock result and remaining exceptions. | Ready, Ready with Exception or Not Ready |

**Notifications and Escalations:**

| Trigger | Recipient | Notification |
|---|---|---|
| Readiness plan created | Operations / Centre | Readiness activities and dates. |
| Centre confirmation pending | Centre User / Operations | Availability and manpower confirmation request. |
| Installation date approaching | Operations / IT | Installation reminder. |
| Installation overdue | Operations / PM | Escalation for delayed installation. |
| Configuration incomplete | Auditor / IT / Operations | Technical readiness reminder. |
| Mock date approaching | Operations / Centre / IT | Mock preparation reminder. |
| Mock failed | Operations / PM / IT | Critical exception alert. |
| Final readiness below threshold | PM | Readiness escalation. |
| Critical exception open | PM / Admin | Immediate blocker notification. |
| Final readiness approved | Operations / Centre | Centre released for exam. |
| Exam readiness not approved | Operations / PM | Exam-release block or escalation. |

**Readiness Dashboard:**

**Key KPIs**

| KPI | Description |
|---|---|
| Total centres | Centres included in the locked allocation. |
| Centres ready | Centres with approved final readiness. |
| Average readiness percentage | Average across applicable centres. |
| Centres below threshold | Centres requiring attention. |
| Installation pending | Centres where installation is not complete. |
| Configuration pending | Centres with technical configuration outstanding. |
| Mock pending | Centres awaiting mock activity. |
| Mock failed | Centres requiring corrective action or repeat mock. |
| Manpower pending | Centres without complete manpower confirmation. |
| Centre availability pending | Centres yet to confirm availability. |
| Critical exceptions | Open issues that may block the exam. |
| Overdue readiness tasks | Tasks past their due date. |
| Exam release status | Ready, Blocked, Escalated or Pending. |

**Dashboard Filters**

| Filter | Examples |
|---|---|
| Client | Client-wise readiness. |
| Project | Project-specific readiness. |
| Exam date | Upcoming examination date. |
| City | City-wise centre readiness. |
| Centre | Specific centre details. |
| Project Manager | Manager-wise status. |
| Operations Executive | Owner-wise readiness tasks. |

| | |
|---|---|
| Installation status | Pending, In Progress or Complete. |
| Mock status | Planned, Complete, Failed or Rework. |
| Final readiness status | Ready, Not Ready, Escalated or Approved. |
| Exception severity | Low, Medium, High or Critical. |
| Due date | Upcoming and overdue activities. |

**Key Business Controls**

| Control | Requirement |
|---|---|
| Locked allocation dependency | Readiness must use the approved and locked allocation version. |
| Centre confirmation | Centre availability and manpower must be confirmed before final readiness. |
| Technical readiness | Installation, configuration, network, power backup and IP scan requirements must be verified. |
| Mock dependency | Mock must be completed where required before final readiness approval. |
| Critical exception block | Critical open exceptions must block or escalate final readiness. |
| Readiness calculation | Percentage must be system-calculated from completed checkpoints. |
| Owner and due date | Every exception and incomplete activity must have an owner and due date. |
| Status history | Readiness status changes must retain user, timestamp and reason. |
| Final approval | Project Manager or authorized user approves final readiness. |
| Exam release | Only approved or authorized Ready with Exception centres may proceed to exam execution. |
| Change control | Changes to locked allocation, date, shift or candidate count must follow change-request workflow. |
| Audit trail | All confirmations, updates, exceptions, approvals and escalations must be recorded. |

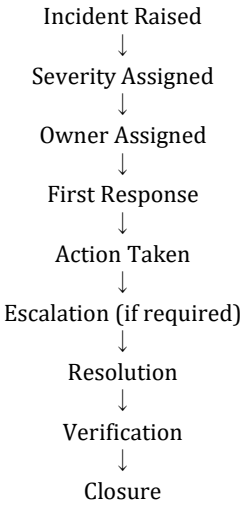
**System Outputs:**

| Output | Description |
|---|---|
| Readiness plan | Centre-wise tasks, dates, owners and dependencies. |
| Centre confirmation | Availability, manpower and operational confirmation. |
| Installation tracker | Installation plan, status, owner and completion evidence. |
| Configuration tracker | Technical configuration status and verification. |
| Mock plan | Mock date, participants, readiness status and result. |
| Exception register | Issue, severity, owner, due date, action and resolution. |
| Readiness percentage | System-calculated project and centre readiness. |
| Final readiness decision | Ready, Ready with Exception, Not Ready or Escalated. |
| Exam release status | Confirmation that the centre is permitted to proceed to exam. |
| Management dashboard | Real-time readiness, exception and overdue-task visibility. |
| MIS update | Readiness status reflected in project reporting and management MIS. |

# M16 – Exam-Day Incidents

| Category | Details |
|---|---|
| Module Name | Exam-Day Incident Management |
| Primary User | Operations |
| Supporting Users | Project Manager, Auditor / IT, Centre User and Admin |
| Purpose | Capture, track, resolve and close exam-day operational incidents with defined severity, ownership, first-response time, escalation and verification. |
| Process Flow | Incident Raised → Severity → Owner → First Response → Action → Escalation → Resolution → Verification → Closure |
| Mandatory / Key Fields | Incident ID; Project; Centre; Exam Date; Shift; Reporter; Raised Time; Category; Severity; Description; Owner; First Response; Action Taken; Resolution Time; Status; Evidence; Closure |
| Business Rule | Operational exam-day incidents must be managed separately from general operational tickets. |
| System Outputs | Incident record; severity classification; owner assignment; first-response timestamp; resolution record; escalation trail; verification; closure; incident dashboard and MIS. |

**Incident Flow:**




**Detailed Process Table:**

| Step | Stage | Responsible Role | Description | System Output |
|---|---|---|---|---|
| 1 | Incident Raised | Reporter | Incident is logged during the exam window. | Incident ID and timestamp |
| 2 | Severity | Reporter / Operations | Severity is assigned based on impact. | Severity classification |
| 3 | Owner | Operations / PM | Incident is assigned to an owner. | Owner and due time |
| 4 | First Response | Owner | Initial acknowledgment and assessment. | First-response timestamp |
| 5 | Action | Owner | Corrective action is executed. | Action record |
| 6 | Escalation | System / PM | Incident is escalated if unresolved or critical. | Escalation record |
| 7 | Resolution | Owner | Issue is resolved and documented. | Resolution timestamp |
| 8 | Verification | Operations / PM | Resolution is verified by an independent user. | Verification status |
| 9 | Closure | Authorized User | Incident is closed after confirmation. | Closed incident record |

**Incident Lifecycle:**

| Status | Meaning | Next Status |
|---|---|---|
| Raised | Incident has been logged. | Severity Assigned |
| Severity Assigned | Severity level has been set. | Owner Assigned |
| Owner Assigned | Incident owner has been designated. | First Response |
| First Response | Initial response has been recorded. | In Progress, Escalated or Resolved |
| In Progress | Action is underway. | Resolved, Escalated or On Hold |
| Escalated | Incident requires higher-level attention. | Resolved, In Progress or On Hold |
| Resolved | Action has been completed. | Verification |
| Verification | Resolution is being verified. | Closed or Reopened |
| Closed | Incident is verified and closed. | Historical record retained |
| Reopened | Issue has recurred or verification failed. | In Progress |
| On Hold | Incident is paused with reason. | In Progress or Closed |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Incident ID | System-generated | Text / ID | Unique identifier for the incident. |
| Project | Yes | Lookup / ID | Project linked to the incident. |
| Centre | Yes | Lookup / ID | Centre where the incident occurred. |
| Exam Date | Yes | Date | Date of the examination. |
| Shift | Yes | Dropdown | Shift during which the incident occurred. |
| Reporter | Yes | User / Contact | Person who raised the incident. |
| Raised Time | System-generated | Date/time | Timestamp when the incident was raised. |
| Category | Yes | Dropdown | Technical, Power, Network, System, Manpower, Security, Centre Facility, Candidate Issue, Logistics or Other. |
| Severity | Yes | Dropdown | Low, Medium, High or Critical. |
| Description | Yes | Long text | Clear description of the incident. |
| Owner | Yes | User Lookup | Person responsible for resolution. |
| First Response | Conditional | Long text / Timestamp | Initial acknowledgment and assessment. |
| Action Taken | Conditional | Long text | Actions performed to resolve the incident. |
| Resolution Time | Conditional | Date/time | Timestamp when resolution was recorded. |
| Status | Yes | Dropdown | Raised, In Progress, Escalated, Resolved, Verification, Closed, Reopened or On Hold. |
| Evidence | Conditional | File / Image / Text | Supporting screenshots, photos, logs or statements. |
| Closure | Conditional | Long text / Decision | Closure remarks and verification decision. |

**Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Incident Title | Yes | Short summary of the incident. |
| Impact | Yes | Candidate impact, exam disruption, time loss or other impact. |
| Affected Systems | Conditional | Number of systems or candidates affected. |
| Downtime Duration | Conditional | Estimated downtime in minutes. |
| Root Cause | Conditional | Identified cause of the incident. |

| | | |
|---|---|---|
| Preventive Action | Conditional | Action to prevent recurrence. |
| Escalation Level | System-generated | Current escalation stage. |
| Escalated To | Conditional | User or role to which the incident is escalated. |
| Escalation Time | Conditional | Timestamp of escalation. |
| Escalation Reason | Conditional | Reason for escalation. |
| Verified By | Conditional | User who verified the resolution. |
| Verification Time | Conditional | Verification timestamp. |
| Closed By | Conditional | User who closed the incident. |
| Closed Time | Conditional | Closure timestamp. |
| Linked Ticket | Conditional | Link to a general ticket if applicable. |
| Linked Audit / Technical Issue | Conditional | Link to audit or technical exception where relevant. |

**Severity Matrix:**

| Severity | Typical Impact | First-Response SLA | Escalation Trigger |
|---|---|---|---|
| Low | Minor issue with no candidate impact. | Within 15–30 minutes. | If not resolved within configured time. |
| Medium | Limited candidate impact or temporary disruption. | Within 10–15 minutes. | If resolution is delayed or impact increases. |
| High | Significant candidate impact or exam disruption. | Within 5–10 minutes. | Immediate escalation to Project Manager. |
| Critical | Exam halt, large-scale system failure or security incident. | Immediate response. | Immediate escalation to senior management. |

**Incident Category Examples:**

| Category | Example Incidents |
|---|---|
| Technical | System freeze, software error, login failure or configuration issue. |
| Power | Power failure, UPS failure, DG failure or voltage fluctuation. |
| Network | Internet outage, slow connectivity, DNS failure or router issue. |
| System | Hardware failure, peripheral issue or OS crash. |
| Manpower | Staff shortage, late arrival or role confusion. |
| Security | Unauthorized access, candidate misconduct or breach of protocol. |
| Centre Facility | Room issue, seating problem, lighting or ventilation failure. |
| Candidate Issue | Candidate complaint, identity issue or seating dispute. |
| Logistics | Material not received, courier delay or inventory mismatch. |
| Other | Any incident not covered by standard categories. |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts / Finance |
|---|---|---|---|---|---|---|
| Raise incident | Full | Full | Technical incidents | Full | Own centre incident | Finance-related incident |
| Assign severity | Full | Full | Technical severity | Full | Suggest severity | No |
| Assign owner | Full | Full | Technical owner | No, unless delegated | No | No |
| Provide first | Full | Monitor | Technical | Full | Provide | No |

| | | | | | | |
|---|---|---|---|---|---|---|
| response | | | response | | information | |
| Execute action | Full | Monitor | Technical action | Full | Support | No |
| Escalate incident | Full | Full | Technical escalation | Request / escalate | Report urgent issue | Finance escalation |
| Resolve incident | Full | Approve / verify | Technical resolution | Operational resolution | Confirm resolution | No |
| Verify resolution | Full | Full | Technical verification | Operational verification | Confirm centre impact | No |
| Close incident | Full | Full approval | Recommend closure | Submit closure | Confirm closure | No |
| View incident history | Full | Full project | Audit / technical scope | Assigned scope | Own centre | Finance-related view |
| Export incident report | Full | Assigned project | Audit scope | Assigned scope | No | No |

**Role Responsibilities**

**Operations**

- Monitor and manage exam-day incidents.
- Assign severity and owner promptly.
- Ensure first response within the configured SLA.
- Coordinate corrective action with Centre User and Auditor / IT.
- Escalate unresolved or critical incidents.
- Verify resolution and close incidents.
- Maintain incident history and evidence.

**Project Manager**

- Monitor critical and escalated incidents.
- Approve high-severity or critical incident closures.
- Allocate additional resources where required.
- Review incident patterns and recurring issues.
- Ensure incidents are reflected in project MIS and closure review.

**Auditor / IT Team Member**

- Handle technical, network, system and infrastructure incidents.
- Provide root-cause analysis and technical corrective action.
- Participate in verification of technical resolutions.
- Cannot independently close incidents without operational verification.

**Centre User**

- Report incidents promptly.
- Provide on-ground information and support.
- Confirm impact and resolution from the centre perspective.
- Assist in evidence collection and corrective action.
- Cannot assign owner, severity or close incidents independently.

**Accounts / Finance**

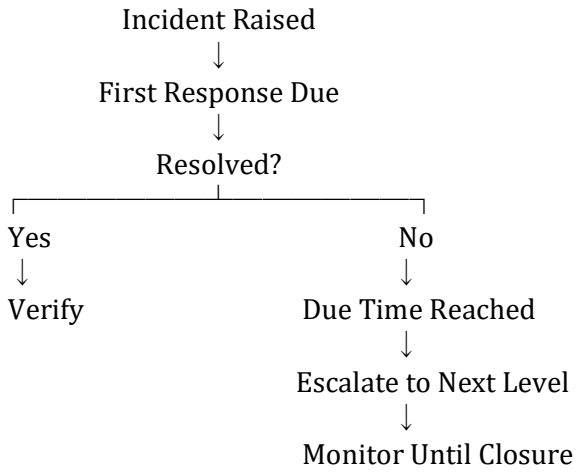
- Handle finance-related incidents such as payment disputes or invoice issues during exam operations.
- No routine authority over operational exam-day incidents.

**Admin**

- Configure incident categories, severities, SLAs, escalation rules and permissions.

- View all incidents and administrative actions.
- Maintain audit logs and system controls.

**Escalation Flow:**




**Escalation Levels:**

| Level | Trigger | Escalated To |
|---|---|---|
| Level 0 | Incident raised. | Assigned Operations owner. |
| Level 1 | First response or resolution SLA missed. | Senior Operations user or Team Lead. |
| Level 2 | High or Critical severity unresolved. | Project Manager. |
| Level 3 | Critical incident or repeated failure. | Senior Management or Admin. |

**Incident vs Ticket:**

| Aspect | Incident (M16) | Ticket (M17) |
|---|---|---|
| Scope | Exam-day operational events. | General operational issues and requests. |
| Timing | Primarily during exam window. | Any time during project lifecycle. |
| Severity | Low, Medium, High, Critical. | Low, Medium, High or configured priority. |
| SLA | First response and resolution time critical. | Resolution SLA based on priority. |
| Owner | Operations, with IT / Centre support. | Operations, IT, Admin or other functional owner. |
| Closure | Requires verification and exam impact assessment. | Requires resolution and requester confirmation. |
| Reporting | Exam-day dashboard and incident MIS. | Operational ticket dashboard and MIS. |

**Notifications and Alerts:**

| Trigger | Recipient | Notification |
|---|---|---|
| Incident raised | Operations / Centre / PM | Incident alert with severity and category. |
| Owner assigned | Assigned owner | Assignment notification with SLA. |
| First response due | Owner | Reminder before SLA deadline. |
| First response missed | Owner / PM | Escalation alert. |
| Incident escalated | Escalated user | Escalation notification with reason. |
| Resolution recorded | Verifier | Verification request. |
| Verification pending | Verifier / PM | Reminder to verify resolution. |
| Incident closed | Reporter / Centre | Closure notification. |
| Critical incident raised | PM / Admin | Immediate critical-incident alert. |
| Repeated incidents at centre | PM / Operations | Pattern alert for recurring issues. |

**Incident Dashboard:**

**Key KPIs**

| KPI | Description |
|---|---|
| Total incidents | All incidents raised in the selected period. |
| Open incidents | Incidents not yet closed. |
| Resolved incidents | Incidents marked resolved but pending verification. |
| Closed incidents | Verified and closed incidents. |
| Critical incidents | Incidents with Critical severity. |
| High-severity incidents | Incidents with High severity. |
| Average first-response time | Time from raised to first response. |
| Average resolution time | Time from raised to resolution. |
| SLA breaches | Incidents where response or resolution SLA was missed. |
| Escalated incidents | Incidents escalated to higher levels. |
| Incidents by category | Distribution across technical, power, network and other categories. |
| Incidents by centre | Centre-wise incident count. |
| Recurring incidents | Repeated issues at the same centre or project. |
| Verification pending | Resolved incidents awaiting verification. |

**Dashboard Filters:**

| Filter | Examples |
|---|---|
| Project | Project-specific incidents. |
| Centre | Centre-wise incident history. |
| Exam date | Incident date range. |
| Shift | Shift-specific incidents. |
| Category | Technical, power, network, manpower or other. |
| Severity | Low, Medium, High or Critical. |
| Status | Open, In Progress, Escalated, Resolved, Verification or Closed. |
| Owner | Incident owner or responsible team. |
| Escalation level | Level 0, 1, 2 or 3. |
| Reporter | Person who raised the incident. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| Exam-day focus | Incidents are primarily raised during the exam window. |
| Severity control | Severity must be assigned and cannot be left blank. |
| Owner assignment | Every incident must have an assigned owner. |
| First-response tracking | First-response timestamp is mandatory for resolution tracking. |
| SLA monitoring | Response and resolution SLAs are automatically tracked. |
| Escalation automation | Unresolved incidents escalate based on severity and time. |
| Verification requirement | Resolution must be verified before closure. |
| Evidence retention | Supporting evidence must be attached where applicable. |
| No silent deletion | Closed incidents remain in history. |
| Reopen control | Reopened incidents must have a reason. |
| Access control | Users can view and update only incidents within their scope. |
| Audit trail | Creation, assignment, response, action, escalation, resolution, verification and closure must be logged. |
| Separation from tickets | Incidents are managed separately from general tickets. |

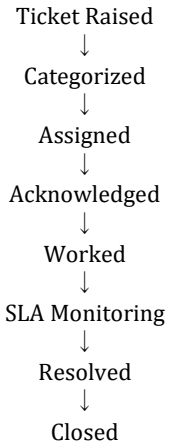
**System Outputs:**

| Output | Description |
|---|---|
| Incident record | Exam-day operational incident with full details. |
| Severity classification | Low, Medium, High or Critical. |
| Owner assignment | Assigned user and SLA. |
| First-response record | Timestamp and initial assessment. |
| Action log | Corrective actions and updates. |
| Escalation trail | Escalation level, reason and timestamps. |
| Resolution record | Resolution details and timestamp. |
| Verification status | Verified or pending verification. |
| Closure record | Closed incident with remarks. |
| Incident dashboard | Real-time incident monitoring for management. |
| MIS update | Incident metrics reflected in project and management reporting. |

# M17 – Ticket Management

| Category | Details |
|---|---|
| Module Name | Ticket Management |
| Primary Users | Operations / Manager |
| Supporting Users | Admin, Auditor / IT, Centre User and Accounts / Finance |
| Purpose | Manage general operational issues and requests across the project lifecycle with defined category, priority, SLA, ownership, response and resolution tracking. |
| Process Flow | Raised → Categorized → Assigned → Acknowledged → Worked → SLA Monitoring → Resolved → Closed |
| Mandatory / Key Fields | Ticket ID; Source; Project; Centre; Raised By; Date/Time; Category; Priority; SLA; Description; Owner; Response Time; Resolution Time; Status; Resolution; Closure Time |
| Business Rule | General operational tickets must be managed separately from exam-day incidents and must include SLA tracking. |
| System Outputs | Ticket record; category and priority; owner assignment; acknowledgment; resolution record; SLA compliance; closure; ticket dashboard and MIS. |

**Ticket Flow:**




**Detailed Process Table:**

| Step | Stage | Responsible Role | Description | System Output |
|---|---|---|---|---|
| 1 | Raised | Reporter | Issue or request is logged. | Ticket ID and timestamp |
| 2 | Categorized | Operations / Manager | Ticket is classified by category and priority. | Category and SLA |
| 3 | Assigned | Manager / Operations | Ticket is assigned to an owner. | Owner and due time |
| 4 | Acknowledged | Owner | Owner acknowledges the ticket. | Acknowledgment timestamp |
| 5 | Worked | Owner | Work is performed to resolve the ticket. | Work log and updates |
| 6 | SLA Monitoring | System | System tracks response and resolution SLA. | SLA status and alerts |
| 7 | Resolved | Owner | Ticket is marked resolved with resolution details. | Resolution timestamp |
| 8 | Closed | Authorized User | Ticket is verified and closed. | Closure timestamp |

**Ticket Lifecycle:**

| Status | Meaning | Next Status |
|---|---|---|
| **New** | Ticket has been raised but not yet assigned. | Assigned or Rejected |
| **Assigned** | Ticket owner has been designated. | Acknowledged |
| **Acknowledged** | Owner has acknowledged the ticket. | In Progress or On Hold |
| **In Progress** | Work is underway. | Resolved, On Hold or Reassigned |
| **On Hold** | Ticket is paused with reason. | In Progress or Closed |
| **Resolved** | Resolution has been recorded. | Closed or Reopened |
| **Closed** | Ticket is verified and closed. | Historical record retained |
| **Reopened** | Issue has recurred or resolution is inadequate. | In Progress |
| **Rejected** | Ticket is invalid or duplicate. | Historical record retained |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Ticket ID | System-generated | Text / ID | Unique identifier for the ticket. |
| Source | Yes | Dropdown | Portal, email, phone, centre user, operations, audit, finance or other configured source. |
| Project | Conditional | Lookup / ID | Project linked to the ticket. |
| Centre | Conditional | Lookup / ID | Centre linked to the ticket. |
| Raised By | Yes | User / Contact | Person who raised the ticket. |
| Date/Time | System-generated | Date/time | Timestamp when the ticket was raised. |
| Category | Yes | Dropdown | Operational, technical, document, allocation, readiness, commercial, finance, logistics, access or other. |
| Priority | Yes | Dropdown | Low, Medium, High or Critical. |
| SLA | System-generated | Time / Date | Configured response and resolution time based on priority and category. |
| Description | Yes | Long text | Clear description of the issue or request. |
| Owner | Yes | User Lookup | Person responsible for resolution. |
| Response Time | Conditional | Date/time | Timestamp of first response or acknowledgment. |
| Resolution Time | Conditional | Date/time | Timestamp when resolution was recorded. |
| Status | Yes | Dropdown | New, Assigned, Acknowledged, In Progress, On Hold, Resolved, Closed, Reopened or Rejected. |
| Resolution | Conditional | Long text | Description of the resolution provided. |
| Closure Time | Conditional | Date/time | Timestamp when the ticket was closed. |

**Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Ticket Title | Yes | Short summary of the ticket. |
| Impact | Conditional | Operational impact description. |
| Expected Resolution Date | Conditional | Expected resolution date. |
| Actual Resolution Date | Conditional | Actual resolution date. |
| SLA Breach | System-generated | Yes / No indicator for SLA breach. |

| | | |
|---|---|---|
| Reassignment Count | System-generated | Number of times the ticket was reassigned. |
| Last Updated By | System-generated | User who last updated the ticket. |
| Last Updated Time | System-generated | Last update timestamp. |
| Linked Records | Conditional | Link to booking, allocation, document, audit, invoice or other records. |
| Attachments | Conditional | Supporting files, screenshots or documents. |
| Internal Notes | Conditional | Internal comments not visible to all users. |
| Requester Comments | Conditional | Additional comments from the ticket requester. |
| Closed By | Conditional | User who closed the ticket. |
| Reopen Reason | Conditional | Reason for reopening a closed ticket. |

**Priority and SLA Matrix:**

| Priority | Typical Impact | Response SLA | Resolution SLA |
|---|---|---|---|
| **Low** | Minor issue with limited operational impact. | Within 4–8 hours. | Within 24–48 hours. |
| **Medium** | Moderate impact on operations or timeline. | Within 2–4 hours. | Within 12–24 hours. |
| **High** | Significant operational impact or blocker. | Within 1–2 hours. | Within 6–12 hours. |
| **Critical** | Critical blocker affecting project delivery. | Within 30–60 minutes. | Within 4–6 hours. |

**Ticket Category Examples:**

| Category | Example Tickets |
|---|---|
| Operational | Centre coordination, scheduling, follow-up or process clarification. |
| Technical | System configuration, network issue, software error or device problem. |
| Document | Document upload, verification, rework or approval request. |
| Allocation | Allocation change, candidate adjustment or centre reassignment. |
| Readiness | Installation, configuration, manpower or mock-related request. |
| Commercial | Rate clarification, cost sheet revision or commercial approval. |
| Finance | Invoice, payment, reconciliation or financial documentation. |
| Logistics | Material dispatch, courier tracking or inventory issue. |
| Access | User access, permission or role-related request. |
| Other | Any ticket not covered by standard categories. |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts / Finance |
|---|---|---|---|---|---|---|
| Raise ticket | Full | Full | Technical tickets | Full | Own centre ticket | Finance-related ticket |
| Categorize ticket | Full | Full | Technical category | Full | Suggest category | Finance category |
| Assign ticket | Full | Full | Technical tickets | No, unless delegated | No | Finance tickets |
| Acknowledge ticket | Full | Monitor | Assigned technical tickets | Assigned tickets | Own ticket | Finance tickets |
| Work ticket | Full | Monitor | Technical tickets | Assigned operational tickets | Provide information | Finance tickets |
| Monitor SLA | Full | Full team scope | Technical scope | Assigned scope | Own ticket | Finance scope |

| | | | | | | |
|---|---|---|---|---|---|---|
| Resolve ticket | Full | Approve / verify | Technical resolution | Operational resolution | Confirm resolution | Finance resolution |
| Close ticket | Full | Full approval | Recommend closure | Submit closure | Confirm closure | Finance closure |
| View ticket history | Full | Full project | Audit / technical scope | Assigned scope | Own centre | Finance-related view |
| Export ticket report | Full | Assigned project | Audit scope | Assigned scope | No | Finance scope |

**Role Responsibilities**

**Operations**

- Monitor and manage operational tickets.
- Assign category, priority and owner.
- Ensure acknowledgment and response within SLA.
- Coordinate resolution with relevant stakeholders.
- Update ticket status and resolution details.
- Close tickets after verification.

**Project Manager**

- Monitor high-priority and escalated tickets.
- Approve critical ticket closures.
- Allocate resources for ticket resolution.
- Review ticket patterns and recurring issues.
- Ensure tickets are reflected in project MIS.

**Auditor / IT Team Member**

- Handle technical and infrastructure-related tickets.
- Provide root-cause analysis and technical resolution.
- Participate in verification of technical resolutions.
- Cannot independently close tickets without operational verification.

**Centre User**

- Raise tickets related to centre operations.
- Provide information and support for resolution.
- Confirm resolution from the centre perspective.
- View only tickets related to the centre's own records.

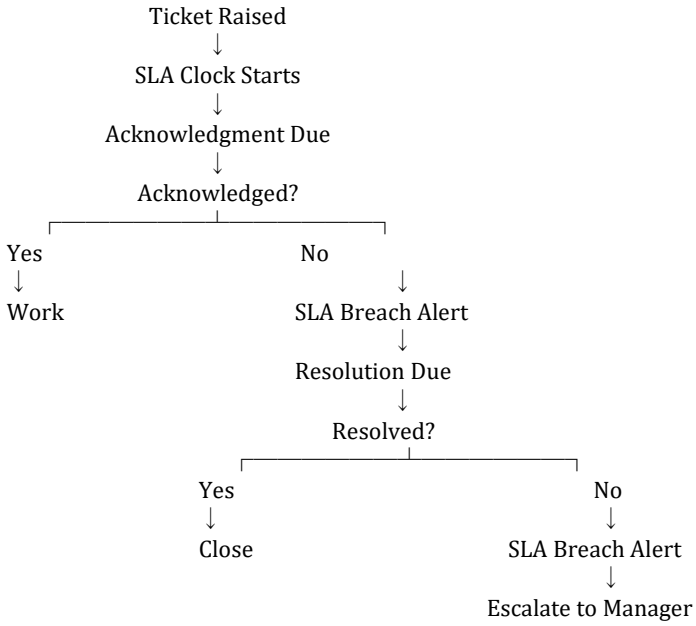
**Accounts / Finance**

- Handle finance-related tickets such as invoice, payment or reconciliation issues.
- Resolve financial queries and documentation requests.
- Close finance tickets after verification.

**Admin**

- Configure ticket categories, priorities, SLAs and permissions.
- View all tickets and administrative actions.
- Maintain audit logs and system controls.

**SLA Monitoring Flow:**




**SLA Breach Rules:**

| Condition | System Action |
|---|---|
| Acknowledgment SLA missed | Alert owner and notify manager. |
| Resolution SLA approaching | Send reminder to owner. |
| Resolution SLA missed | Mark SLA breach and escalate. |
| Critical ticket SLA missed | Immediate escalation to senior management. |
| Repeated SLA breaches | Pattern alert for process improvement. |

**Notifications and Alerts:**

| Trigger | Recipient | Notification |
|---|---|---|
| Ticket raised | Owner / Operations | Ticket assignment alert. |
| Acknowledgment due | Owner | Reminder before SLA deadline. |
| Acknowledgment missed | Owner / Manager | Escalation alert. |
| Resolution due | Owner | Reminder before resolution SLA. |
| Resolution missed | Owner / Manager | SLA breach alert. |
| Ticket escalated | Escalated user | Escalation notification with reason. |
| Ticket resolved | Requester / Manager | Resolution notification. |
| Ticket closed | Requester | Closure notification. |
| Critical ticket raised | Manager / Admin | Immediate critical-ticket alert. |
| Repeated tickets at centre | Manager / Operations | Pattern alert for recurring issues. |

**Ticket Dashboard:**

**Key KPIs**

| KPI | Description |
|---|---|
| Total tickets | All tickets raised in the selected period. |
| Open tickets | Tickets not yet closed. |
| Resolved tickets | Tickets marked resolved but pending closure. |
| Closed tickets | Verified and closed tickets. |
| Critical tickets | Tickets with Critical priority. |
| High-priority tickets | Tickets with High priority. |
| Average response time | Time from raised to acknowledgment. |

| | |
|---|---|
| Average resolution time | Time from raised to resolution. |
| SLA breaches | Tickets where response or resolution SLA was missed. |
| Escalated tickets | Tickets escalated to higher levels. |
| Tickets by category | Distribution across operational, technical, document and other categories. |
| Tickets by centre | Centre-wise ticket count. |
| Recurring tickets | Repeated issues at the same centre or project. |
| Closure pending | Resolved tickets awaiting closure. |

**Dashboard Filters:**

| Filter | Examples |
|---|---|
| Project | Project-specific tickets. |
| Centre | Centre-wise ticket history. |
| Category | Operational, technical, document, finance or other. |
| Priority | Low, Medium, High or Critical. |
| Status | New, Assigned, In Progress, Resolved or Closed. |
| Owner | Ticket owner or responsible team. |
| Source | Portal, email, phone or other source. |
| Raised By | Person who raised the ticket. |
| SLA Status | Within SLA, Breached or Pending. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| **Category control** | Category must be assigned and cannot be left blank. |
| **Priority control** | Priority must be assigned to determine SLA. |
| **Owner assignment** | Every ticket must have an assigned owner. |
| **SLA tracking** | Response and resolution SLAs are automatically tracked. |
| **Acknowledgment requirement** | Owner must acknowledge the ticket within SLA. |
| **Resolution documentation** | Resolution details must be recorded before closure. |
| **No silent deletion** | Closed tickets remain in history. |
| **Reopen control** | Reopened tickets must have a reason. |
| **Access control** | Users can view and update only tickets within their scope. |
| **Audit trail** | Creation, assignment, acknowledgment, resolution and closure must be logged. |
| **Separation from incidents** | General tickets are managed separately from exam-day incidents. |

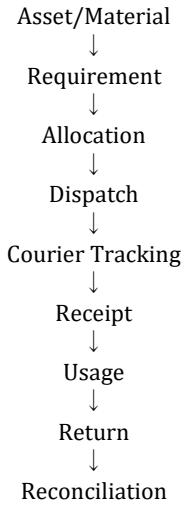
**System Outputs:**

| Output | Description |
|---|---|
| Ticket record | General operational ticket with full details. |
| Category and priority | Classification for SLA determination. |
| Owner assignment | Assigned user and SLA. |
| Acknowledgment record | Timestamp of first response. |
| Resolution record | Resolution details and timestamp. |
| SLA compliance | SLA status and breach indicators. |
| Closure record | Closed ticket with remarks. |
| Ticket dashboard | Real-time ticket monitoring for management. |
| MIS update | Ticket metrics reflected in project and management reporting. |

# M18 – Inventory & Logistics

| Category | Details |
|---|---|
| Module Name | Inventory & Logistics |
| Primary User | Operations |
| Supporting Users | Project Manager, Centre User, Admin and Accounts / Finance |
| Purpose | Track project assets and materials from requirement through allocation, dispatch, courier tracking, receipt, usage, return and final reconciliation. |
| Process Flow | Asset/Material → Requirement → Allocation → Dispatch → Courier Tracking → Receipt → Usage → Return → Reconciliation |
| Mandatory / Key Fields | Asset ID; Item Type; Serial/Barcode; Quantity; Project; Centre; Issued By; Issue Date; Courier; Tracking Number; Dispatch Date; Receipt Date; Return Date; Condition; Status |
| Business Rule | All project materials must be tracked end-to-end with complete reconciliation between issued, received, used and returned items. |
| System Outputs | Asset record; material requirement; allocation log; dispatch record; courier tracking; receipt confirmation; usage log; return record; reconciliation report; inventory dashboard. |

**Inventory Flow:**




**Detailed Process Table:**

| Step | Stage | Responsible Role | Description | System Output |
|---|---|---|---|---|
| 1 | Asset/Material | Operations | Define asset types and material catalog. | Asset master record |
| 2 | Requirement | Operations | Identify material needs for project or centre. | Requirement record |
| 3 | Allocation | Operations | Assign materials to project or centre. | Allocation record |
| 4 | Dispatch | Operations | Prepare and dispatch materials. | Dispatch record |
| 5 | Courier Tracking | Operations | Track shipment in transit. | Tracking updates |
| 6 | Receipt | Centre / Operations | Confirm material receipt at destination. | Receipt confirmation |
| 7 | Usage | Centre / Operations | Record material deployment and usage. | Usage log |
| 8 | Return | Centre / Operations | Initiate and track material return. | Return record |
| 9 | Reconciliation | Operations / Accounts | Match issued, received and returned items. | Reconciliation report |

**Asset Lifecycle:**

| Status | Meaning | Next Status |
|---|---|---|
| **Available** | Asset is in inventory and available for allocation. | Allocated or Under Maintenance |
| **Allocated** | Asset has been assigned to a project or centre. | Dispatched or Returned to Inventory |
| **Dispatched** | Asset is in transit to the destination. | In Transit or Delivered |
| **In Transit** | Asset is being tracked during shipment. | Received or Exception |
| **Received** | Asset has been received at the destination. | In Use or Stored |
| **In Use** | Asset is actively being used at the centre. | Returned or Damaged |
| **Returned** | Asset has been returned from the centre. | Reconciled or Under Maintenance |
| **Reconciled** | Asset has been verified and matched with records. | Available or Disposed |
| **Under Maintenance** | Asset requires repair or servicing. | Available or Disposed |
| **Damaged** | Asset is damaged and unusable. | Disposed or Replaced |
| **Lost** | Asset cannot be located. | Investigation or Disposed |
| **Disposed** | Asset has been removed from inventory. | Historical record retained |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Asset ID | System-generated | Text / ID | Unique identifier for the asset or material. |
| Item Type | Yes | Dropdown | System, peripheral, network device, UPS, DG, furniture, consumable or other configured type. |
| Serial/Barcode | Conditional | Text | Unique serial number or barcode identifier. |
| Quantity | Yes | Number | Number of items in the record. |
| Project | Conditional | Lookup / ID | Project linked to the asset. |
| Centre | Conditional | Lookup / ID | Centre where the asset is deployed. |
| Issued By | Yes | User Lookup | Person who issued the asset. |
| Issue Date | Yes | Date | Date when the asset was issued. |
| Courier | Conditional | Text | Courier or logistics provider name. |
| Tracking Number | Conditional | Text | Shipment tracking number. |
| Dispatch Date | Conditional | Date | Date when the asset was dispatched. |
| Receipt Date | Conditional | Date | Date when the asset was received. |
| Return Date | Conditional | Date | Date when the asset was returned. |
| Condition | Yes | Dropdown | New, Good, Fair, Damaged, Under Maintenance or Lost. |
| Status | Yes | Dropdown | Available, Allocated, Dispatched, In Transit, Received, In Use, Returned, Reconciled, Under Maintenance, Damaged, Lost or Disposed. |

**Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Asset Name | Yes | Descriptive name of the asset. |
| Category | Yes | Hardware, software, furniture, consumable or other. |
| Specification | Conditional | Technical or physical specifications. |

| | | |
|---|---|---|
| Vendor | Conditional | Supplier or manufacturer details. |
| Purchase Date | Conditional | Date of asset purchase. |
| Warranty Expiry | Conditional | Warranty expiration date. |
| Location | Conditional | Current physical location of the asset. |
| Assigned User | Conditional | Person currently responsible for the asset. |
| Expected Return Date | Conditional | Expected date for asset return. |
| Actual Return Date | Conditional | Actual date of asset return. |
| Dispatch Address | Conditional | Shipping destination address. |
| Received By | Conditional | Person who received the asset. |
| Returned By | Conditional | Person who returned the asset. |
| Reconciliation Status | Yes | Matched, Mismatch, Pending or Exception. |
| Remarks | Conditional | Additional notes about the asset. |
| Attachments | Conditional | Photos, invoices or supporting documents. |
| Depreciation Value | Conditional | Current asset value for accounting. |
| Replacement Asset ID | Conditional | Link to replacement asset if applicable. |

**Inventory Transaction Types:**

| Transaction Type | Description | Example |
|---|---|---|
| **Requirement** | Material need identified for project or centre. | 50 systems required for new project. |
| **Allocation** | Assets assigned from inventory to project. | 50 systems allocated to Project A. |
| **Dispatch** | Assets shipped to destination. | Systems dispatched via courier. |
| **Transit** | Assets in transit with tracking updates. | Package location updates. |
| **Receipt** | Assets received and verified at destination. | Centre confirms 50 systems received. |
| **Usage** | Assets deployed and in active use. | Systems installed at exam centre. |
| **Return** | Assets returned from centre to inventory. | Systems returned after exam completion. |
| **Reconciliation** | Issued, received and returned items matched. | All 50 systems accounted for. |
| **Maintenance** | Asset sent for repair or servicing. | Damaged system under repair. |
| **Disposal** | Asset removed from inventory. | Obsolete system disposed. |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts / Finance |
|---|---|---|---|---|---|---|
| Create asset master | Full | Review | Technical input | Suggest | No | No |
| Raise requirement | Full | Approve | Technical requirement | Create | Request | Commercial impact view |
| Allocate asset | Full | Approve | Technical allocation | Create | View | No |
| Dispatch asset | Full | Monitor | Technical dispatch | Create | No | No |
| Update tracking | Full | View | Technical tracking | Update | View | No |
| Confirm receipt | Full | View | Technical verification | Coordinate | Confirm | No |
| Record usage | Full | View | Technical usage | Update | Confirm | No |
| Initiate return | Full | Monitor | Technical return | Create | Request | No |
| Perform reconciliation | Full | Review | Technical verification | Execute | Support | Financial reconciliation |
| View inventory status | Full | Project scope | Technical scope | Assigned scope | Own centre | Financial view |
| Approve disposal | Full | Approve | Technical recommendation | Recommend | No | Financial approval |
| Export | Full | Assigned | Audit scope | Assigned scope | No | Financial scope |

| | | | | | | |
|---|---|---|---|---|---|---|
| inventory report | | project | | | | |

**Role Responsibilities**

**Operations**
- Maintain asset master and inventory records.
- Identify material requirements for projects and centres.
- Allocate and dispatch assets according to project needs.
- Track courier shipments and delivery status.
- Coordinate receipt confirmation with centres.
- Monitor asset usage and deployment.
- Manage asset returns and reconciliation.
- Handle damaged, lost or maintenance items.

**Project Manager**
- Approve material requirements and allocations.
- Monitor inventory status across projects.
- Review reconciliation reports and exceptions.
- Approve asset disposal or write-offs.
- Ensure inventory costs are reflected in project budgets.

**Auditor / IT Team Member**
- Verify technical specifications and asset conditions.
- Participate in technical reconciliation.
- Recommend maintenance or replacement actions.
- Validate IT asset deployment and returns.

**Centre User**
- Request materials needed for centre operations.
- Confirm receipt of dispatched materials.
- Record asset usage and condition at the centre.
- Initiate return requests for unused or excess materials.
- Report damaged or lost items promptly.

**Accounts / Finance**
- Review inventory value and depreciation.
- Participate in financial reconciliation.
- Approve disposal and write-off transactions.
- Track inventory costs and recoveries.

**Admin**
- Configure asset types, categories, statuses and workflows.
- View all inventory records and transactions.
- Maintain system controls and audit logs.

**Courier Tracking Flow:**

Dispatch Initiated
↓
Courier Assigned
↓
Tracking Number Generated
↓
Shipment In Transit
↓

Location Updates
↓
Out for Delivery
↓
Delivered
↓
Receipt Confirmed

**Tracking Status Options:**

| Status | Description |
|---|---|
| **Dispatched** | Package has left the warehouse. |
| **In Transit** | Package is moving through the courier network. |
| **Reached Hub** | Package has reached a sorting facility. |
| **Out for Delivery** | Package is on the final delivery vehicle. |
| **Delivered** | Package has been delivered to the recipient. |
| **Exception** | Delivery issue such as address problem or delay. |
| **Returned** | Package is being returned to sender. |

**Reconciliation Process:**

| Reconciliation Step | Action | Output |
|---|---|---|
| **Issue Verification** | Match issued items with allocation records. | Issue confirmation |
| **Receipt Verification** | Match received items with dispatch records. | Receipt confirmation |
| **Usage Verification** | Confirm items deployed at centres. | Usage log |
| **Return Verification** | Match returned items with centre records. | Return confirmation |
| **Condition Check** | Verify asset condition upon return. | Condition report |
| **Exception Handling** | Identify and investigate mismatches. | Exception report |
| **Final Reconciliation** | Complete matching and closure. | Reconciliation certificate |

**Notifications and Alerts:**

| Trigger | Recipient | Notification |
|---|---|---|
| Asset allocated | Operations / Centre | Allocation confirmation. |
| Asset dispatched | Operations / Centre | Dispatch notification with tracking. |
| Tracking update | Operations | Shipment status update. |
| Delivery exception | Operations / Manager | Delivery issue alert. |
| Receipt pending | Centre / Operations | Receipt confirmation reminder. |
| Return due | Centre / Operations | Return deadline reminder. |
| Return overdue | Centre / Manager | Overdue return escalation. |
| Reconciliation pending | Operations | Reconciliation deadline alert. |
| Asset damaged | Operations / Manager | Damage report notification. |
| Asset lost | Operations / Manager | Loss investigation alert. |
| Low inventory | Operations / Manager | Restock requirement alert. |

**Inventory Dashboard:**

**Key KPIs**

| KPI | Description |
|---|---|
| Total assets | Total items in inventory. |
| Available assets | Items ready for allocation. |
| Allocated assets | Items assigned to projects or centres. |
| In transit | Items currently being shipped. |
| Received assets | Items delivered and confirmed. |
| In use | Items actively deployed at centres. |

| | |
|---|---|
| Returned assets | Items returned from centres. |
| Reconciled assets | Items verified and matched. |
| Under maintenance | Items being repaired or serviced. |
| Damaged assets | Items reported as damaged. |
| Lost assets | Items cannot be located. |
| Pending returns | Items due for return. |
| Overdue returns | Items past return deadline. |
| Reconciliation pending | Items awaiting reconciliation. |
| Inventory value | Total monetary value of assets. |

**Dashboard Filters:**

| Filter | Examples |
|---|---|
| Project | Project-specific inventory. |
| Centre | Centre-wise asset deployment. |
| Item Type | Systems, peripherals, furniture or consumables. |
| Status | Available, allocated, dispatched or returned. |
| Condition | New, good, damaged or under maintenance. |
| Date Range | Issue, dispatch or return date range. |
| Courier | Logistics provider filter. |
| Reconciliation Status | Matched, mismatch or pending. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| **Asset identification** | Every asset must have a unique ID or barcode. |
| **Transaction logging** | All movements must be recorded with timestamps. |
| **Receipt confirmation** | Assets must be confirmed received at destination. |
| **Return tracking** | All issued assets must be tracked for return. |
| **Condition reporting** | Asset condition must be recorded at each stage. |
| **Reconciliation requirement** | Issued and returned items must be reconciled. |
| **Exception handling** | Mismatches must be investigated and resolved. |
| **Access control** | Users can view and update only assigned inventory. |
| **Audit trail** | All transactions must be logged with user details. |
| **No silent deletion** | Historical records must be retained. |
| **Disposal approval** | Asset disposal requires authorized approval. |
| **Financial linkage** | Inventory value must link to financial records. |

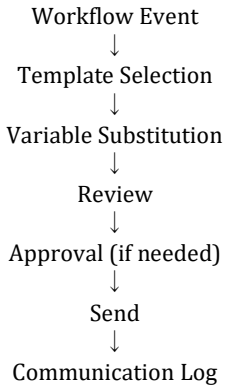
**System Outputs:**

| Output | Description |
|---|---|
| Asset master | Complete inventory catalog with specifications. |
| Requirement record | Material needs for projects and centres. |
| Allocation log | Asset assignment history. |
| Dispatch record | Shipment details and tracking information. |
| Receipt confirmation | Delivery verification at destination. |
| Usage log | Asset deployment and utilization records. |
| Return record | Asset return tracking and condition. |
| Reconciliation report | Issued, received and returned item matching. |
| Exception report | Mismatches and investigation status. |
| Inventory dashboard | Real-time inventory monitoring. |
| MIS update | Inventory metrics in project reporting. |

# M19 – Communication Management

| Category | Details |
|---|---|
| Module Name | Communication Management |
| Primary Users | Operations / Manager |
| Supporting Users | Admin, Project Manager, Centre User and Accounts / Finance |
| Purpose | Standardize, control and log all official project communications through approved templates, variables and workflows. |
| Process Flow | Workflow Event → Template → Variables → Review → Approval if needed → Send → Communication Log |
| Mandatory / Key Fields | Communication ID; Project; Centre; Template; Template Version; Recipient; Subject; Body; Attachment; Approved By; Sent By; Sent Time; Delivery Status |
| Business Rule | All official emails must be standardized through approved templates and fully logged for audit and compliance. |
| System Outputs | Communication record; template selection; variable substitution; approval trail; sent communication; delivery status; communication log and dashboard. |

**Communication Flow:**




**Detailed Process Table:**

| Step | Stage | Responsible Role | Description | System Output |
|---|---|---|---|---|
| 1 | Workflow Event | System | System or user triggers a communication requirement. | Event trigger |
| 2 | Template Selection | Operations | Select appropriate approved template for the event. | Template selection |
| 3 | Variable Substitution | System | Auto-populate template with project, centre and recipient data. | Draft communication |
| 4 | Review | Operations | Review content for accuracy and completeness. | Reviewed draft |
| 5 | Approval | Manager | Approve communication if required by policy. | Approval record |
| 6 | Send | System / Operations | Send communication through configured channel. | Sent communication |
| 7 | Communication Log | System | Log all details for audit and tracking. | Communication record |

**Communication Lifecycle:**

| Status | Meaning | Next Status |
|---|---|---|
| **Draft** | Communication is being prepared. | Review or Cancelled |
| **Review** | Communication is under review. | Approved, Rejected or Revised |
| **Approved** | Communication has been approved for sending. | Sent or Cancelled |
| **Sent** | Communication has been dispatched. | Delivered, Failed or Bounced |
| **Delivered** | Communication has been successfully delivered. | Read or Historical |
| **Failed** | Communication delivery failed. | Retry or Cancelled |
| **Bounced** | Communication bounced back. | Corrected or Cancelled |
| **Read** | Recipient has opened the communication. | Historical |
| **Cancelled** | Communication was cancelled before sending. | Historical record retained |

**Field-Level Specification:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Communication ID | System-generated | Text / ID | Unique identifier for the communication. |
| Project | Conditional | Lookup / ID | Project linked to the communication. |
| Centre | Conditional | Lookup / ID | Centre linked to the communication. |
| Template | Yes | Lookup / ID | Approved template used for the communication. |
| Template Version | System-generated | Number / Text | Version of the template used. |
| Recipient | Yes | Email / Contact | Recipient email address or contact. |
| Subject | Yes | Text | Email subject line. |
| Body | Yes | Long text | Email body content with variables substituted. |
| Attachment | Conditional | File | Attached documents or files. |
| Approved By | Conditional | User Lookup | User who approved the communication. |
| Sent By | System-generated | User / System | User or system that sent the communication. |
| Sent Time | System-generated | Date/time | Timestamp when communication was sent. |
| Delivery Status | System-generated | Dropdown | Draft, Sent, Delivered, Failed, Bounced or Read. |

**Additional Fields:**

| Field | Mandatory | Description |
|---|---|---|
| Communication Type | Yes | Email, SMS, portal notification or letter. |
| Priority | Yes | Normal, High or Urgent. |
| Trigger Event | Yes | Workflow event that initiated the communication. |
| CC Recipients | Conditional | Additional recipients copied. |
| BCC Recipients | Conditional | Hidden recipients. |
| Reply To | Conditional | Reply-to email address. |
| Variable Data | System-generated | JSON or structured data of substituted variables. |
| Review Comments | Conditional | Reviewer comments and suggestions. |
| Approval Comments | Conditional | Approver comments and conditions. |
| Retry Count | System-generated | Number of delivery retry attempts. |
| Last Retry Time | System-generated | Last retry timestamp. |
| Opened Time | Conditional | Timestamp when recipient opened the communication. |
| Clicked Links | Conditional | Links clicked within the communication. |
| Bounce Reason | Conditional | Reason for communication bounce. |
| Failure Reason | Conditional | Reason for delivery failure. |
| Linked Record | Conditional | Link to booking, allocation, invoice or other records. |
| Category | Yes | Booking, consent, document, audit, allocation, readiness, invoice or payment. |

**Template Categories:**

| Category | Example Templates |
|---|---|
| **Booking** | Centre booking invitation, booking confirmation, booking cancellation or replacement. |
| **Consent** | Consent request, consent reminder, consent acceptance or consent rejection. |
| **Document** | Document pending, document rework, document approval or document rejection. |
| **Audit** | Audit assignment, audit schedule, audit approval, audit rejection or audit findings. |
| **Allocation** | Allocation communication, allocation change, allocation confirmation or allocation lock. |
| **Readiness** | Readiness reminder, installation schedule, mock schedule or final readiness confirmation. |
| **Invoice** | Invoice request, invoice reminder, invoice approval or invoice rejection. |
| **Payment** | Payment confirmation, payment reminder, payment rejection or payment closure. |
| **Incident** | Incident notification, incident update or incident closure. |
| **General** | General announcement, policy update or other communications. |

**Role and Responsibility Matrix:**

| Function | Admin | Project Manager | Auditor / IT | Operations Executive | Centre User | Accounts / Finance |
|---|---|---|---|---|---|---|
| Create template | Full | Approve | Technical template input | Suggest | No | Finance template input |
| Edit template | Full | Approve | Technical input | Suggest | No | Finance input |
| Approve template | Full | Full approval | No | No | No | Finance approval |
| Select template | Full | Full | Technical templates | Full | View | Finance templates |
| Review content | Full | Full | Technical content | Full | View own communications | Finance content |
| Approve communication | Full | Full approval | Technical approval | Submit | No | Finance approval |
| Send communication | Full | Full | Technical communications | Full | Receive | Finance communications |
| View sent communications | Full | Full project | Technical scope | Assigned scope | Own communications | Finance communications |
| View delivery status | Full | Full project | Technical scope | Assigned scope | Own communications | Finance communications |
| Retry failed communication | Full | Full | Technical retry | Full | No | Finance retry |
| Export communication log | Full | Assigned project | Audit scope | Assigned scope | No | Finance scope |

**Role Responsibilities:**

**Operations**

- Select appropriate templates for workflow events.
- Review and verify communication content accuracy.
- Ensure correct variable substitution and recipient details.
- Send communications through approved channels.
- Monitor delivery status and handle failures.
- Maintain communication logs and audit trails.

**Project Manager**

- Approve high-priority or critical communications.

- Review communication templates and content quality.
- Monitor communication effectiveness and delivery rates.
- Handle escalated communication issues.
- Ensure compliance with communication policies.

**Auditor / IT Team Member**
- Provide technical content for audit and technical communications.
- Review technical accuracy of system-generated communications.
- Approve technical communications within their scope.
- Cannot approve non-technical communications outside their domain.

**Centre User**
- Receive official communications through the portal.
- View communication history for their centre.
- Respond to communications as required.
- Cannot send official communications on behalf of the organization.

**Accounts / Finance**
- Create and approve finance-related communications.
- Send invoice, payment and commercial communications.
- Monitor finance communication delivery and responses.
- Maintain finance communication records.

**Admin**
- Configure communication templates, variables and workflows.
- Approve template changes and new templates.
- View all communications and administrative actions.
- Maintain system controls and audit logs.

**Template Variable System:**

**Standard Variables**

| Variable | Description | Example |
|---|---|---|
| {ProjectName} | Project name | "SSC CGL 2026" |
| {ExamName} | Examination name | "Combined Graduate Level Examination" |
| {ExamDate} | Examination date | "15 October 2026" |
| {CentreName} | Centre name | "ABC Computer Institute" |
| {CentreAddress} | Centre address | "123 Main Street, City" |
| {ContactPerson} | Centre contact person | "Mr. John Smith" |
| {CandidateCount} | Number of candidates | "250" |
| {Shift} | Examination shift | "Morning Shift (9:00 AM - 12:00 PM)" |
| {BookingID} | Booking reference | "BKG-2026-001" |
| {AllocationID} | Allocation reference | "ALL-2026-001" |
| {InvoiceNumber} | Invoice number | "INV-2026-001" |
| {Amount} | Amount | "₹50,000" |
| {DueDate} | Due date | "30 September 2026" |
| {UserName} | Recipient name | "Mr. John Smith" |
| {CurrentDate} | Current date | "04 September 2026" |

**Template Example**

**Subject:** Booking Confirmation – {ProjectName} at {CentreName}
**Body:**
Dear {ContactPerson},

This is to confirm your centre booking for {ProjectName} at {CentreName}.

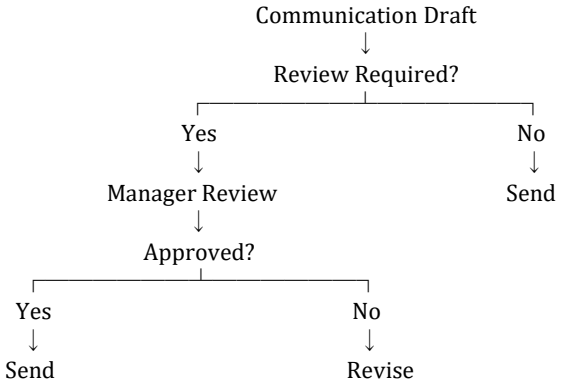
Exam Details:
- Examination: {ExamName}
- Date: {ExamDate}
- Shift: {Shift}
- Candidates: {CandidateCount}
- Booking ID: {BookingID}

Centre Address:
{CentreAddress}

Please confirm your availability and readiness by {DueDate}.

Best Regards,
Operations Team

**Approval Workflow:**




**Approval Rules:**

| Communication Type | Approval Required | Approver |
|---|---|---|
| Booking confirmation | No | Operations |
| Booking cancellation | Yes | Project Manager |
| Consent request | No | Operations |
| Document rejection | Yes | Project Manager |
| Audit findings | Yes | Audit Manager |
| Allocation change | Yes | Project Manager |
| Invoice rejection | Yes | Accounts Manager |
| Payment confirmation | No | Accounts |
| Critical incident | Yes | Senior Management |
| Policy announcement | Yes | Admin / Management |

**Notifications and Alerts:**

| Trigger | Recipient | Notification |
|---|---|---|
| Communication sent | Sender | Delivery confirmation. |
| Communication delivered | Sender | Delivery status update. |
| Communication failed | Sender / Manager | Failure alert with reason. |
| Communication bounced | Sender | Bounce notification with reason. |
| Communication read | Sender | Read receipt notification. |
| Approval required | Approver | Approval request notification. |
| Approval overdue | Approver / Manager | Approval reminder. |
| Template expired | Operations | Template renewal reminder. |
| Communication retry | System | Automatic retry for failed delivery. |
| High-priority communication | Manager | Critical communication alert. |

**Communication Dashboard:**

**Key KPIs**

| KPI | Description |
|---|---|
| Total communications | All communications sent in the selected period. |
| Successful deliveries | Communications successfully delivered. |
| Failed deliveries | Communications that failed to deliver. |
| Bounced communications | Communications that bounced back. |
| Read rate | Percentage of communications opened by recipients. |
| Pending approvals | Communications awaiting approval. |
| Template usage | Most frequently used templates. |
| Communication by category | Distribution across booking, consent, document and other categories. |
| Average delivery time | Time from send to delivery. |
| Retry success rate | Percentage of successful retries after failure. |

**Dashboard Filters:**

| Filter | Examples |
|---|---|
| Project | Project-specific communications. |
| Centre | Centre-wise communication history. |
| Category | Booking, consent, document, audit or other. |
| Status | Draft, sent, delivered, failed or bounced. |
| Date Range | Communication date range. |
| Template | Specific template filter. |
| Recipient | Recipient email or contact filter. |
| Delivery Status | Delivered, failed, bounced or read. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| **Template approval** | All templates must be approved before use. |
| **Version control** | Template versions must be tracked and retained. |
| **Variable validation** | Variables must be validated before substitution. |
| **Approval workflow** | Required communications must go through approval. |
| **Delivery tracking** | All communications must track delivery status. |
| **Audit trail** | Creation, review, approval, sending and delivery must be logged. |
| **Access control** | Users can only send communications within their scope. |
| **No silent deletion** | All communication records must be retained. |

| | |
|---|---|
| **Retry mechanism** | Failed communications must be retried automatically. |
| **Bounce handling** | Bounced communications must be corrected and resent. |
| **Confidentiality** | Sensitive communications must be restricted by role. |
| **Compliance** | All communications must comply with organizational policies. |

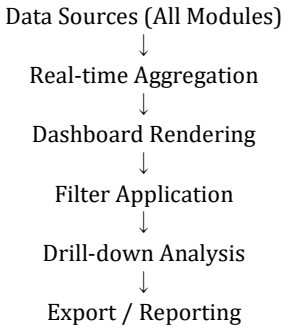
**System Outputs:**

| Output | Description |
|---|---|
| Communication record | Complete communication with all details. |
| Template selection | Approved template used for communication. |
| Variable substitution | Auto-populated content with project data. |
| Approval trail | Review and approval history. |
| Sent communication | Actual communication sent to recipient. |
| Delivery status | Real-time delivery tracking. |
| Communication log | Complete audit trail of all communications. |
| Dashboard metrics | Communication performance and effectiveness. |
| MIS update | Communication metrics in project reporting. |

## M20 – Management MIS

| Category | Details |
|---|---|
| Module Name | Management MIS & Dashboard |
| Primary User | Management (CEO, Senior Management, Project Managers) |
| Supporting Users | Admin, Operations, Accounts / Finance and Auditor / IT |
| Purpose | Provide real-time visibility, analytics and decision-support across all projects, centres and operational workflows through configurable dashboards and reports. |
| Process Flow | Real-time Data Aggregation → Dashboard Rendering → Filter Application → Drill-down Analysis → Export / Reporting |
| Mandatory / Key Fields | Filters: Client; Project; Exam; Date; City; Centre; Project Manager; Booking Executive; Status; Stage |
| Business Rules | Dashboards must display real-time data from all operational modules. Management must be able to drill down from summary metrics to transaction-level details. |
| System Outputs | Executive dashboard; project dashboard; centre dashboard; operational metrics; financial summary; exception reports; MIS exports; alert notifications. |

**Dashboard Architecture:**




**Dashboard Hierarchy:**

| Level | Dashboard | Primary Audience | Focus |
|---|---|---|---|
| 1 | Executive Dashboard | CEO, Senior Management | Organization-wide KPIs and health metrics. |
| 2 | Project Dashboard | Project Managers, Operations | Project-specific operational status. |
| 3 | Centre Dashboard | Operations, Centre Users | Centre-wise readiness and performance. |
| 4 | Functional Dashboard | Operations, Finance, Audit | Module-specific metrics (booking, audit, finance, etc.). |
| 5 | Exception Dashboard | Management, Operations | Overdue, blocked and escalated items. |

**Key Dashboard Metrics:**

**Executive Dashboard**

| Metric | Description | Calculation |
|---|---|---|
| **Total Projects** | All projects in the system. | Count of all project records. |
| **Active Projects** | Projects currently in execution. | Count of projects with status = Active. |
| **Centres Required** | Total centres needed across all projects. | Sum of centre requirements from allocations. |
| **Centres Booked** | Centres successfully booked. | Count of confirmed bookings. |
| **Pending Confirmations** | Centres awaiting consent or confirmation. | Count of bookings with pending consent. |
| **Pending Documents** | Centres with incomplete documentation. | Count of centres with pending mandatory documents. |

| | | |
|---|---|---|
| **Audit Pending** | Centres awaiting audit or with expired audit. | Count of centres requiring audit. |
| **Allocation Pending** | Projects awaiting client allocation. | Count of projects without approved allocation. |
| **Readiness %** | Overall readiness percentage across centres. | Average of all centre readiness percentages. |
| **Exam-Day Issues** | Active incidents during exam period. | Count of open incidents. |
| **Post-Exam Pending** | Projects with pending post-exam activities. | Count of projects in post-exam stage. |
| **Invoice Pending** | Centres with pending invoice submission or approval. | Count of pending invoices. |
| **Payment Outstanding** | Total unpaid amount across all projects. | Sum of outstanding payment amounts. |
| **Closure-Ready Projects** | Projects ready for final closure. | Count of projects with all stages complete. |

**Project Dashboard:**

| Metric | Description |
|---|---|
| Project stage | Current project lifecycle stage. |
| Centre coverage | Percentage of required centres booked. |
| Booking status | Booked, pending or cancelled centres. |
| Consent status | Accepted, pending or rejected consents. |
| Document compliance | Percentage of approved documents. |
| Audit status | Approved, pending or re-audit centres. |
| Allocation status | Allocation uploaded, validated or locked. |
| Readiness percentage | Centre-wise and overall readiness. |
| Mock status | Mock completed, pending or failed. |
| Incident count | Open, resolved and closed incidents. |
| Invoice status | Submitted, approved or pending invoices. |
| Payment status | Due, paid and outstanding amounts. |
| Project timeline | Planned vs actual milestone dates. |
| Exception summary | Overdue, blocked or escalated items. |

**Centre Dashboard:**

| Metric | Description |
|---|---|
| Centre status | Active, inactive or restricted. |
| Project assignments | Projects associated with the centre. |
| Booking history | Current and historical bookings. |
| Consent status | Current consent decision and history. |
| Document status | Approved, pending or rejected documents. |
| Audit status | Current audit validity and status. |
| Allocation details | Candidate count, date and shift. |
| Readiness status | Installation, configuration and mock status. |
| Incident history | Past and current incidents. |
| Invoice status | Submitted and approved invoices. |
| Payment status | Received, pending or outstanding payments. |
| Performance score | Centre performance rating based on historical data. |

**Filter Configuration:**

**Mandatory Filters**

| Filter | Description | Options |
|---|---|---|
| **Client** | Filter by client organization. | Multi-select dropdown. |
| **Project** | Filter by specific project. | Multi-select dropdown. |
| **Exam** | Filter by examination name or code. | Multi-select dropdown. |
| **Date** | Filter by date range. | Date picker with from/to. |
| **City** | Filter by city location. | Multi-select dropdown. |
| **Centre** | Filter by specific centre. | Multi-select dropdown. |
| **Project Manager** | Filter by responsible PM. | Multi-select user selection. |
| **Booking Executive** | Filter by booking owner. | Multi-select user selection. |
| **Status** | Filter by record status. | Multi-select status options. |
| **Stage** | Filter by project or workflow stage. | Multi-select stage options. |

**Optional Filters:**

| Filter | Description |
|---|---|
| **Priority** | Filter by priority level (Low, Medium, High, Critical). |
| **Severity** | Filter by severity (for incidents and exceptions). |
| **Overdue** | Filter only overdue items. |
| **Escalated** | Filter only escalated items. |
| **My Items** | Filter items assigned to the logged-in user. |
| **Team Items** | Filter items assigned to the user's team. |
| **Custom Date Range** | User-defined date range for analysis. |
| **Amount Range** | Filter by financial amount ranges. |

**Dashboard Components:**

**Summary Cards**

| Component | Purpose | Example |
|---|---|---|
| **KPI Cards** | Display key metrics with trend indicators. | Total Projects: 45 (+5 from last month). |
| **Progress Bars** | Show completion percentage. | Readiness: 78% complete. |
| **Status Indicators** | Color-coded status display. | Green = On Track, Red = Blocked. |
| **Trend Charts** | Show metric trends over time. | Booking trend over last 6 months. |

**Charts and Graphs**

| Chart Type | Use Case |
|---|---|
| **Bar Chart** | Compare metrics across categories (e.g., centres by city). |
| **Pie Chart** | Show distribution (e.g., booking status distribution). |
| **Line Chart** | Display trends over time (e.g., project completion trend). |
| **Stacked Bar** | Show composition and total (e.g., booking by status and project). |
| **Gauge Chart** | Display percentage against target (e.g., readiness %). |
| **Heat Map** | Show intensity or concentration (e.g., incidents by centre). |

**Data Tables:**

| Feature | Description |
|---|---|
| **Sortable Columns** | Click to sort by any column. |
| **Filterable Rows** | Quick filter within the table. |
| **Drill-down Links** | Click to view detailed records. |
| **Export Options** | Export to Excel, PDF or CSV. |
| **Pagination** | Navigate through large datasets. |
| **Column Customization** | Show/hide columns as needed. |

**Drill-Down Capability:**

**Drill-Down Hierarchy**




**Example Drill-Down Path:**

| Level | View | Example |
|---|---|---|
| 1 | Executive Dashboard | Total Projects: 45 |
| 2 | Project List | Click on "SSC CGL 2026" |
| 3 | Project Dashboard | Centres Required: 150 |
| 4 | Centre List | Click on "Centre A - Delhi" |
| 5 | Centre Details | Booking, documents, audit, readiness |
| 6 | Transaction Records | Specific booking ID, invoice ID |
| 7 | Supporting Documents | Uploaded files and approvals |

**Alert and Notification System:**

**Alert Types**

| Alert Type | Trigger | Recipient |
|---|---|---|
| **Threshold Alert** | Metric crosses defined threshold. | Project Manager, Operations. |
| **SLA Breach** | SLA deadline missed. | Owner, Manager. |
| **Overdue Alert** | Task or milestone overdue. | Owner, Manager. |
| **Exception Alert** | Critical exception identified. | Manager, Senior Management. |
| **Milestone Alert** | Key milestone approaching or reached. | Project Team. |
| **Financial Alert** | Payment overdue or budget exceeded. | Finance, Project Manager. |

**Alert Configuration**

| Parameter | Description |
|---|---|
| **Threshold Value** | Numeric value that triggers the alert. |
| **Comparison Operator** | Greater than, less than, equal to. |
| **Time Window** | Time period for evaluation. |
| **Recipients** | Users or roles to notify. |
| **Notification Channel** | Email, SMS, portal notification. |
| **Escalation Rule** | Escalation path if not addressed. |
| **Repeat Frequency** | How often to repeat the alert. |

**Report Types:**

**Standard Reports**

| Report Name | Purpose | Frequency |
|---|---|---|
| **Project Status Report** | Overall project health and progress. | Weekly. |
| **Centre Readiness Report** | Centre-wise readiness status. | Daily during exam period. |
| **Booking Summary Report** | Booking status across all projects. | Weekly. |
| **Audit Compliance Report** | Audit status and validity. | Monthly. |
| **Document Compliance Report** | Document submission and approval status. | Weekly. |
| **Incident Report** | Incident summary and resolution status. | Daily/Weekly. |
| **Financial Summary Report** | Invoice, payment and outstanding summary. | Monthly. |
| **Payment Outstanding Report** | Detailed outstanding payment analysis. | Weekly. |
| **Closure Readiness Report** | Projects ready for closure. | Monthly. |
| **Exception Report** | Overdue, blocked and escalated items. | Daily. |

**Custom Reports**

| Feature | Description |
|---|---|
| **Ad-hoc Queries** | Build custom queries on demand. |
| **Saved Views** | Save frequently used report configurations. |
| **Scheduled Reports** | Automatically generate and email reports. |
| **Pivot Tables** | Dynamic data summarization and analysis. |
| **Cross-Module Reports** | Combine data from multiple modules. |
| **Trend Analysis** | Historical trend comparison. |

**Role-Based Dashboard Access:**

| Role | Dashboard Access |
|---|---|
| **CEO / Senior Management** | Executive Dashboard, all projects, all centres, financial summary. |
| **Project Manager** | Assigned projects, all centres in projects, project financials. |
| **Operations Executive** | Assigned projects and centres, operational metrics. |
| **Accounts / Finance** | Financial dashboards, invoice and payment reports. |
| **Auditor / IT** | Audit dashboards, technical readiness, incident reports. |
| **Centre User** | Own centre dashboard, assigned tasks and communications. |
| **Admin** | All dashboards, system metrics, user activity. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| **Real-time data** | Dashboards must reflect current system state. |
| **Data accuracy** | Metrics must be calculated from verified transaction data. |
| **Access control** | Users can only view data within their authorized scope. |
| **Drill-down security** | Drill-down must respect role-based permissions. |
| **Audit trail** | Dashboard access and report exports must be logged. |
| **Data refresh** | Automatic data refresh at configured intervals. |
| **Performance** | Dashboards must load within acceptable time limits. |
| **Export control** | Report exports must be restricted by role. |
| **Data retention** | Historical data must be retained per policy. |
| **Alert management** | Alerts must be configurable and actionable. |
| **Mobile access** | Dashboards must be accessible on mobile devices. |
| **Customization** | Users must be able to customize their dashboard views. |

**System Outputs:**

| Output | Description |
|---|---|
| Executive Dashboard | Organization-wide KPIs and health metrics. |
| Project Dashboard | Project-specific operational status. |
| Centre Dashboard | Centre-wise performance and readiness. |
| Exception Reports | Overdue, blocked and escalated items. |
| Financial Reports | Invoice, payment and outstanding analysis. |
| Compliance Reports | Audit, document and process compliance. |
| Trend Analysis | Historical trends and projections. |
| Alert Notifications | Real-time alerts and escalations. |
| MIS Exports | Excel, PDF and CSV report exports. |
| Mobile Dashboards | Mobile-optimized dashboard views. |

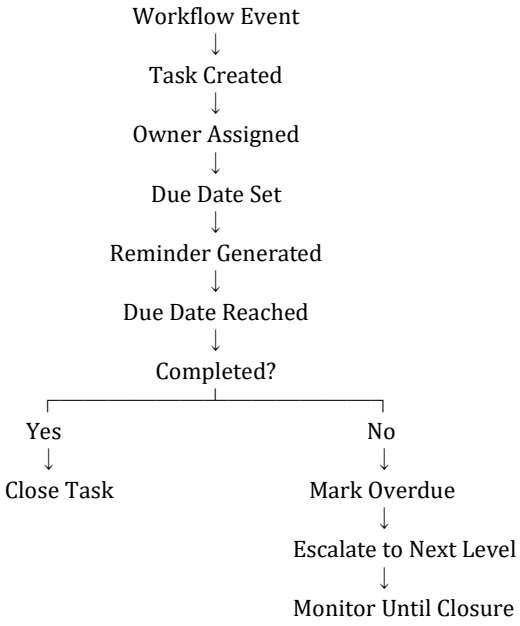
**Dashboard Performance Requirements:**

| Requirement | Target |
|---|---|
| Dashboard Load Time | < 5 seconds for standard dashboards. |
| Data Refresh Rate | Real-time or near real-time (max 5-minute delay). |
| Concurrent Users | Support 100+ concurrent dashboard users. |
| Report Generation | < 30 seconds for standard reports. |
| Export Performance | < 60 seconds for large exports. |
| Mobile Responsiveness | Full functionality on mobile devices. |
| Data Accuracy | 100% accuracy in metric calculations. |
| Uptime | 99.5% dashboard availability. |

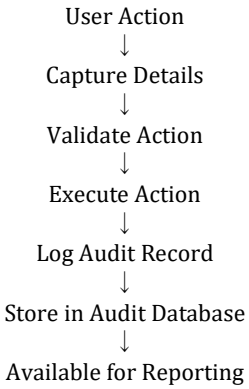
# M21 – Notification, Escalation & Audit Trail

| Category | Details |
|---|---|
| **Module Name** | Notification, Escalation & Audit Trail |
| **Primary Users** | All Users / System |
| **Supporting Users** | Admin, Project Manager, Operations, Accounts / Finance, Auditor / IT and Centre User |
| **Purpose** | Provide system-wide notification, task assignment, reminder, escalation and complete audit trail for all critical actions and workflow events. |
| **Process Flow** | Event → Task → Owner → Due Date → Reminder → Overdue → Escalation |
| **Mandatory / Key Fields** | WHO; WHAT; WHEN; OLD VALUE; NEW VALUE; REASON; APPROVER |
| **Business Rules** | Every critical action must be logged with complete audit trail. Overdue work must be automatically escalated according to configured rules. |
| **System Outputs** | Event log; task assignment; notification; reminder; escalation record; audit trail; compliance report; dashboard alerts. |

**Notification & Escalation Flow:**




**Audit Trail Flow:**



**Event Types:**

**Workflow Event**

| Event Category | Example Events |
|---|---|
| **Project Events** | Project created, project stage changed, project status changed, project closed. |
| **Booking Events** | Centre selected, soft lock created, booking confirmed, booking cancelled, replacement booking created. |
| **Consent Events** | Consent request sent, consent accepted, consent rejected, consent discussion requested. |
| **Document Events** | Document uploaded, document verified, document approved, document rejected, document rework requested. |
| **Audit Events** | Audit assigned, audit visit completed, audit submitted, audit approved, audit rejected, re-audit initiated. |
| **Allocation Events** | Allocation uploaded, allocation validated, allocation approved, allocation locked, allocation changed. |
| **Commercial Events** | Cost sheet created, cost sheet approved, cost sheet revised, commercial override approved. |
| **Invoice Events** | Invoice request created, invoice uploaded, invoice verified, invoice approved, invoice rejected. |
| **Payment Events** | Payment due, payment received, payment verified, payment cleared, payment overdue. |
| **Readiness Events** | Installation completed, configuration completed, mock completed, final readiness approved. |
| **Incident Events** | Incident raised, incident escalated, incident resolved, incident closed, incident reopened. |
| **Ticket Events** | Ticket raised, ticket assigned, ticket resolved, ticket closed, SLA breached. |
| **Inventory Events** | Asset allocated, asset dispatched, asset received, asset returned, asset reconciled. |
| **Communication Events** | Communication sent, communication delivered, communication failed, communication approved. |
| **User Events** | User created, user activated, user deactivated, role changed, permission changed. |

**Notification Types:**

| Notification Type | Channel | Priority | Example |
|---|---|---|---|
| System Notification | Portal | Normal | Task assigned, status updated. |
| Email Notification | Email | Normal | Booking confirmation, document approval. |
| SMS Notification | SMS | High | Critical incident, payment overdue. |
| Escalation Notification | Email + SMS | High | SLA breach, critical exception. |
| Digest Notification | Email | Low | Daily summary, weekly report. |
| Immediate Alert | Portal + Email + SMS | Critical | System downtime, security incident. |

**Field-Level Specification:**

**Notification Fields**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Notification ID | System-generated | Text / ID | Unique identifier for the notification. |
| Event Type | Yes | Dropdown | Type of event triggering the notification. |
| Event ID | Yes | Lookup / ID | ID of the event or record triggering notification. |
| Notification Type | Yes | Dropdown | System, email, SMS, escalation or digest. |
| Recipient | Yes | User / Email / | Person or group receiving the notification. |

| | | | |
|---|---|---|---|
| | | Contact | |
| Subject | Yes | Text | Notification subject or title. |
| Message | Yes | Long text | Notification content and details. |
| Priority | Yes | Dropdown | Low, Medium, High or Critical. |
| Status | Yes | Dropdown | Pending, Sent, Delivered, Failed or Read. |
| Sent Time | System-generated | Date/time | Timestamp when notification was sent. |
| Delivered Time | Conditional | Date/time | Timestamp when notification was delivered. |
| Read Time | Conditional | Date/time | Timestamp when notification was read. |
| Channel | Yes | Dropdown | Portal, email, SMS or push notification. |
| Retry Count | System-generated | Number | Number of delivery retry attempts. |
| Failure Reason | Conditional | Text | Reason for delivery failure. |

**Escalation Fields:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Escalation ID | System-generated | Text / ID | Unique identifier for the escalation. |
| Original Task ID | Yes | Lookup / ID | ID of the original task or record. |
| Escalation Level | Yes | Number | Current escalation level (1, 2, 3, etc.). |
| Escalated From | Yes | User Lookup | User or role from which escalation originated. |
| Escalated To | Yes | User / Role Lookup | User or role to which escalation is made. |
| Escalation Reason | Yes | Long text | Reason for escalation. |
| Escalation Time | System-generated | Date/time | Timestamp of escalation. |
| Due Date | Yes | Date/time | New due date after escalation. |
| Status | Yes | Dropdown | Pending, In Progress, Resolved or Closed. |
| Resolution | Conditional | Long text | Resolution details after escalation. |
| Resolved Time | Conditional | Date/time | Timestamp when escalation was resolved. |

**Audit Trail Fields:**

| Field | Mandatory | Field Type | Description |
|---|---|---|---|
| Audit ID | System-generated | Text / ID | Unique identifier for the audit record. |
| WHO | Yes | User Lookup | User who performed the action. |
| WHAT | Yes | Text | Action performed (create, update, delete, approve, etc.). |
| WHEN | System-generated | Date/time | Timestamp of the action. |
| OLD VALUE | Conditional | JSON / Text | Previous value before change. |
| NEW VALUE | Conditional | JSON / Text | New value after change. |
| REASON | Conditional | Long text | Reason or justification for the action. |
| APPROVER | Conditional | User Lookup | User who approved the action, if applicable. |
| Module | Yes | Dropdown | Module where action occurred (booking, audit, allocation, etc.). |
| Record ID | Yes | Lookup / ID | ID of the affected record. |
| Record Type | Yes | Text | Type of record (booking, centre, project, etc.). |
| IP Address | System-generated | Text | IP address from which action was performed. |
| Session ID | System-generated | Text | User session identifier. |
| Action Status | Yes | Dropdown | Success, Failed, Rejected or Pending Approval. |

| | | | |
|---|---|---|---|
| Failure Reason | Conditional | Text | Reason for action failure, if applicable. |

**Escalation Rules:**

**Escalation Matrix**

| Priority | Level 1 | Level 2 | Level 3 | Level 4 |
|---|---|---|---|---|
| Low | Operations Executive (24 hours) | Senior Operations (48 hours) | Project Manager (72 hours) | Senior Management (96 hours) |
| Medium | Operations Executive (12 hours) | Senior Operations (24 hours) | Project Manager (36 hours) | Senior Management (48 hours) |
| High | Operations Executive (4 hours) | Senior Operations (8 hours) | Project Manager (12 hours) | Senior Management (24 hours) |
| Critical | Operations Executive (1 hour) | Project Manager (2 hours) | Senior Management (4 hours) | CEO (8 hours) |

**Escalation Triggers**

| Trigger | Condition | Action |
|---|---|---|
| Overdue Task | Due date passed without completion. | Escalate to next level. |
| SLA Breach | Response or resolution SLA missed. | Immediate escalation. |
| Critical Incident | Critical severity incident raised. | Escalate to senior management. |
| Payment Overdue | Payment past due date. | Escalate to finance manager. |
| Audit Exception | Critical audit finding identified. | Escalate to audit committee. |
| Security Incident | Security breach or unauthorized access. | Immediate escalation to admin. |
| System Failure | System downtime or critical failure. | Escalate to IT and management. |
| Compliance Violation | Regulatory or policy violation. | Escalate to compliance officer. |

**Audit Trail Categories:**

**Data Change Audit**

| Action | Logged Details |
|---|---|
| Create | WHO, WHAT (create), WHEN, NEW VALUE, Module, Record ID. |
| Update | WHO, WHAT (update), WHEN, OLD VALUE, NEW VALUE, REASON. |
| Delete | WHO, WHAT (delete), WHEN, OLD VALUE, REASON, APPROVER. |
| View | WHO, WHAT (view), WHEN, Record ID (for sensitive data). |
| Export | WHO, WHAT (export), WHEN, Record Type, Export Format. |

**Workflow Audit**

| Action | Logged Details |
|---|---|
| Submit | WHO, WHAT (submit), WHEN, Record ID, Status change. |
| Approve | WHO, WHAT (approve), WHEN, Record ID, APPROVER, Comments. |
| Reject | WHO, WHAT (reject), WHEN, Record ID, APPROVER, REASON. |
| Reassign | WHO, WHAT (reassign), WHEN, From User, To User, REASON. |
| Escalate | WHO, WHAT (escalate), WHEN, From Level, To Level, REASON. |
| Override | WHO, WHAT (override), WHEN, REASON, APPROVER, OLD VALUE, NEW VALUE. |

**Access Audit**

| Action | Logged Details |
|---|---|
| Login | WHO, WHEN, IP Address, Session ID, Login Status. |
| Logout | WHO, WHEN, Session ID, Session Duration. |
| Password Change | WHO, WHEN, Password Reset (yes/no), Method. |
| Permission Change | WHO, WHAT, WHEN, OLD PERMISSIONS, NEW PERMISSIONS, APPROVER. |
| Role Change | WHO, WHAT, WHEN, OLD ROLE, NEW ROLE, APPROVER. |
| Failed Login | WHO, WHEN, IP Address, Failure Reason, Attempt Count. |

## Notification Configuration:

### Template-Based Notifications

| Component | Description |
|---|---|
| Template ID | Unique identifier for notification template. |
| Event Trigger | Workflow event that triggers notification. |
| Recipient Rule | Rule to determine notification recipient. |
| Subject Template | Subject line with variables. |
| Body Template | Message body with variables. |
| Channel Rule | Channel selection based on priority or type. |
| Timing Rule | When to send (immediate, scheduled, digest). |
| Escalation Rule | Escalation path if not acknowledged. |

### Variable Substitution

| Variable | Description | Example |
|---|---|---|
| {UserName} | Recipient name | "Mr. John Smith" |
| {ProjectName} | Project name | "SSC CGL 2026" |
| {CentreName} | Centre name | "ABC Computer Institute" |
| {TaskType} | Task type | "Document Verification" |
| {DueDate} | Due date | "15 October 2026" |
| {OverdueDays} | Number of overdue days | "3 days" |
| {EscalationLevel} | Current escalation level | "Level 2" |
| {ActionRequired} | Required action | "Please approve the document" |
| {LinkURL} | Direct link to record | "https://portal.com/booking/123" |

## Role-Based Notification Rules:

| Role | Notification Scope |
|---|---|
| CEO / Senior Management | Critical escalations, project health alerts, financial exceptions, security incidents. |
| Project Manager | All project notifications, team escalations, milestone alerts, budget alerts. |
| Operations Executive | Assigned tasks, booking alerts, document reminders, centre communications. |
| Accounts / Finance | Invoice alerts, payment reminders, outstanding notifications, financial approvals. |
| Auditor / IT | Audit assignments, technical exceptions, incident alerts, compliance notifications. |
| Centre User | Centre-specific notifications, task assignments, document requests, payment status. |
| Admin | System alerts, user management, security notifications, configuration changes. |

## Audit Trail Reporting:

### Standard Audit Reports

| Report Name | Purpose | Frequency |
|---|---|---|
| User Activity Report | All actions performed by users. | On-demand / Scheduled. |
| Data Change Report | All create, update, delete operations. | Daily / Weekly. |
| Workflow Audit Report | All workflow transitions and approvals. | Daily / Weekly. |
| Access Report | Login, logout and access attempts. | Daily. |
| Exception Report | Failed actions and rejected operations. | Daily. |
| Escalation Report | All escalations and resolutions. | Weekly. |
| Compliance Report | Regulatory and policy compliance audit. | Monthly. |
| Security Report | Security-related events and incidents. | Daily / Weekly. |

**Audit Query Capabilities**

| Query Type | Description |
|---|---|
| User-wise Query | All actions by specific user. |
| Date-range Query | Actions within specific date range. |
| Module-wise Query | Actions in specific module. |
| Record-wise Query | All actions on specific record. |
| Action-type Query | Specific action types (create, update, delete). |
| Exception Query | Failed or rejected actions. |
| Approval Query | All approval and rejection actions. |
| Access Query | Login, logout and access attempts. |

**Key Business Controls:**

| Control | Requirement |
|---|---|
| Mandatory audit | All critical actions must generate audit trail. |
| Immutable logs | Audit records cannot be modified or deleted. |
| Complete capture | WHO, WHAT, WHEN, OLD VALUE, NEW VALUE, REASON, APPROVER must be captured. |
| Real-time logging | Audit records must be created immediately after action. |
| Secure storage | Audit data must be stored securely with access control. |
| Retention policy | Audit records must be retained per organizational policy. |
| Escalation automation | Overdue work must automatically escalate. |
| Notification reliability | Notifications must be delivered through multiple channels. |
| Acknowledgment tracking | Notification read and acknowledgment must be tracked. |
| Escalation hierarchy | Clear escalation path must be defined and enforced. |
| Access control | Audit trail access must be restricted by role. |
| Export control | Audit exports must be authorized and logged. |
| Performance | Audit logging must not impact system performance. |
| Compliance | Audit trail must meet regulatory and compliance requirements. |

**System Outputs:**

| Output | Description |
|---|---|
| Event Log | Complete record of all workflow events. |
| Task Assignment | Automated task creation and assignment. |
| Notification | Multi-channel notification delivery. |
| Reminder | Automated reminders for pending tasks. |
| Escalation Record | Complete escalation trail and history. |
| Audit Trail | Immutable record of all critical actions. |
| Compliance Report | Regulatory and policy compliance reports. |
| Dashboard Alerts | Real-time alerts on management dashboards. |
| User Activity Report | Comprehensive user activity tracking. |
| Exception Report | Failed actions and system exceptions. |
| Access Report | Login, logout and access attempt logs. |
| MIS Update | Notification and audit metrics in reporting. |

## Notification & Escalation Dashboard

**Key Metrics**

| Metric | Description |
|---|---|
| Total Notifications | All notifications sent in selected period. |
| Delivered Notifications | Successfully delivered notifications. |
| Failed Notifications | Notifications that failed to deliver. |
| Read Rate | Percentage of notifications read by recipients. |
| Average Response Time | Time from notification to action. |
| Total Escalations | All escalations in selected period. |
| Escalation Resolution Rate | Percentage of resolved escalations. |
| Average Escalation Time | Time from escalation to resolution. |
| Overdue Tasks | Tasks past their due date. |
| SLA Breaches | Tasks where SLA was breached. |
| Critical Alerts | Critical priority notifications and escalations. |
| Audit Trail Volume | Number of audit records created. |

**Dashboard Filters**

| Filter | Examples |
|---|---|
| Date Range | Notification or audit date range. |
| User | Specific user or role. |
| Module | Booking, audit, allocation, finance, etc. |
| Priority | Low, Medium, High or Critical. |
| Status | Pending, Sent, Delivered, Failed or Read. |
| Escalation Level | Level 1, 2, 3 or 4. |
| Action Type | Create, update, delete, approve, reject, etc. |
| Event Type | Workflow event category. |

This M21 module ensures that every critical action across the system is logged with complete audit trail (WHO, WHAT, WHEN, OLD VALUE, NEW VALUE, REASON, APPROVER), and every overdue task is automatically escalated through the configured hierarchy, providing complete accountability and transparency across all operations.

## 8. Role-Based Operating Model

| Role | End-to-End Responsibility |
|---|---|
| **Admin** | Users, roles, permissions, scopes, master data, eligibility & blacklist, configuration, emergency controls, audit logs, template management, system-wide reporting and cross-module administrative actions. |
| **Project Manager** | Client requirement validation, project creation, team assignment, booking oversight, audit oversight, allocation approval, readiness monitoring, exam-day escalation, post-exam activities, commercial approval, invoice oversight, payment monitoring and project closure. |
| **Operations Executive** | Centre search, eligibility check, booking lock, centre contact, consent follow-up, document coordination, allocation execution, calling tasks, readiness coordination, exam-day operations, incident management, ticket handling, inventory coordination and post-exam activities. |
| **Audit Manager** | Auditor assignment, audit monitoring, technical scan oversight, audit review, rework/re-audit approval, audit validity management, compliance monitoring and audit reporting. |
| **Auditor / IT Team Member** | Centre visit, checklist completion, evidence collection, photos, documents verification, technical/IP scan, observations, audit submission and technical exception resolution; cannot approve own audit. |
| **Accounts / Finance** | Commercial review and approval, cost sheet management, invoice verification, amount due tracking, payment processing, verification, clearance, financial closure, reconciliation and finance reporting. |
| **Centre User** | Invitation acceptance, centre details submission, consent response, document upload, audit support, allocation confirmation, manpower confirmation, readiness support, mock participation, exam-day support and invoice submission. |

## 9. Permission Matrix

| Function | Admin | Manager | Operations | Auditor | Accounts | Centre |
|---|---|---|---|---|---|---|
| **User Management** | Full | View | No | No | No | No |
| **Centre Master** | Full | Approve/View | View/Suggest | View | No | Submit |
| **Blacklist** | Full | Approve | View status | View | No | No |
| **Booking** | Full | Approve | Create/Edit | View | View | Consent |
| **Documents** | Full | Approve | Verify | Audit docs | Commercial docs | Upload |
| **Audit** | Full | Review | View | Create/Submit | No | Support |
| **Allocation** | Full | Approve | Create/Edit | View | View | View |
| **Cost Sheet** | Full | Approve | Prepare | No | Edit after approval | View if allowed |
| **Payment** | Full | View | View | No | Full | View where allowed |
| **Calling** | Full | Assign/Monitor | Execute | No | No | Respond |
| **Incident** | Full | Escalate | Create/Handle | View audit issues | No | Report/Respond |
| **Ticket** | Full | Assign/Monitor | Create/Handle | Technical tickets | Finance tickets | Raise/Respond |
| **Inventory** | Full | Approve | Create/Manage | View technical | View financial | Confirm receipt |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Communication** | Full | Approve | Send | Technical | Finance | Receive |
| **MIS** | Full | Full project | Own scope | Audit scope | Finance scope | Own centre |
| **Notification & Audit** | Full | Full project | Own activity | Audit activity | Finance activity | Own activity |

## 10. Cross-Cutting Business Rules

- No silent deletion: Cancelled bookings, rejected documents, old cost sheets and old allocations remain in history.
- Submission is distinct from verification and approval: Centre uploads do not equal approval; internal verification is mandatory.
- Final allocation follows Draft → Validated → Approved → Locked: Later changes require a reasoned change request and approval.
- Commercial records are versioned: Approved V1 → Revised V2 → Re-approval.
- Soft booking locks prevent parallel work: Database constraints provide final concurrency protection.
- Confidential restriction reasons are visible only to authorized roles: Eligibility status visible, confidential reasons restricted.
- All important transactional records have unique ID, creator, timestamps and status: System-generated fields mandatory.
- Approval workflows capture submitter, reviewer, decision, dates, rejection/rework reason and approver: Complete approval trail required.
- Important modifications capture old value, new value, changed by, changed time and reason: Audit trail for all changes.
- Audit independence: Auditor cannot approve own audit.
- Payment segregation: Operations views payment status; Accounts creates/edits financial information.
- Incident vs Ticket separation: Exam-day incidents managed separately from general operational tickets.
- Readiness dependency: Final readiness requires completed installation, configuration, mock and resolved exceptions.
- Communication standardization: All official emails use approved templates and are fully logged.

## 11. Non-Functional and Data Principles

- **Role-based access control**: Enforced at module, project, client, centre and financial-data scope.
- **Timestamped workflows**: Every workflow transition timestamped and attributable to user or system event.

- **File versioning**: Files require versioning, metadata, expiry tracking and controlled access.
- **Transactional dashboards**: Dashboards use transactional data rather than manually prepared reports.
- **Integration-ready design**: Support future WhatsApp, email-to-ticket and telephony integrations through APIs/events.
- **Database separation**: Separate master, transaction, workflow, document, audit and reporting concerns.
- **Real-time data**: Dashboards reflect current system state with minimal delay.
- **Audit immutability**: Audit records cannot be modified or deleted.
- **Performance standards**: Dashboards load within 5 seconds; data refresh in near real-time.
- **Mobile accessibility**: Full functionality on mobile devices.
- **Data retention**: Historical data retained per organizational policy.
- **Security compliance**: Audit trail meets regulatory and compliance requirements.
- **Scalability**: Support 100+ concurrent dashboard users.
- **Export control**: Report exports restricted by role and logged.

## 12. Delivery Roadmap

| Release | Scope | Exit Condition |
|---|---|---|
| **Release 1 – MVP** | M01 (User Management), M02 (Project Management), M03 (Centre Master), M04 (Eligibility), M05 (Booking & Locking), M06 (Centre Consent), M07 (Document Management), M11 (Commercial), M12 (Invoice Request), M20 (MIS core), M21 (Notification & Audit core) | Projects and bookings operate from a controlled portal with history, approvals, documents, commercial control and management visibility. |
| **Release 2 – Operational Control** | M08 (Audit), M09 (Technical Scan), M10 (Allocation), M13 (Accounts & Payment), M14 (Calling), M15 (Mock & Readiness), M16 (Incidents), M17 (Tickets), M18 (Inventory), M19 (Communication enhanced), M20 (MIS enhanced) | Audit, allocation, readiness, incidents, tickets, finance and accountability are integrated. |
| **Release 3 – Integrations** | WhatsApp API, email-to-ticket, telephony integration, advanced analytics, predictive reporting | External communication and telephony are connected without changing core records. |

## 13. Developer and QA Acceptance Criteria

**Booking & Centre Management**

- Two users cannot successfully create conflicting bookings.
- Cancelled bookings retain history.
- Replacement bookings are linked to cancelled bookings.
- Centre Master maintains old/new data history.
- Centre submissions require internal verification.
- Soft locks expire automatically after configured time.
- Database-level validation prevents duplicate bookings.

**Document & Audit Management**

- Mandatory documents cannot be skipped.
- Document versions are retained; rejected documents remain in history.
- Audit expiry is automatically tracked.
- Duplicate MAC addresses are automatically flagged.
- Auditor cannot approve own audit.
- Audit validity is configurable by type, client and exam.

**Allocation & Commercial Management**

- Allocation files can be imported and validated.
- Final allocation can be locked.
- Allocation changes after lock require change request and approval.
- Commercial revisions create new versions.
- Approved cost sheets cannot be directly edited.
- Commercial approval requires Finance review.

**Finance & Payment Management**

- Accounts data is protected from unauthorized operational editing.
- Invoice matching with approved cost sheet is enforced.
- Payment records require verification before clearance.
- Outstanding amounts are system-calculated.
- Partial payments are supported with balance tracking.

**Accountability & Communication**

- Calling tasks have owner, due date and outcome.
- Callback tasks are automatically tracked.
- Management can see pending and overdue tasks.
- Communication templates are standardized and versioned.
- All official communications are logged.

**Incident, Ticket & Readiness**

- Incidents have owner, response and resolution timestamps.
- Tickets have SLA tracking.
- Critical incidents escalate automatically.
- Readiness percentage is system-calculated.
- Final readiness blocks on critical exceptions.

**Management & Audit**

- Management can see complete project stage.
- Payment status is visible according to permission.
- Every critical action has an audit trail.
- Audit records capture WHO, WHAT, WHEN, OLD VALUE, NEW VALUE, REASON and APPROVER.
- Dashboards display real-time data from all modules.

## 14. Management View

The CEO and management should see a single-screen view that connects Operations, Finance and Compliance to the Centre Master and Centre/Venue Portal. The primary dashboard should expose:

**Executive Dashboard Metrics**

- **Total Projects** and **Active Projects**
- **Centres Required** and **Centres Booked**
- **Pending Confirmations** (consent, documents, audit)
- **Document and Audit Status** (approved, pending, rejected)
- **Allocation Status** (uploaded, validated, locked)
- **Readiness Percentage** (overall and centre-wise)
- **Exam-Day Issues** (open incidents, critical escalations)
- **Post-Exam Pending** (invoices, payments, closure activities)
- **Invoice Status** (submitted, approved, rejected)
- **Payment Outstanding** (total, overdue, by project)
- **Closure-Ready Projects** (all stages complete)

**Dashboard Filters**

- Client
- Project
- Exam
- Date Range
- City
- Centre
- Project Manager
- Booking Executive
- Status
- Stage

**Drill-Down Capability**

- Executive Summary → Project List → Project Details → Centre List → Centre Details → Transaction Records → Supporting Documents

**Alert Types**

- Threshold alerts (metrics crossing defined limits)
- SLA breach alerts
- Overdue task alerts
- Critical exception alerts
- Milestone alerts
- Financial alerts (payment overdue, budget exceeded)

## 15. Recommended Submission Structure

**Page 1: Executive Summary and Business Objective**

- Current problems (Excel dependency, duplicate bookings, scattered data, document gaps, commercial confusion, weak accountability, limited visibility)
- Proposed solution (centralized Operations Management Portal)
- Business benefits (single source of truth, controlled workflows, complete history, management visibility)
- Operating principle: **CAPTURE ONCE → VALIDATE → APPROVE → EXECUTE → TRACK → ESCALATE → CLOSE → RETAIN HISTORY**

**Page 2: Current Problem → Proposed Solution → Business Benefit**

| Current Problem | Proposed Solution | Business Benefit |
|---|---|---|
| **Duplicate centre bookings** | Centre locking with soft/hard locks | No conflicting bookings |
| **Scattered centre information** | Single Centre Master | Authoritative centre database |
| **Document gaps and follow-up** | Centralized document management | Complete document trail |
| **Commercial misunderstandings** | Versioned cost sheets with approval | Clear commercial terms |
| **Invoice and payment follow-up** | Integrated invoice and payment workflow | Automated tracking |
| **Excel dependency** | Centralized portal with workflows | Single source of truth |
| **Weak accountability** | Task assignment with owner and due date | Clear ownership |
| **Limited management visibility** | Real-time MIS and dashboards | Complete project visibility |

**Page 3: Complete End-to-End Process Flow**

CLIENT REQUIREMENT → PROJECT CREATION → EXAM/DATE/SHIFT/CITY/CANDIDATE → CENTRE IDENTIFICATION → ELIGIBILITY + BLACKLIST CHECK → CENTRE LOCK → BOOKING → CENTRE CONSENT → DOCUMENT COLLECTION → AUDIT CHECK → AUDIT/TECHNICAL VERIFICATION → FINAL CENTRE APPROVAL → CLIENT ALLOCATION → ALLOCATION VALIDATION → FINAL ALLOCATION LOCK → COMMERCIAL/COST SHEET → APPROVAL → READINESS PLAN → CALLING/FOLLOW-UP → INSTALLATION/CONFIGURATION → MOCK DAY → FINAL READINESS → EXAM DAY → INCIDENT/TICKET MANAGEMENT → POST-EXAM → INVOICE REQUEST → ACCOUNTS/PAYMENT → PROJECT MIS → PROJECT CLOSURE

**Cancellation Flow:**

CANCELLED BOOKING → Cancellation Reason → Original Record Retained → Replacement Centre → New Booking → Both Records Linked

**Page 4: Module Architecture**

| Module | Purpose | Primary User |
|---|---|---|
| M01 | Authentication, Users & Roles | Admin |
| M02 | Client & Project Management | PM / Operations |
| M03 | Centre Master | Operations / Admin |
| M04 | Eligibility & Blacklist | Manager / Admin |
| M05 | Booking & Locking | Operations |
| M06 | Centre Consent | Operations / Centre |
| M07 | Document Management | Operations / Centre |
| M08 | Audit & Compliance | Auditor / Manager |
| M09 | Technical / IP Scan | Auditor / IT |
| M10 | Allocation | Operations / PM |
| M11 | Commercial & Cost Sheet | Operations / Finance |
| M12 | Invoice Request | Operations / Centre |
| M13 | Accounts & Payment | Accounts |
| M14 | Calling & Accountability | Operations |
| M15 | Mock & Readiness | Operations / Centre |
| M16 | Exam-Day Incidents | Operations |
| M17 | Ticket Management | Operations / Manager |
| M18 | Inventory & Logistics | Operations |

| | | |
|---|---|---|
| M19 | Communication | Operations / Manager |
| M20 | MIS & Dashboard | Management |
| M21 | Notification, Escalation & Audit Trail | All / System |

## 16. Final Project Outcome

When implemented against this blueprint, the portal will become the organization's operational system of record:

- Each project will have clear stage, status, team, timeline and completion criteria.
- Each centre will have authoritative master data, eligibility status, booking history and performance record.
- Each booking will have controlled workflow, consent trail, document compliance and audit history.
- Each approval will have submitter, reviewer, decision, timestamp and reason.
- Each document will have version history, verification status and expiry tracking.
- Each audit will have independent review, technical scan and compliance record.
- Each allocation will be validated, locked and change-controlled.
- Each readiness activity will have owner, status, percentage and exception tracking.
- Each incident will have response time, resolution and closure verification.
- Each invoice will match approved commercials and trigger payment workflow.
- Each payment will have verification, clearance and outstanding tracking.
- Each communication will use approved templates and maintain complete log.
- Each action will have complete audit trail with WHO, WHAT, WHEN, OLD VALUE, NEW VALUE, REASON and APPROVER.

Management will have single-screen visibility into project stage, centre coverage, pending confirmations, document and audit status, allocation and readiness, exam-day issues, invoices, payment outstanding and closure readiness.

**The portal will eliminate:**

- Duplicate centre bookings
- Scattered centre information
- Document gaps and follow-up chaos
- Commercial misunderstandings
- Invoice and payment confusion
- Excel dependency
- Weak accountability
- Limited management visibility

**The portal will provide:**

- Single source of truth
- Controlled workflows
- Complete history
- Clear ownership
- Automated escalations
- Real-time visibility
- Audit compliance
- Operational excellence